A discrete portfolio learner: what it is, how it compares with gradient-trained networks, and what it learns

Claude (Anthropic)*
prompted by Kaarel Hänni

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Abstract

We describe v3, the best learner of a study that compares discrete learners with gradient-descent-trained networks at no more FLOPs than the networks use. v3 is a stacked portfolio in which no feature is fitted by gradient descent. On Boolean inputs its main member, DFG, admits Fourier characters and indicators of signed sums of inputs by union-bound tests against the residual. On images its main members are boosted trees over sums of pixels in sets chosen from pixel correlations, so the pixel layout is never read.

On a fresh 12-target Boolean suite drawn from a generator written blind to the sum-feature design (v3 was developed on two other draws), v3 needs clearly less data than the per-target best of three pre-registered network variants on 10 targets and more on none (2.8× less, geometric mean). Its predecessor v2 does the same, and on five of the ten, default boosting or logistic regression also beats the networks. On two fresh 10-class QuickDraw sets with permuted pixels, v3 has 12.8% and 12.6% test error at $n = 60000$. It clearly beats an ensemble of three MLPs at all six (set, n) pairs, but it beats a CPU-matched ensemble of wider MLPs, and v2, by less than 10%, which the pre-registered rule calls a tie. A small CNN given the pixel layout has lower error wherever both ran (clearly at one of three). The held-out sets are small, with one seed at $n = 60000$.

Verified descriptive analyses (seed 0) show what v3 learns. DFG's model contains the target's exact signed block on 49 of 54 signed-sum targets. For targets that are functions of one signed sum, its test would admit the block at $n = 128$, but the block is found later the less of the target's variance sits at Fourier degrees 1–2 (Spearman -0.86); residue classes mod 3, with almost none there, are never found. On QuickDraw and MNIST the pixel sets are patches of the true grid, and a booster on "bars" along recovered lines has the lowest out-of-fold error of all members in every analysed fit.

*Written by an AI system in a series of sessions prompted and steered by Kaarel Hänni. Not reviewed by a human. The interpretability analyses were checked by independent AI verifiers, and the paper's claims by independent AI checkers against the result files and records, as described in Appendix D; the comparisons were not re-audited for leakage, compute or reference strength. Code, result files, the pre-registration and the underlying records are on the `claude/nifty-mccarthy-j0lw2a` branch of https://github.com/kaarelh/AI-works, folder `gd-circuit-growth/`.

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1 Introduction

The question. Can a learner without continuous feature parameters do at least as well as gradient-descent-trained networks, with the same training data and no more FLOPs? This paper reports the latest learner of a ten-part study, whose Parts 6–10 have asked this question, and what that learner does.

The learner. v3 is a stacked portfolio: a convex stacker combines the class probabilities of several members, and the input type decides which members run (Section 2). On Boolean inputs the member that matters is DFG (discrete feature growth). It enumerates Fourier characters and indicators of signed sums of inputs, admits each only when its correlation with the current residual passes a union-bound test over its whole class, and reads the admitted features out linearly. On images the main members are gradient-boosted trees over a fixed bank of pixel subset sums, whose pixel sets are chosen from pixel correlations without reading the layout; v3's new member, the *bar booster* `cgbt`, adds sums along short lines recovered from those correlations. In familiar terms, DFG is a greedy, test-gated version of the low-degree algorithm, close to forward stepwise regression, and v3's image side is boosting on a fixed, data-derived feature bank (Section 2.1). No member fits a feature direction by gradient descent (Section 2.5). Appendix A lists every name used in this paper.

Part 10. Part 10 was pre-registered (Section 3.4). Three builders working on development data proposed components; a fixed rule adopted two of them, a wider sum-feature search for DFG and the bar booster; and a second fixed rule chose v3 over its predecessor v2 as the subject of this paper. v3 was frozen before any of its held-out results were read.

What was found. Table 1 gives the held-out results. By the pre-registered verdict rules, v3 cannot be told apart from v2 on any held-out set. What Part 10 adds shows only as consistently small image gains (v3's error is lower than v2's in all 14 seed-paired held-out fits, a description made after the results were in, but by less than the 10% threshold at every cell) and on development suites. On the toy suite v3's margin over the networks is v2's; on images v3 is clearly better than the control `mlp_pool`, which v2 only ties.

- **The other data** [Numerical] (Sections 4.2 and 4.4). On the nine other toy suites with network results, all development data for v3, its suite means against the per-suite best network range from -1.50 to -2.44, and a network is clearly better on one target only (h6_00, a residue class mod 3). On 24 of the 26 tabular datasets v3 runs v1's code.
- **What DFG learns** [Numerical] (seed 0; Section 6.1). On 49 of 54 targets built from signed sums, DFG's model comes to hold the target's exact signed blocks. On the 25 functions of one signed sum the test would admit the true block at $n = 128$ every time, and the further data needed to find it grows as the target's share of variance at Fourier degrees 1–2 shrinks (Spearman $\rho = -0.86$; Figure 4). The three residue classes mod 3 have almost none ($W_1 + W_2 \leq 0.0002$); the search never proposes their block, although at $n = 16384$ it would pass the test at 134–155 times the threshold.
- **What the image members encode** [Numerical] (Section 6.2). On QuickDraw and MNIST the pooling sets are local neighbourhoods of the true pixel grid; on Fashion-MNIST some include the mirror pixel. The bar booster has the lowest out-of-fold error of all members in every analysed fit, and holds most of the boosters' stacking weight in the analysed QuickDraw and MNIST fits.

held-out set	reference	v3's result	v3 against v2	limits
`dlheld7`: 12 Boolean targets, $d = 30$ (Section 4.1)	`nets_best`, the per-target best of three pre-registered network variants (hindsight)	mean $\log_2 n^*$ ratio -1.50 over the 11 targets both learn (about $2.8\times$ less data; per-target s.d. 0.80, s.e. of the mean 0.24); clearly better on 10, worse on 0; wins from -0.53 (f7_10) to -2.93 (f7_00)	mean -0.004; no target differs clearly	one target generator; on 5 of the 10 wins `hgb` or `lin` also beats `nets_best`; the verdict is a fixed ± 0.25 threshold with no noise term
`qd_c`, `qd_d`: 10-class QuickDraw, permuted pixels (Section 5.1)	`mlp_xl`, three 1024-wide MLPs, CPU-matched, 9.2 budgets ($n = 60000$); `mlp_strong` at every n	12.76% and 12.64% error at $n = 60000$, 9.6% and 8.6% below `mlp_xl` (two ties); clearly better than `mlp_strong` at 6 of 6 cells	six ties, lower at all six (mean $\log_2$ ratio -0.10)	one seed at $n = 60000$ (binomial s.e. about 0.34 points); a CNN that sees the layout (seed 0) has lower error at all three cells where both ran, clearly at one
HELDOUT2: 10 tabular datasets (Section 4.4)	`mlp_best`, the per-cell best Part 7–8 MLP (hindsight)	mean $\log_2$ error ratio -0.168; lower mean error on 8 of 10 datasets (sign test $p = 0.11$)	the same code	v3 runs v1's stack there, unchanged since Part 8; DFG runs on none of these datasets

Table 1: Held-out results at a glance [Numerical]. A cell is a (set, n) pair; "clearly better" and "tie" are the pre-registered verdicts of Section 3.3. The per-target s.d. and the s.e. of the mean on `dlheld7` treat the 11 targets as independent draws from the generator. Source: `study/part10/LEARNER_RESULTS.json`.

Scope.

- *What is held out.* For v3 as a whole only `dlheld7`, `qd_c` and `qd_d` (Table 4). `dlheld7`'s targets are fresh, but v3's toy builder had seen two other draws from the same generator (Section 3.1); the image sets come from one source, with one seed at $n = 60000$. The earlier held-out sets measure v3 but do not test it. On tabular data the held-out claim is about v1's code path, which v3 runs there.
- *The references* are standard-recipe networks without augmentation or architecture search, not tuned per target (Section 3.2); the per-target and per-cell minima over their variants are hindsight choices that favour them. The CNN is context, not a reference.
- *"Discrete"* means that no member fits a feature direction by gradient descent (Section 2.5). Boosting is functional gradient descent over trees, so the comparison is "with gradient-descent-trained networks", not "with gradient descent".

Claims are labelled [Numerical] (measured here, with the source file named) or [Heuristic] (a reading or a back-of-envelope argument, not tested here). There are no theorems in this paper.

Background: Parts 1–9. From Part 6 on, each part froze its learners before running them on the held-out data that it reports as its main evidence.

- *Parts 1–5* [6] found that rich-regime networks on Boolean targets grow features greedily and pay in samples: for a k-bit parity on d inputs their n^*, the training-set size at 10% test error, grows like $d^{1.0}$, $d^{1.9}$ and $d^{2.4}$ for $k = 2, 3, 4$, close to the leap and correlational-SQ rate d^{k-1} [1, 2] for $k \le 3$, somewhat below it for $k = 4$, and far from the $\approx k \log d$ of an Occam learner.
- *Part 6* built DFG [5], which on a blind 7-target suite needed less data than the networks on

every target (median factor 3.1), with at most 0.26 of one network run's FLOPs per fit.

- *Part 7* took DFG to standard tasks, where it was about level with a plain MLP on tabular data, behind a regularized one, and clearly behind on images at large n.
- *Part 8* stacked DFG with LightGBM, ExtraTrees, nearest-neighbour votes and logistic regression (v1), which needed clearly less data than the best network variant on 9 of 10 fresh toy targets and had lower mean error than the hindsight-best MLP on 8 of 10 fresh tabular datasets ($p = 0.11$), but was behind a regularized MLP ensemble on images at $n \geq 30000$.
- *Part 9* added sum blocks to DFG and permutation-invariant image members (v2), which needed clearly less data than the best network variant on 10 of 12 targets of a suite written blind to that design and more on none; on two fresh QuickDraw sets it beat a strong MLP ensemble at $n \geq 10000$, but a CPU-matched ensemble of wider MLPs and an MLP given v2's pooled features tied it at $n = 60000$, and a small CNN that sees the pixel layout was clearly better at $n = 10000$.

Contributions.

1. What v3 is, how it relates to familiar learners and what in it is discrete (Section 2), and how it was built and evaluated (Section 3).
2. A held-out test of v3 on a fresh toy suite and two fresh image sets against the pre-registered references (Sections 4.1 and 5.1).
3. A comparison with every reference run on the study's other sets (MNIST excepted), with each set's status for v3 (Sections 4.2, 4.4 and 5.2; in-sample detail in Appendix B).
4. A descriptive, independently verified account of what v3's fitted members contain (Section 6; per-target detail in Appendix C).
5. What kind of learner v3 turned out to be, why it works where it does, and its limits (Section 7); how Part 10 was run (Appendix D).

2 What kind of learner it is

2.1 In one paragraph

v3 (`hyb_portfolio3` in the code) is a *stacked portfolio*. Each member produces class probabilities. A convex stacker, fitted on the members' out-of-fold log-probabilities, combines them. Which members run is decided by the input type, n and the budget (Table 2).

- *On Boolean inputs* (the toy suites, and binary tabular data with two classes), the member that matters is *DFG*: discrete feature growth. DFG enumerates a structured dictionary of discrete features (Fourier characters $\chi_S(x) = \prod_{i \in S} x_i$ and indicator functions of integer signed sums $\sum_{i \in S} s_i x_i$). It admits a feature only when that feature's correlation with the current residual passes a statistical test, with a union bound over the whole class the feature was drawn from. It then reads the admitted features out by least squares, refined by logistic regression.
- *On images*, the members that carry most stacking weight at $n \geq 10^4$ are *gradient-boosted trees over a fixed bank of subset sums of pixels* (at $n = 1000$, nearest-neighbour votes on the same sums carry more). The subsets are built from the training data alone: blobs of the pixels most correlated with each pixel, and short "bars" along lines recovered from the correlations. The pixel layout is never read.

- *Everywhere*, LightGBM, ExtraTrees, k-nearest-neighbour votes and multinomial logistic regression run as well.

No member fits a continuous feature direction by gradient descent. The fitted continuous quantities are readout weights (including those of a logistic regression on the inputs), tree leaf values and stacking weights. Two components are borderline, and they are discussed in Section 2.5.

How v3 relates to familiar learners [Heuristic]. The portfolio as a whole is an AutoML-style dispatch by input type [5]. Its parts compare with known learners as follows.

- *Gradient boosting.* Like boosting, DFG fits the residual greedily and stagewise. Unlike boosting, its candidates come from an explicit finite dictionary (characters and signed-sum blocks), not from fitted trees. A feature enters only if it passes a union-bound significance test over its dictionary class; that test, not shrinkage or early stopping, is the regulariser. And all coefficients are refitted jointly by least squares after each round, as in orthogonal matching pursuit or forward stepwise regression. On images, by contrast, v3's main members *are* gradient-boosted trees (LightGBM). What is new there is only their input, a fixed bank of pixel subset sums chosen by correlation ranks. The calibration reference `hgb`, default boosting on the raw pixels (Section 5.1), shows what the bank and the stack add.
- *A neural network.* A network fits continuous feature directions by SGD and pays in samples to find a sparse feature. DFG fits no direction by gradient descent: it searches a combinatorial dictionary, rounds a coarse spectral estimate to an exact signed subset, and tests that subset. Its sample cost is therefore logarithmic in the dictionary size, and it pays in enumeration FLOPs instead (Section 7.2). On images the bank plays the role of a first layer whose receptive fields are chosen from pixel correlations instead of learned, with no weight sharing.
- *Fourier learners.* The low-degree algorithm [20] estimates every Fourier coefficient up to degree k. DFG is a greedy, budgeted, test-gated and adaptive version of it: it extends admitted characters and Walsh-transforms the inputs already in use. It adds a second dictionary, indicators of signed sums. These represent a symmetric function of k signed inputs with about $k+1$ group indicators, where a Fourier expansion needs up to 2^k characters of degree up to k.

2.2 The portfolio and its budget

Stacking [27]. Let $L_m(x) \in \mathbb{R}^C$ be member m's out-of-fold log-probabilities. The score is

$$s_c(x) = b_c + \textstyle\sum_m w_m L_m(x)_c, \qquad w_m \geq 0,$$

fitted by L2-penalized multinomial log-loss, which is convex. Out-of-fold predictions come from 10-fold cross-validation for $n \leq 500$, 5-fold for $n \leq 3000$, and one stratified holdout of $\min(n/5, 6000)$ points above that. Leave-one-out votes serve for nearest-neighbour members and out-of-bag votes for ExtraTrees. Members with positive weight are refitted on all n, except the boosters on inputs with $nd > 10^7$, where the holdout-trained booster is kept.

Budget guard. Every fit has a FLOP budget (Section 3.3), and every member's FLOPs are counted: the image boosters by the compute audit's per-tree formula, and Part 8's `gbt` member by its original formula, which Part 9's audit found low by 1.14–1.46× at $n = 3000$. A member is skipped when its projected cost would take the total above 90% of the budget (95% for logistic regression). DFG's out-of-fold fits share 35% of the budget and its final fits at most 50%. Prediction is not charged.

member	what it is	when it runs
`dfg`, `dfg_forced`	DFG v1.3 (Section 2.3); `dfg_forced` is its small-n variant, which forces the best candidate when nothing passes	2 classes, every column binary or categorical, ≤ 256 bits; `dfg_forced` at $n \leq 512$
`ggbt`, the blob booster	LightGBM on [x, pooled subset sums $p_i^{(k)}(x) = \sum_{j \in S_i(k)} x_j$ over the k-pixel pooling sets, $k = 4, 9, 16, 300$ prototype-pair projections] (Section 2.4)	images (every column continuous, $d \geq 256$)
`cgbt`, the bar booster	LightGBM on [x, the same pooled sums, bar sums $\sum_{j \in \text{bar}} x_j$ for bar half-lengths 2 and 4]	images
`gknn_k`	nearest-neighbour votes with 1, 3, 9 or 27 neighbours on the 9-pixel pooled sums	images
`gbt`, `et`	`gbt`: LightGBM (100 rounds) on the raw columns (categoricals as categories; rows 0.5, columns 0.25 per tree when $nd > 10^7$); `et`: ExtraTrees (100 trees) on the standardized encoding	always
`knn_<metric>_k`	nearest-neighbour votes on standardized columns (and on a common scale when column scales differ)	always
`lin_C`	multinomial L2 logistic regression on the standardized encoding, $C \in \{0.05, 1\}$ ($C = 1$ only for $n > 10^4$)	always

Table 2: The members of v3. On toy inputs the image members do not run. On images, DFG does not run. Any member is skipped when the budget guard (Section 2.2) says so. Versions: v1 (Part 8) is DFG v1.1 with the classical members (`gbt`, `et`, `knn`, `lin_C`); v2 (Part 9) has DFG v1.2, which added sum blocks, and the image members `gknn` and `ggbt`; v3 (Part 10) has DFG v1.3, which adds weighted blocks, forced blocks and a beam search, and the bar booster `cgbt`.

2.3 DFG v1.3 on Boolean inputs

DFG works with the residual $r = y - \hat{y}$ of a least-squares fit of the ± 1 labels on the current features, and with $s = \text{rms}(r)$. It runs in rounds. Each round searches several candidate classes and admits everything that passes its test. The fit is then redone, and the next round starts from the new residual. Training stops when a round admits nothing, when every training sign is right, after 8 rounds, when the feature count reaches $n/2$, or once 75% of DFG's budget share is spent.

The hypothesis class. DFG predicts $\text{sign}\, F(x)$ with

$$F(x) = \sum_{S \in \mathcal{S}} a_S \, \chi_S(x) + \sum_{B \in \mathcal{B}} \sum_{v} b_{B,v} \, \chi_{T_B}(x) \, [t_B(x) \in G_{B,v}] + \sum_{u \in \mathcal{U}} c_u ((u \cdot x)^2 - |u|^2) + \sum_{i=1}^{d} g_i x_i. \quad (1)$$

Here $\mathcal{S}$ is the set of admitted characters (the constant included), $\mathcal{B}$ the set of admitted sum blocks, with statistic t_B and value groups $G_{B,v}$ (below), and $\mathcal{U}$ the set of admitted spectral directions (below). $\chi_{T_B} = 1$ unless B is modulated, and a product block uses the joint cells of its two statistics. The last term is the kernel background, $g_i = \frac{1}{n} \sum_t r_t x_{t,i}$ on the final residual: one Hebbian step over all d inputs. The coefficients a, b and c are fitted by least squares on the p admitted features, then refined by logistic Newton steps (below).

Characters. The candidates are every χ_S with $|S| \leq 4$ over all d inputs (degree 3 after the first round; degrees 5–6 on a budget-limited subsample), extensions $S \cup \{j\}$ and $S \cup \{j, l\}$ of admitted

characters, and every subset of the at most 12 inputs already in use (a Walsh transform). A character from a class of size N is admitted when

$$\left|\tfrac{1}{n}\textstyle\sum_t r_t \chi_S(x_t)\right| \;>\; s\sqrt{2\ln(4KN/\delta)/n}, \qquad K = 4,\ \delta = 0.1. \tag{2}$$

Here $K = 4$ is the maximum degree of the global search, and a *class* is a set of candidates enumerated together: all characters of one degree, the extensions of the current features, or the subsets of one size of the inputs in use. N is its size. The small classes (extensions and local subsets) have low thresholds. Features on inputs already in use are therefore cheap: the discrete counterpart of the composition effect seen in the study's networks.

Sum blocks. A block is a subset S with signs $s_i \in \{\pm 1\}$ and the integer statistic $t(x) = \sum_{i\in S} s_i x_i$, or $|t(x)|$ when the block is *folded*. The training values of t are cut into contiguous groups of at least $\max(3, 0.005n)$ points. The block's features are the group indicators $[t(x) \in \text{group } v]$. Every function of t, so every symmetric function of the signed inputs in S, is a linear combination of them, up to merged rare values. Four variants extend this:

- *modulated* blocks multiply the indicators by an admitted character χ_T, with $T \cap S = \emptyset$;
- *product* blocks take the joint cells of two blocks on disjoint supports;
- *split* blocks replace a mixed-sign candidate by the product block of its positive-sign part and a block searched on its negative-sign part (the joint cells of t_A and t_B);
- v1.3's *weighted* blocks use $u = t + a\,\tau(x)$ with $a \in \{\pm 2, \pm 3\}$ and τ a literal, a parity of two inputs or an AND of two literals.

A block is admitted when the residual's projection on its indicators passes a χ^2 tail bound [17] with a union bound over its whole class:

$$T = \textstyle\sum_v n_v\, \bar{r}_v^2/\hat{\sigma}^2 \;>\; m + 2\sqrt{m\,\tau_k} + 2\tau_k, \tag{3}$$

where n_v and $\bar{r}_v$ are the size and mean residual of group v, m is the number of occupied groups minus one, $\tau_k = \ln\!\big(4K\binom{d}{k}2^{k-1}\cdot 2\,(1 + \#\text{modulators})\,k_{\max}/\delta\big)$ for a block on k inputs, and $k_{\max} = 24$. Weighted blocks pay a further $\ln(2(4d + 20\binom{d}{2})) \approx 9.8$ nats for their inner term. The bound covers every (S, s) of every size, both fold variants and every modulator, so the adaptive search does not invalidate it. It is not free: at $d = 30$ no 7-input block can pass below $n \approx 67$, and below $n \approx 75$ with three or more value groups.

Spread signal. Two block moves catch signal that no single character carries. The *linear block* admits all remaining degree-1 characters at once when $n\sum_i c_i^2/s^2$, with $c_i = \frac{1}{n}\sum_t r_t x_{t,i}$ over those inputs, exceeds the χ^2 tail at level δ; it carries halfspaces. *Spectral features* take the eigenvectors u of the residual pair matrix (below), at most four, whose eigenvalues clear the noise edge $2s\sqrt{d/n}\,(1 + \sqrt{2\ln(1/\delta)/d})$ in absolute value (only for $n \geq 4d$), and add $(u \cdot x)^2 - |u|^2$ as features.

The block search uses only the training residual. Candidates come from five sources:

- prefixes of the inputs ordered by the top eigenvectors of the residual pair matrix $M = \frac{1}{n}\sum_t r_t x_t x_t^\top$ with its diagonal removed. If $f = g(\sum_{i\in S} s_i x_i)$ with g even, then $\mathbb{E}[f\, x_i x_j] = c\, s_i s_j$ for distinct $i, j \in S$ and 0 for every other pair, so M is about $c\, ss^\top$ on S and its top eigenvector gives both the support and the signs. If c is near 0 (no Fourier weight at degree 2), this seed sees nothing; Section 6.1 relates DFG's late and failed discoveries to this;

- prefixes ordered by the degree-1 coefficients $\frac{1}{n}\sum_t r_t x_{t,i}$;
- greedy growth from the 3 strongest pairs and the 2 strongest single inputs;
- in the additive search, v1.3's weighted candidates: weighted eigenvector prefixes, an inner term on the strongest pair, and extensions of the two best unweighted candidates;
- at $n \le 128$, in the first additive search of a round (v1.3), a width-32 beam over (S, s).

The best candidates are refined by single add, remove and sign-flip moves, then tested on all n. A block must also pass the test on the residual after the round's admitted characters are projected out.

Readout and small n. DFG predicts the sign of F in (1): the least-squares fit on its p features, refined by 15 Newton steps of L2-regularized logistic regression when $n \ge 8p$, plus the kernel background. With few points the union bound passes nothing. The forced variant then takes the best character, or (v1.3, $n \ge 48$) the best block, anyway. The stacker decides by cross-validation whether to trust it.

Why this is sample-efficient where it works [Heuristic]. With ρ the correlation of a feature with the residual in units of s, the test (2) admits it and no null competitor once $n \gtrsim 2\ln(4KN/\delta)/\rho^2$. For a k-sparse character $N \le d^k$, so the sample cost is about $2k \ln d/\rho^2$: the Occam rate. Enumeration costs about nN FLOPs, which at $d = 30$ fits inside one network run up to degree 4. Gradient-trained networks pay the other way round, close to d^{k-1} samples for a k-parity with $k \le 3$ (somewhat fewer at $k = 4$) [2, 6].

Where it cannot work. The test bounds detection, but the search must also *propose* the right candidate. Block candidates are seeded by degree-1 and degree-2 statistics of the residual. A target whose block has almost no Fourier weight at degrees 1 and 2 is invisible to the seeds at every n of the study's grid (≤ 16384); for h6_00 and f7_05 the seeds' statistics are zero in expectation. A residue class mod 3 of a 10–14-input count is such a target (Section 6.1).

2.4 The image members

The comparator on images is an MLP, which is invariant to a fixed permutation of its input pixels. So the image members may not use the pixel layout, and the QuickDraw sets are stored with their columns permuted. The members recover what they need of the layout from the training data.

Pooling sets. For every non-constant column i, $S_i(k)$, the k-pixel pooling set of i, is the set of the k columns with the largest correlation with column i on the training rows (i itself first). Ties are broken by a column fingerprint, never by index, so permuting the columns permutes the sets exactly. The pooled feature is the subset sum $p_i^{(k)}(x) = \sum_{j \in S_i(k)} x_j$. On images the most correlated pixels are the nearest ones, so $S_i(k)$ is in effect a small blob around pixel i (Section 6.2).

Bars (v3). The 8 most correlated pixels of i are paired greedily into 4 axes, least correlated pair first. On a grid, opposite neighbours are the least correlated pair. A bar of half-length L is i, the two ends of an axis, and $L-1$ further steps in each direction. Each step goes "straight on": from j, reached from i', it moves to the partner of i' in j's own pairing. A bar is a subset sum along a recovered line through i, of up to 5 ($L = 2$) or 9 ($L = 4$) pixels, with 4 orientations per pixel. *Example*: on an exact grid the eight neighbours of (r, c) pair into the four axes $\{(r-1, c), (r+1, c)\}$, $\{(r, c-1), (r, c+1)\}$ and the two diagonals. The $L = 2$ bar on the vertical axis is $(r-2, c), \ldots, (r+2, c)$: from $(r+1, c)$,

object	how it is chosen	continuous parameters?
characters χ_S, blocks $(S, s, \text{fold}, a, \tau,$ modulator)	enumeration, greedy or beam search, statistical tests	no
group boundaries of a block, histogram bins	training quantiles / counts	no
pooling sets, bars	correlation ranking of pixels on training rows	no
DFG readout weights	least squares, then logistic Newton steps	yes (linear readout)
logistic regression `lin_C`: weights on the standardized inputs	convex log-loss (L-BFGS)	yes (linear readout on the inputs)
tree splits; leaf values	greedy split search on histograms; Newton steps (boosting)	leaf values
stacking weights	convex log-loss	yes (≤ 30 weights)
DFG's spectral features $(u \cdot x)^2 - \lvert u \rvert^2$	top eigenvectors of the residual pair matrix	borderline (directions u)
`ggbt`'s prototype pairs $x_a - x_b$	two random training points of different classes	borderline (directions from data)

Table 3: What v3 chooses discretely and what it fits continuously.

reached from (r, c), "straight on" is the partner of (r, c) in $(r + 1, c)$'s own pairing, namely $(r + 2, c)$. The orientation labels are canonical per pixel but not consistent across pixels. The bars are therefore per-position features: nothing is shared across positions, as a convolution would share it.

The boosters. `ggbt`, the blob booster, and `cgbt`, the bar booster, are LightGBM [15] with 31 leaves, learning rate 0.1, 63 histogram bins, and per-tree column fractions 0.12 and 0.05. Below 5000 training points they run 100 rounds. Above that they stop early on the error of an inner validation split (patience 50, 150–400 rounds). Every split is a gate $[f(x) > \theta]$, with f a pixel, a blob sum, a bar sum or (in `ggbt`) a prototype-pair projection $(x_a - x_b) \cdot x$ for two random training points of different classes, and θ a histogram-bin boundary. A tree is a conjunction of such gates along each path. The ensemble is a sum of trees fitted by functional gradient descent (boosting).

2.5 What is discrete, and what is not

Table 3 lists what v3 fits.

- *No feature is learned by gradient descent.* Features are enumerated or built from data statistics, and admitted by tests or by greedy split search. The continuous fits are linear readouts and leaf values. The one readout on the inputs themselves, the member `lin_C`, is a convex logistic regression fitted by a quasi-Newton method (L-BFGS).
- *Boosting is functional gradient descent.* So the image members are best described as "a fixed, data-derived feature hierarchy read out by gradient-boosted trees".
- *The comparison is with gradient-descent-trained networks, not with gradient descent.* On images, the pooling sets and bars recover the pixel grid from correlations (Section 6.2). The members use locality in spirit, though not in the letter. A comparator that is told the layout, such as a small CNN, is reported as context only.

3 How it is evaluated

This section gives the tasks, the references, the verdict rules and the budgets, and what is held out for v3. The names of learners, members, references, sets and targets are collected in Appendix A.

3.1 Tasks

Toy Boolean targets. Inputs are uniform on $\{\pm 1\}^d$ with $d = 30$, and labels are ± 1 values of a target f. There are 11 suites, 138 targets in all, summarized in Table 4 and defined in `study/boolfuncs.py`.

- `designed` (21) and `random` (20). Parities, majorities, halfspaces, staircases, compositions, a multiplexer, a DNF, a decision list and random small formulas.
- `dlheld`, `dlheld2`, `dlheld3` (12, 7, 10). Dense halfspaces and majorities, χ_5, multiplexers, symmetric functions such as $[|x_1 + \cdots + x_9| \leq 1]$, mixtures of parities with gates, and random formulas.
- `dlsymdev` (8). Symmetric and signed-sum targets used for development.
- `dlheld4`, `dlheld5`, `dlheld6` (12 each). Signed sums with threshold, band, interval, residue-class and arbitrary accept sets, some mixed with parities and gates, plus 2–3 non-symmetric controls per suite.
- `dlfamdev` and `dlheld7` (12 each). Two fresh draws (seeds 61616 and 70707) from the blind family below.

The blind family is a random generator of signed-sum targets, written in Part 9 by an agent that had not seen DFG's sum-block design (source: `study/part9/heldout6/FAMILY.md`). It has 9 structured slots and 3 controls: sums, sums times a parity, sums with a gate, and compositions of sums. Each block's accept set is drawn uniformly from five types, one of them a residue class mod 3 or 4, except in the inner-term form $u = s + a\tau$, which uses a threshold or a band. `dlheld6`, `dlfamdev` and `dlheld7` are three draws from it. v3's toy builder developed on `dlheld6` and `dlfamdev`, so `dlheld7` tests v3 on fresh targets from a distribution its designer had seen.

The *sum-structured targets* are the 54 targets of `dlsymdev`, `dlheld4`–`dlheld7` and `dlfamdev`, minus their 14 non-symmetric controls; Tables 5, 17 and 18 define them. Every suite puts the relevant inputs first. The discrete learners therefore see the columns in a fresh random order in every fit, so that tie-breaking by column index cannot help them.

Tabular data. There are 26 PMLB datasets [23]: 5 for development, 11 held out in Part 7 (including mushroom), and 10 fresh in Part 8 (HELDOUT2). They have continuous, categorical, binary and mixed features and 2–26 classes.

Images. MNIST [19], Fashion-MNIST [28] and five 10-class sets of QuickDraw bitmaps [13]. The QuickDraw sets use 50 categories drawn at random from the 345 (seed 9191) and split into five disjoint sets of ten:

- `qd_dev`, development;
- `qd_a` and `qd_b`, held out in Part 9;

set	size	status for v3	network references
toy `designed`, `dlsymdev`, `dlfamdev`	41	development; in-sample for the toy component	variants run there (none on `dlfamdev`)
toy `random`	20	development check for DFG in Part 6, evaluated with Part 8's held-out suites; development data in Part 10 (not used by the toy builder)	`nets`, `nets_wd`
toy `dlheld`–`dlheld5`	53	held out for earlier versions; development data in Part 10 (not used by the toy builder)	`nets`, `nets_wd`; `nets_cewd` from `dlheld3` on
toy `dlheld6`	12	held out in Part 9; used by the Part 10 toy builder	all five variants
toy `dlheld7`	12	**held out**	`nets_wd`, `nets_cewd0.3`, `nets_cewd1.0`
tabular (26 sets)	26	v3 runs v1's code there, except on kr_vs_kp and mushroom, where DFG v1.3 replaces v1's DFG v1.1 (kr_vs_kp: the toy builder's development check); HELDOUT2 is held out for that code (Part 8)	MLP variants
MNIST, Fashion-MNIST, `qd_dev`	3	development	MLP variants
`qd_a`, `qd_b`	2	held out in Part 9; development data in Part 10	`mlp_strong`, `mlp_wide`, `mlp_xl`
`qd_c`, `qd_d`	2	**held out**	`mlp_strong`, `mlp_wide`, `mlp_xl`

Table 4: Every set and its status for v3. "Held out" sets were fixed in a commit before any Part 10 design work, and no result of a Part 10 learner on them was read before v3 was frozen.

- `qd_c` and `qd_d`, held out in Part 10.[1]

Each set has 6000 training and 1000 test drawings per class, so a 60000-point training pool and a 10000-point test set. The 784 pixel columns are scaled to $[0, 1]$ and permuted by one fixed permutation (seed 9192). Every reference MLP is invariant to that permutation.

3.2 References

Networks on the toy targets. These are the companion paper's μP two-hidden-layer ReLU MLPs [6, 29]: width 128, Adam at learning rate 10^{-3}, batch 64, 20000 steps on a fixed training set, read at the final step. There are five variants:

- `nets`: square loss;
- `nets_wd`: decoupled weight decay 0.1;
- `nets_cewd`, `nets_cewd0.3`, `nets_cewd1.0`: logistic loss with weight decay 0.1, 0.3 and 1.0.

Not every variant was run on every suite (Table 4). `nets_best` is, per target, the variant with the smallest n^* among those run: a hindsight choice that favours the networks. `lin` (cross-validated L2 logistic regression on the raw inputs; a separate run, not v3's member `lin_C`) and `hgb` (scikit-learn's histogram gradient boosting, default settings [24]) are calibration references. The networks are a fixed recipe, not tuned per target or suite. On `dlheld7` a learner that searches no features (`hgb`

[1] `qd_c`: hospital, submarine, telephone, bathtub, hockey stick, key, snake, bowtie, hat, binoculars. `qd_d`: basketball, purse, train, pencil, remote control, camel, onion, harp, squiggle, beard.

or `lin`) already beats `nets_best` on 5 of the 12 targets (Section 4.1). The comparison is therefore with a standard gradient-trained network, not with the best network for each target (Section 7.3).

MLPs on standard tasks. All are ReLU MLPs with softmax cross-entropy:

- `mlp`: 256-256, Adam, learning rate and stopping point chosen on validation;
- `mlp_reg`: the same with an L2 penalty chosen on validation;
- `mlp_strong`: quantile-transformed inputs, AdamW with cosine decay, weight decay in $\{0, 0.1, 1\}$ chosen on validation, dropout 0.1, an ensemble of 3;
- `mlp_wide`: the same at width 512 with weight decay in $\{0.1, 1, 3\}$;
- `mlp_xl`: width 1024 at $n = 60000$, at the weight decay `mlp_wide` chose there. Its CPU time per fit is close to the discrete learners'.

`mlp_best` is the per-cell hindsight minimum over the Part 7–8 variants, and `mlp_best+` adds the wider ones. Two further learners are *not* references:

- `mlp_pool`, `mlp_strong` given v2's correlation-pooled sums as extra inputs, is a control for how much of the image result is representation;
- `cnn`, a small convolutional network on the *unpermuted* pixels, is context: it uses the layout that every other learner here is denied. It is one network, not an ensemble: conv 3×3 with 16 channels and ReLU, max-pool 2, conv 3×3 with 32 channels and ReLU, max-pool 2, dense 128 with ReLU and dropout 0.1; AdamW at learning rate 10^{-3} with cosine decay, decoupled weight decay fixed at 0.1, batch 128, $\min(10^4, \max(2000, 50 \text{ epochs}))$ steps, no augmentation, seed 0 only (`study/standard_tasks/cnn_ref.py`). It ran on `qd_c` and `qd_a` at $n = 1000$ and 10^4.

Earlier versions. v1 (Part 8) and v2 (Part 9) are run as references. v2 is run on every toy suite and at every image cell of v3. v1 is run at every image cell of v3 and on every toy suite except `dlfamdev` and `dlheld7`. On tabular data without DFG, v1's, v2's and v3's code paths are the same.

3.3 Metrics, verdicts and budgets

Toy targets. n^* is the training-set size at which the seed-mean test error reaches 0.1. It is interpolated in $\log n$ on the grid $n = 16, 32, \ldots, 16384$, with 3 seeds (2 on `random`) and one 4096-point test set per target. A comparison is the $\log_2$ ratio $\log_2(n^*_{\text{learner}}/n^*_{\text{ref}})$. A target is *clearly better* or *clearly worse* when the ratio is below -0.25 or above $+0.25$, or when only one of the two reaches 10% error. Suite means are taken over targets both learn. Unlike the cell rule below, this verdict has no noise term: it does not compare the difference with a standard error.

Standard tasks. Test error is measured on a fixed test set at $n \in \{100, 300, 1000, 3000, 10000, 30000, 60000\}$ (up to the pool size). There are 3 seeds, except on images at $n \geq 30000$, where every learner runs seed 0 only, except the baseline `mlp`, which has 3 seeds at every n. The control `mlp_pool` and the context `cnn` ran seed 0 only at every n, and `mlp_xl` at $n = 60000$ only. A *cell* is a (set, n) pair. A cell is *better* or *worse* when the seed-mean errors differ by more than 10% relative and by more than 2 standard errors. The s.e. comes from the seed spread of both learners and is at least the test set's binomial noise; otherwise the cell is a *tie*. Errors are floored at 0.002 in $\log_2$ ratios. At $n = 60000$ on images the s.e. is the test set's binomial noise alone, about 0.34 points at

an error of 13%, so 2 s.e. of a difference is about 1.0 point. The 10% criterion asks for 1.3 points there, so it is the binding one.

Budgets. Compute is counted in FLOPs, everything a learner does in a fit included, prediction excluded. The budget caps the discrete learners only.

- *Toy fit:* one training run of one toy network, 1.62×10^{11} FLOPs at $d = 30$. Each network variant spends exactly one budget, and `nets_best` is a hindsight minimum over up to five such runs.
- *Real-data fit:* three training runs of the baseline `mlp`, which is the baseline's own cost including its learning-rate search: 1.24, 2.42 and 6.21×10^{12} FLOPs at $n = 1000$, 10^4 and 6×10^4 on the image sets.

The references run their standard recipes uncapped. On the held-out image cells `mlp_strong` spends 2.0 budgets per fit, `mlp_wide` 5.0, `mlp_pool` 5.9 and `mlp_xl` 9.2 (Section 5.4). "No more FLOPs" therefore means that the discrete learner spends no more than the network it is compared with. Because equal FLOPs are far from equal CPU time, CPU time per fit is reported separately, and `mlp_xl` is the reference matched to v3's CPU time.

3.4 How v3 was built and what is held out for it

Pre-registration. The Part 10 protocol (`study/part10/PROTOCOL.md`) was committed before any design work (`dc70266c`). It fixed six things:

- the fresh held-out sets `dlheld7`, `qd_c` and `qd_d`, and a development draw `dlfamdev`;
- the budgets;
- three builder tasks on development data only: a toy component (DFG's sum-feature search) and two image components from different angles;
- an *integration rule.* The toy component is adopted if its mean $\log_2 n^*$ ratio against v2 on `designed`, `dlsymdev` and `dlfamdev` is below -0.05, with no target worse by more than 0.5. The image component with the lower mean $\log_2$ error ratio against v2 over 8 development cells is adopted if that mean is below -0.05 and, at `qd_a` with $n = 60000$, its error is within 5% of v2's, its CPU time is at most 1.5× v2's, and it stays within the FLOP budget;
- a *best-learner rule.* v3 replaces v2 as the learner of this paper if its mean $\log_2$ error ratio against v2 on `qd_c` and `qd_d` at $n \geq 10000$ is below 0, and its mean $\log_2 n^*$ ratio against v2 on `dlheld7` is at most $+0.1$;
- the held-out reference runs.

The builders froze their files, recorded md5 sums and ran the fixed development evaluation. The rules were then applied mechanically (Appendix D.2). v3 was assembled and checked identical to its sources on every input type, then frozen (md5 `b045fafe`). Only then were held-out results of any Part 10 learner read.

What is in-sample. v3 is the last of several pre-registered steps, and every step used the earlier held-out sets as development data. For v3 as a whole, only `dlheld7`, `qd_c` and `qd_d` are untouched (Table 4). The other suites still measure the learner, but they are not held out for its design: the toy builder developed on `dlheld6`, and the image builders on validation rows of `qd_dev`, `qd_a`, `qd_b` and Fashion-MNIST (and MNIST for topology diagnostics). On tabular data the claim is narrower

and cleaner: v3's code path there is v1's. HELDOUT2 was fixed before any Part 8 design work (commit `04a215b2`), and v1 was frozen before it was run there, so HELDOUT2 is held out for that code.

4 How well it does: toy targets and tabular data

This section compares v3 with every reference on the Boolean toy suites (Sections 4.1 to 4.3) and on the 26 tabular datasets (Section 4.4); Appendix B has the in-sample detail, and Section 5 the image sets. Numbers come from `study/part10/LEARNER_RESULTS.json` unless another file is named, and Appendix D.5 says which result files stand for v3 and v2. The verdict rules are those of Section 3.3, and the discrete learners see the inputs permuted afresh in every fit.

4.1 The held-out suite `dlheld7`

`dlheld7` is the only toy suite held out for v3. It has 12 targets, a fresh draw (seed 70707) from the blind family (Section 3.1), and it was fixed in a commit before any Part 10 design work. No result of a Part 10 learner on it was read before v3 was frozen (Section 3.4). Table 5 gives every target's definition and n^*.

Against the networks [Numerical].

- *Against `nets_best`:* mean $\log_2 n^*$ ratio -1.50 over the 11 targets that both learn, about $2.8\times$ less data. v3 is clearly better on 10 of 12 targets and clearly worse on none. The 11 per-target ratios have s.d. 0.80; treating the targets as independent draws from the family, the mean has a standard error of about 0.24.
- *The two that are not wins.* f7_01, a threshold of a 13-input count, is a tie: 106 against 119. f7_05, a residue class mod 3 of a 14-input count, is learned by no learner (Section 4.3).
- *Against each variant alone:* -1.83 against `nets_wd` (11 of 12 targets clearly better), -1.59 against `nets_cewd0.3` (10 of 12) and -1.58 against `nets_cewd1.0` (10 of 12). No variant is clearly better than v3 on any target.
- *Largest margins.* f7_00, an OR of conditions on two signed sums: 203 against 1545 ($\log_2$ ratio -2.93). f7_08, an arbitrary accept set of an 11-input count: 634 against 2932. f7_03, a multiplexer between a residue class mod 4 of a 6-input count and a condition on $|s|$ for a 4-input count s: 352 against 1599. f7_09, a threshold of a signed 10-input sum times the parity $x_{11}x_{12}$: 215 against 883.
- *Smallest margins.* The two halfspaces: f7_01 (106 against 119) and f7_10, $[x_1+\cdots+x_5+2x_6 \leq -1]$ (82 against 119).

Which wins need no feature search [Numerical]. `lin` reaches 10% error on only 4 of the 12 targets, and `hgb` on 9. v3's mean $\log_2$ ratio is -0.70 against `lin` (over those 4) and -1.64 against `hgb` (over those 9); it is clearly better than `hgb` on 9 targets and than `lin` on 9, and clearly worse than neither on any. The wins over the networks fall into two groups.

- *Five wins need no feature search.* A learner that searches no features is already clearly better than `nets_best` on five targets: `hgb` on the three controls (f7_02, a DNF; f7_04, a decision list; f7_06, a read-once formula) and on the multiplexer f7_11, and `lin` on the halfspace

target	definition	v3	v2	nets_wd	nets_ cewd0.3	nets_ cewd1.0	nets_best	lin	hgb
f7_00	OR([|s1| ≤ 0], [s2 == −1]), s1 = x1+x2 −x3−x4, s2 = −x5−x6−x7+x8+x9+x10+ x11	203	207	2418	1545	1565	1545	never	6208
f7_01	[s ≤ 1], s = x1+...+x13	106	107	205	121	119	119	111	684
f7_02	OR(AND(x2,−x6,−x8), AND(x1,−x2,−x4, −x5), AND(x3,−x5,−x7))	106	106	337	341	348	337	never	120
f7_03	−x11 ? [((s1+6)/2) mod 4 != 2] : [|s2| > 2], s1 = x1+...+x6, s2 = x7+...+x10	352	342	1710	1665	1599	1599	never	2199
f7_04	if AND(−x6,−x9) → +1; elif −x8 → −1; elif −x4 → +1; elif −x2 → −1; elif x7 → −1; elif x5 → −1; elif AND(x3,x1) → −1; else +1	70	74	164	175	181	164	never	84
f7_05	[((s+14)/2) mod 3 != 0], s = x1+...+x14	never	never	never	never	never	never	never	never
f7_06	OR(AND(AND(−x1,x2),AND(AND(−x3, x4),−x5)),AND(OR(−x6,x7),x8))	44	43	108	124	124	108	127	59
f7_07	[−2 ≤ s ≤ 8] * x13, s = −x1+x2−x3+x4+ x5+x6+x7+x8−x9+x10−x11+x12	174	174	445	443	415	415	never	1373
f7_08	[s in {−11,−9,−3,1,9,11}], s = x1+...+x11	634	634	5843	3239	2932	2932	never	never
f7_09	[s ≤ 0] * x11 * x12, s = x1−x2+x3−x4+ x5−x6+x7+x8+x9+x10	215	215	1000	883	928	883	never	never
f7_10	[u ≤ −1], u = s + 2*(x6), s = x1+...+x5	82	82	210	120	119	119	87	114
f7_11	−x11 ? [s1 in {−4,4}] : [|s2| ≤ 4], s1 = x1+x2−x3+x4, s2 = −x5−x6+x7−x8 −x9+x10	55	55	114	144	141	114	119	63

Table 5: The held-out toy suite `dlheld7`: n^*, the training-set size at 10% test error (mean of 3 seeds, interpolated in $\log n$), for v3, v2, the three pre-registered network variants, their per-target minimum `nets_best`, and the calibration references `lin` and `hgb` [Numerical]. v3 and v2 see the inputs permuted per fit. "never": 10% error is not reached by $n = 16384$. In the definitions, $[\cdot]$ is +1 when the condition holds and −1 otherwise; `*` multiplies ±1 values; `c ? a : b` is a multiplexer (a when $c = +1$); a literal `x5` or `−x5` inside AND/OR, a decision-list rule or a selector stands for $x_5 = +1$ or $x_5 = -1$. f7_02, f7_04 and f7_06 are the family's non-symmetric controls. Source: `study/part10/LEARNER_RESULTS.json`.

f7_10. On these five, v3 is ahead of the better calibration reference by factors of 1.06–1.35 only ($\log_2$ ratios −0.09 to −0.43).

- *Five wins need the feature search.* On the other seven targets neither `lin` nor `hgb` beats the networks. `hgb` needs 684–6208 samples or never reaches 10%, and `lin` learns only f7_01. There v3's mean against `nets_best` is −1.80: clearly better on 5 (f7_00, f7_03, f7_07, f7_08, f7_09, all sum-structured), a tie on f7_01, and f7_05 learned by none.

Against v2 [Numerical]. The mean $\log_2 n^*$ ratio is −0.004. No target is clearly better or worse, and the largest per-target change is 0.09 in $\log_2$ (f7_04: 70 against 74). So the Part 10 toy component changes nothing measurable on the held-out suite. The best-learner rule only asked that v3 not be worse than v2 there (mean ≤ +0.1). v3 became the learner of this paper through its image side (Section 5). Over all 138 toy targets, v3's n^* differs clearly from v2's on 14, 12 lower and 2 higher, none of them on `dlheld7`; Appendix B.2 lists them with what the models show.

suite	status	targets	v3 vs `nets_best`	v3 vs v2	v3 vs `lin`	v3 vs `hgb`
`designed`	dev	21	−2.42 (21/0)	−0.03 (1/0)	−0.85 (19/0)	–
`random`	dev check (P6)[c]	20	−2.26 (20/0)	−0.02 (1/0)	−1.29 (20/0)	−1.78 (20/0)
`dlheld`	held out (P6)	12	−2.03 (12/0)	+0.02 (0/1)	−0.62 (10/0)	−2.13 (10/0)
`dlheld2`	held out (P6)	7	−1.99 (7/0)	−0.14 (1/0)	−0.58 (5/0)	−2.79 (6/0)
`dlheld3`	held out (P8)[d]	10	−1.80 (10/0)	−0.04 (0/0)	−0.99 (9/0)	−1.82 (9/0)
`dlsymdev`	dev (P9)	8	−2.22 (8/0)	−0.18 (2/0)	n/a (8/0)	–
`dlheld4`	held out (P9)[a]	12	−1.87 (12/0)	−0.19 (3/0)	−1.54 (12/0)	−2.85 (11/0)
`dlheld5`	held out (P9)	12	−2.44 (10/0)	−0.04 (1/0)	n/a (11/0)	−4.11 (11/0)
`dlheld6`	held out (P9), dev (P10)[b]	12	−1.50 (11/1)	−0.25 (2/0)	−1.13 (11/0)	−2.21 (10/0)
`dlfamdev`	dev (P10)	12	–	−0.05 (1/1)	–	–
`dlheld7`	**held out (P10)**	12	−1.50 (10/0)	−0.00 (0/0)	−0.70 (9/0)	−1.64 (9/0)

Table 6: v3 on every toy suite [Numerical]. Each entry is the mean $\log_2 n^*$ ratio of v3 to the reference over the targets both learn, with the numbers of targets on which v3 is clearly better / clearly worse in brackets. "n/a": the reference learns none of the targets, so no mean exists; "–": the reference was not run on the suite. `nets_best` is the per-target minimum over the network variants run on the suite, which differ between suites (Table 4). Status: the Part (P6–P10) in which the suite was held out or used for development. Only `dlheld7` is held out for v3; for v3 every other suite is development data. [a]The Part 9 sum-feature builder saw the source of `dlheld4`. [b]The Part 10 toy builder developed on `dlheld6`; weighted blocks were designed after v2's failure on its target h6_10. [c]`random` was a development check for DFG in Part 6 (run and read with intermediate versions) and was evaluated with the held-out suites in Part 8. [d]v1's failure on sym9 (`dlheld3`) motivated v2's sum blocks. Source: `study/part10/LEARNER_RESULTS.json`.

Why nothing moved on `dlheld7` [Heuristic]. The gains of v1.3 are concentrated on a few targets: 12 of 138, 11 of which v3 now learns at $n \leq 256$. A 12-target draw can easily contain none of them. The three draws of the blind family agree with this: v3's mean against v2 is −0.05 on `dlfamdev` (development), −0.25 on `dlheld6` (used by the builder) and −0.004 on `dlheld7`. `dlheld7` has one weighted-sum target, f7_10. It is a halfspace that v2 already learns at the same n^* (82), although at $n = 256$ only v3's model holds the exact weighted block.

Compute [Numerical]. Every v3 fit on `dlheld7` used at most 0.53 of one network training run's FLOPs (f7_04). Over all 11 suites (every v3 toy fit) the maximum is 0.83 (h5s7 on `dlheld5`). CPU time per fit is at most 46 s on the shared 4-core machine (`designed`, thr10). Source: the `flops` and `cpu_seconds` fields of the v3 result files.

4.2 Every toy suite

Table 6 summarizes all 11 suites, and Figure 1 shows every target with a network reference. Table 4 gives each suite's status for v3: only `dlheld7` is held out, and the footnotes of Table 6 mark the suites that shaped v3 or an earlier version.

Results [Numerical]. On the 126 targets of Figure 1, v3 is clearly better than `nets_best` on 121 and needs at least 4× less data on 54 (`figures/fig_toy_nstar.json`). Five targets are not wins:

- h6_00 (`dlheld6`) is clearly worse: only the most regularized network learns it, at 12304;
- f7_01 (`dlheld7`) and h5s7 (`dlheld5`) are ties;
- f7_05 (`dlheld7`) and h5s1 (`dlheld5`) are learned by no learner.

Section 4.3 discusses them. The suite means against `nets_best` range from −1.50 to −2.44 (Table 6).

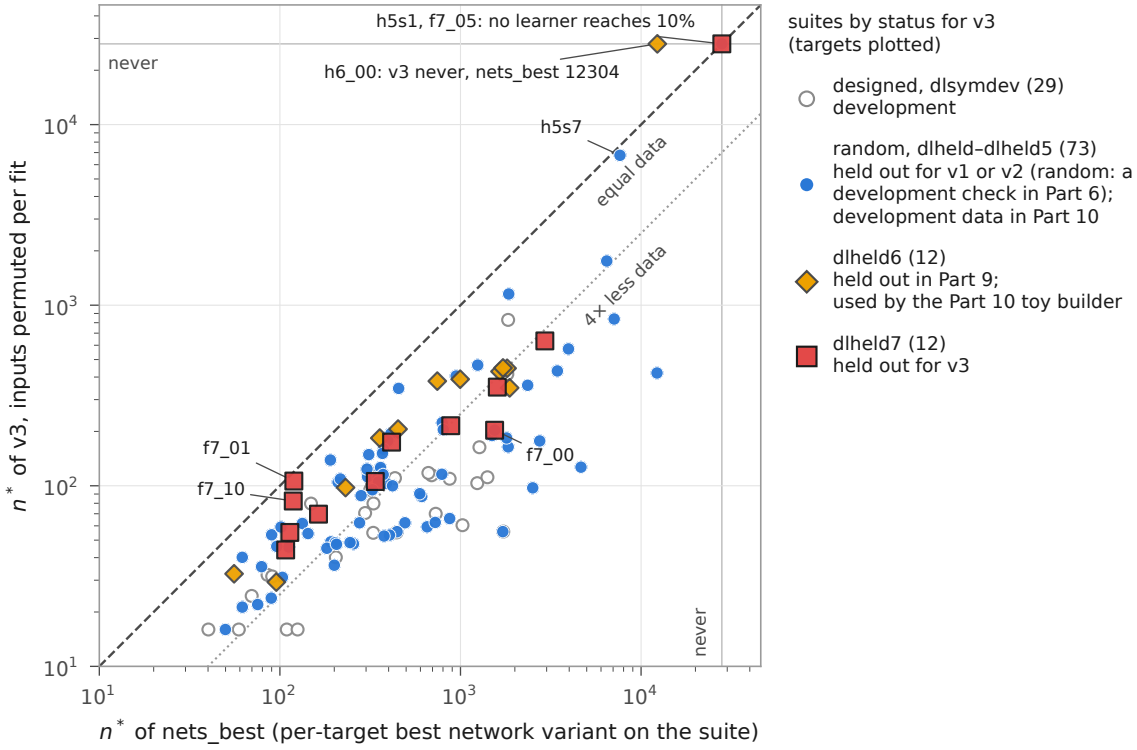


Figure 1: n^* of v3 (inputs permuted per fit) against `nets_best` for every target of every toy suite with network references, one point per target [Numerical]. Colour and shape give the status of the suite for v3 (Table 4); the held-out suite `dlheld7` is drawn as large outlined squares. "never": 10% error not reached by $n = 16384$, drawn at the edge. Points below the dashed line need less data than the best network variant, and points below the dotted line at least 4× less. `nets_best` is a minimum over 2 variants on `designed`, `random`, `dlheld` and `dlheld2`, 3 on `dlsymdev` and `dlheld3`–`dlheld5`, 5 on `dlheld6` and the 3 pre-registered ones on `dlheld7`. `dlfamdev` has no network runs and is not shown. Source: `study/part10/LEARNER_RESULTS.json`.

Against the one variant run on every suite [Numerical]. The suite means against `nets_best` are not comparable across suites, because `nets_best` minimizes over different sets of variants. `nets_wd` is the one variant run on every suite with network results. Against it, v3's mean $\log_2 n^*$ ratio is:

- development suites: −2.42 on `designed` and −2.53 on `dlsymdev`; the Part 6 development check `random`: −2.26;
- earlier held-out suites: −2.03 on `dlheld`, −1.99 on `dlheld2`, −2.07 on `dlheld3`, −2.25 on `dlheld4` and −2.66 on `dlheld5`;
- the two draws of the blind family: −1.94 on `dlheld6` and −1.83 on `dlheld7`.

The two draws of the blind family give the two smallest margins, but `dlheld2` is within 0.05 of `dlheld6`, and differences of this size on suites of 7 to 12 targets are within noise. If the gap is real, two explanations fit it, and the data do not separate them [Heuristic]. Several of the other suites informed the design of v3 or of an earlier version, so some selection in v3's favour is expected there. And the suites differ in kind: `dlheld`–`dlheld5` were written within the study, mostly from families close to its development targets, while the blind family was written by an agent that had not seen

the sum-feature design.

Wins that need no feature search, over the held-out suites [Numerical]. Excluding the 30 of the 77 targets of the seven held-out suites (`dlheld`–`dlheld7`) on which `lin` or `hgb` is already clearly better than `nets_best`, v3's mean against `nets_best` is -2.11 over the 44 that both learn, with 42 targets clearly better and 1 clearly worse (`figures/fig_toy_nstar.json`). The 30 are of two kinds: 6 halfspace-like targets on which `lin` wins (v3 is within the tie band of `lin` or below it on each), and 24, mostly formulas, decision lists, multiplexers and controls, on which `hgb` wins; Appendix B.1 lists them. So the margins do not come from targets that are easy for trees or linear models. They come from the targets with parity or signed-sum structure, where both calibration references fail, and Part 8's ablations located the toy wins in DFG [5]. The pooled summary over the seven held-out suites, most of them development data for v3, is in Appendix B.1.

4.3 Where v3 is not better

Five of the 126 targets with network references are not wins (Figure 1).

- *Learned by no learner.* f7_05 $= [\frac{1}{2}(s+14) \bmod 3 \neq 0]$, $s = x_1 + \cdots + x_{14}$ (`dlheld7`), and h5s1, a signed 9-input count with accept set $\{-9, -5, -1, 3, 9\}$ times the parity $x_{10}x_{11}$ (`dlheld5`). No network variant, `lin` or `hgb` reaches 10% error on either by $n = 16384$ (3 seeds).
- *The one loss.* h6_00, a residue class mod 3 of a 13-input count (`dlheld6`), is learned only by the most regularized network (`nets_cewd1.0`, $n^* = 12304$). It is the only target of any suite on which a network is clearly better than v3, and it was v2's loss as well. The third residue class mod 3, fd_10 on 10 inputs (`dlfamdev`), has no network runs. v3 and v2 learn none of the three by $n = 16384$.
- *Ties.* The halfspace f7_01 $= [x_1 + \cdots + x_{13} \leq 1]$: v3 (106), `lin` (111) and the logistic-loss networks (119–121) are all within the tie band, and a linear readout of the inputs alone already does about as well. And h5s7, an arbitrary accept set of a signed 13-input count: 6760 against 7641 (`nets_cewd`).

The interpretability analysis (seed 0) locates the failures on the residue classes in DFG's search, not in its test: their true blocks would pass the test from $n = 128$, but the targets have almost no Fourier weight at degrees 1–2, from which every seed of the search starts (Section 6.1.3; Section 2.3 explains why such a block is invisible to the seeds). h5s7 is a weaker case of the same mechanism, found at $n = 8192$ on seed 0, and h5s1's bottleneck is detecting its modulating parity (Appendix C.3). The next-smallest margins are listed in Appendix B.1.

4.4 Tabular data

What v3 runs on tabular data. DFG runs only when there are 2 classes and every column is binary or categorical (≤ 256 bits), and the image members only on images. So on 24 of the 26 datasets v3 runs v1's code, the stack of LightGBM, ExtraTrees, k-NN and logistic regression (its predictions were checked identical to v2's on one optdigits fit, and v2 runs v1's code path there), and its results there are v1's Part 8 result files. Only two datasets qualify for DFG: kr_vs_kp (development) and mushroom (Part 7 held-out set). On mushroom v3's own runs give v1's seed-mean errors at every n. On kr_vs_kp they differ from v1's at 3 of 4 sizes, but at no cell clearly (Appendix B.3).

split	reference	cells (better / tie / worse)	mean $\log_2$ ratio	datasets with lower mean error (sign test p)
Part 7 held-out set	`mlp`	59 (22/36/1)	−0.331	10 of 11 (0.012)
	`mlp_reg`	59 (16/40/3)	−0.160	9 of 11 (0.065)
	`mlp_strong`	54 (22/30/2)	−0.339	10 of 11 (0.012)
	`mlp_best`	59 (12/44/3)	−0.114	8 of 11 (0.23)
	`hgb`	59 (20/38/1)	−0.259	10 of 11 (0.012)
Part 8 fresh set (HELDOUT2)	`mlp`	53 (15/36/2)	−0.445	9 of 10 (0.021)
	`mlp_reg`	53 (16/35/2)	−0.272	8 of 10 (0.11)
	`mlp_strong`	53 (15/36/2)	−0.284	8 of 10 (0.11)
	`mlp_best`	53 (10/41/2)	−0.168	8 of 10 (0.11)
	`hgb`	53 (12/40/1)	−0.250	9 of 10 (0.021)
development	`mlp`	24 (4/20/0)	−0.379	5 of 5 (0.062)
	`mlp_reg`	24 (3/21/0)	−0.267	5 of 5 (0.062)
	`mlp_best`	24 (1/23/0)	−0.244	5 of 5 (0.062)
	`hgb`	24 (4/20/0)	−0.270	5 of 5 (0.062)

Table 7: v3 on the tabular datasets against the MLP references and `hgb` [Numerical]. "Cells": the (dataset, n) pairs compared, each a mean over 3 seeds, with the verdict rule of Section 3.3. "Mean $\log_2$ ratio": of v3's test error to the reference's, over cells, errors floored at 0.002. The last column counts the datasets whose mean ratio over n is below 0, with the two-sided sign-test p. `mlp_best` is the per-cell best of the Part 7–8 MLP variants (hindsight). `mlp_strong` was run on the Part 7 set only at $n \leq 10000$ (54 of 59 cells) and not on the development sets. On 24 of the 26 datasets v3's code path is v1's, so these are Part 8's numbers, except for v3's own runs on mushroom (identical errors) and kr_vs_kp. Source: `study/part10/LEARNER_RESULTS.json`.

What is held out. HELDOUT2 (10 datasets) was fixed in a commit before any Part 8 design work (`04a215b2`). v1 was developed on the 5 development datasets only, and its code path has not changed since. HELDOUT2 contains no dataset on which DFG runs, so it is held out for this code path. The Part 7 held-out set is held out in the weaker sense that the Part 8 designers had read its MLP references. The 5 development datasets are in-sample.

Results (Table 7) [Numerical].

- *HELDOUT2, against `mlp_best`:* mean $\log_2$ error ratio −0.168, a geometric-mean error about 11% lower. 10 of 53 cells are better and 2 worse. The mean error is lower on 8 of 10 datasets, which is not significant (sign test $p = 0.11$).
- *HELDOUT2, against single references:* the margins are larger: −0.445 against the baseline `mlp` (9 of 10 datasets, $p = 0.021$), −0.284 against `mlp_strong` (8 of 10, $p = 0.11$) and −0.250 against `hgb` (9 of 10, $p = 0.021$).
- *The Part 7 set:* −0.114 against `mlp_best` (12 better, 3 worse, 8 of 11 datasets, $p = 0.23$) and −0.259 against `hgb` (10 of 11, $p = 0.012$).
- *Development:* −0.244 against `mlp_best`, lower on 5 of 5 ($p = 0.062$).

The per-dataset picture is in Appendix B.3.

Where the tabular result comes from [Heuristic]. The tabular numbers are those of a classical stack, and they say little about discrete feature search: in Part 8, DFG did not carry them, no single member family matched the stack, and default `hgb` alone did not beat the MLPs [5] (Appendix B.3). This agrees with the tabular-learning literature. Tree ensembles tend to beat plain MLPs on

medium-sized tabular data [12, 25]. Stacking diverse models adds to that [10, 27]. And careful regularization and tuning of the MLP narrow the gap [11, 14]. The MLP references here are standard recipes without tabular-specific architectures, so the margins would likely shrink against a tuned tabular network [Heuristic]. The tabular evidence for v3 is therefore modest: it is not worse than the MLPs on average on fresh tabular data (2 of 10 datasets have higher mean error), it is better on average without statistical significance, and the reason is the classical stack, unchanged since Part 8.

5 How well it does: images

On images v3 runs its image members and the classical ones (Table 2); DFG does not run. So on images v3 is v2 with one more member, the bar booster `cgbt`, and a stacker refitted over all members (Section 2.4). This section gives v3's test error on the two QuickDraw sets held out for it (Section 5.1) and on the image sets it was developed on (Section 5.2). It then asks where the gain over v2 comes from (Section 5.3) and what a fit costs (Section 5.4). Errors, verdicts, stacking weights, FLOPs and CPU seconds come from `study/part10/LEARNER_RESULTS.json` and from the result files and `.info.json` records in `study/standard_tasks/results/`; Appendix D.5 says which file stands for v3 on each set. Cells and verdicts are those of Section 3.3, where the noise at $n = 60000$ is discussed.

5.1 The held-out sets `qd_c` and `qd_d`

`qd_c` and `qd_d` are ten fresh QuickDraw classes each (Section 3.1). They were fixed in a commit before any Part 10 design work. No result of a Part 10 learner on them was read before v3 was frozen (Section 3.4).

- *Runs.* v3, v2, v1, `hgb`, `mlp_strong` and `mlp_wide` ran 3 seeds at $n = 1000$ and $n = 10000$ and seed 0 at $n = 60000$. `mlp_xl` ran at $n = 60000$ only. The control `mlp_pool` ran on `qd_c` at $n \in \{10000, 60000\}$, seed 0, and the context `cnn` on `qd_c` at $n \in \{1000, 10000\}$, seed 0. These are the pre-registered cells. The Part 7 `mlp`, which enters `mlp_best`, was also run, with 3 seeds at every n (not pre-registered).
- *Where to look.* Table 8 gives every held-out cell, Table 9 the verdicts, and Figure 2 the learning curves.

Against the MLP references [Numerical]. At all six held-out cells v3's error is lower than every reference's and the control's (the context `cnn`'s is lower at both `qd_c` cells where it ran).

- `mlp_strong` (ensemble of three 256-256 networks): v3 is clearly better at all 6 cells. Its error is 17–23% lower, and the mean $\log_2$ ratio is -0.31 (-0.32 at $n \geq 10000$).
- `mlp_wide` (three 512-512 networks): clearly better at 5 cells. The sixth, `qd_c` at $n = 60000$, is a tie: 12.76% against 14.12%, 9.6% lower, just inside the 10% threshold. Mean -0.25.
- `mlp_xl` (three 1024-1024 networks, the CPU-matched reference; $n = 60000$ only): two ties. v3 is 9.6% lower on `qd_c` (12.76% against 14.12%) and 8.6% lower on `qd_d` (12.64% against 13.83%). Mean -0.14.
- `mlp_best` equals `mlp_strong` at all six held-out cells, so v3 is clearly better at all six (mean -0.31). `mlp_best+` adds the wider ensembles: 4 cells better and 2 ties, the two $n = 60000$

set	n	v3	v2	v1	hgb	mlp_ strong	mlp_ wide	mlp_ xl	mlp_ best+	mlp_ pool	cnn
held out for v3 (Part 10): 3 seeds at $n \leq 10000$, seed 0 at $n = 60000$											
qd_c	1000	**26.58**	27.48°	31.68	37.52	33.01	33.02	–	33.01	–	24.36°
qd_c	10000	**16.17**	17.79°	20.05	22.76	20.94	20.41	–	20.41	19.37	14.84°
qd_c	60000	**12.76**	13.63°	15.77	18.22	16.08	14.12°	14.12°	14.12°	14.48	–
qd_d	1000	**26.12**	26.71°	30.21	35.33	31.62	31.40	–	31.40	–	–
qd_d	10000	**16.38**	17.82°	19.70	21.85	20.30	19.89	–	19.89	–	–
qd_d	60000	**12.64**	13.99°	15.97	17.48	15.34	14.31	13.83°	13.83°	–	–

Table 8: Test error (%) at every held-out image cell [Numerical]. Seed mean of 3 at $n \leq 10000$ and seed 0 at $n = 60000$. `mlp_xl`, `mlp_pool` and `cnn` are seed 0 everywhere. Bold: the lowest error among v3, v2, v1, `hgb`, `mlp_strong`, `mlp_wide` and `mlp_xl`. A reference's entry carries ° where the cell is a tie with v3 and * where the reference is clearly better than v3. An unmarked entry is clearly worse than v3. "–": not run. `mlp_best+` is the per-cell minimum over the Part 7–9 MLP variants (`mlp_pool` excluded); it needs the Part 7 `mlp`, which has run at every cell here. `mlp_pool` is `mlp_strong` given v2's pooled sums (a control). `cnn` sees the unpermuted pixels (context, not a reference). The development cells are in Table 20. Source: `study/part10/LEARNER_RESULTS.json`; markers by the cell rule (`compare_hybrid.cells`).

	all n		$n \geq 10000$	
reference	cells (better/tie/worse)	mean $\log_2$ ratio	cells (better/tie/worse)	mean $\log_2$ ratio
held out for v3 (Part 10): qd_c, qd_d				
mlp_strong	6 (6/0/0)	−0.314	4 (4/0/0)	−0.324
mlp_wide	6 (5/1/0)	−0.253	4 (3/1/0)	−0.235
mlp_xl	2 (0/2/0)	−0.138	2 (0/2/0)	−0.138
mlp_best	6 (6/0/0)	−0.314	4 (4/0/0)	−0.324
mlp_best+	6 (4/2/0)	−0.245	4 (2/2/0)	−0.223
hgb	6 (6/0/0)	−0.471	4 (4/0/0)	−0.473
v2	6 (0/6/0)	−0.097	4 (0/4/0)	−0.125
v1	6 (6/0/0)	−0.281	4 (4/0/0)	−0.305
mlp_pool (control)	2 (2/0/0)	−0.222	2 (2/0/0)	−0.222
cnn (context)	2 (0/2/0)	+0.125	1 (0/1/0)	+0.124

Table 9: v3 against each image reference on the held-out sets [Numerical]. "Cells" counts the (set, n) pairs compared, with v3 better / tied / worse by the cell rule (Section 3.3). "Mean $\log_2$ ratio" is the mean over those cells of $\log_2$(v3's error / the reference's error); negative means v3 is lower. `mlp_best` is the per-cell minimum over the Part 7–8 MLP variants (`mlp`, its regularized versions, `mlp_strong` and its single network `mlp_strong1`). `mlp_best+` adds `mlp_wide`, its single network `mlp_wide1`, and `mlp_xl`. Both need the Part 7 `mlp`, which has run at every cell here. `mlp_xl`, `mlp_pool`, `cnn` and all $n = 60000$ cells are seed 0. The development sets are in Table 21. Source: `study/part10/LEARNER_RESULTS.json`.

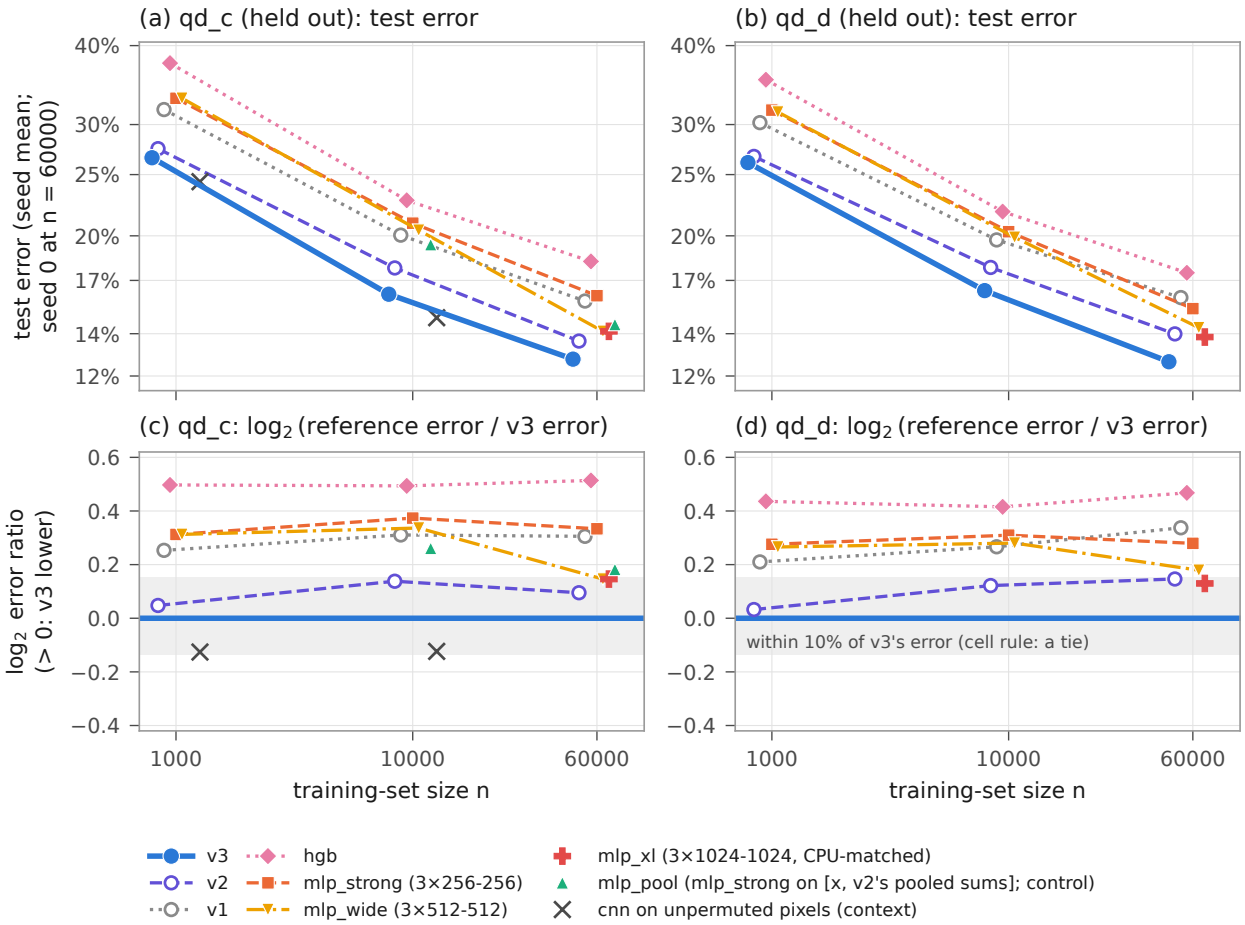


Figure 2: The held-out QuickDraw sets [Numerical]. Top: test error against n (log–log; seed mean of 3 at $n \leq 10000$, seed 0 at $n = 60000$; `mlp_xl`, `mlp_pool` and `cnn` seed 0). Bottom: each reference's error relative to v3's at the same n, as $\log_2$(reference error / v3 error), so points above 0 are cells where v3 is lower. In the grey band the two errors are within 10% of each other, and the cell rule calls the cell a tie whatever the noise. Markers are shifted slightly sideways so that they do not hide each other. Source: the result files in `study/standard_tasks/results/`.

cells (mean −0.25). v3 is also clearly better than the Part 7 baseline `mlp` at all six cells (mean −0.53).

So at $n \leq 10000$ v3 is clearly better than every MLP reference. At $n = 60000$ the two widest ensembles come within 10% of it on `qd_c`, and `mlp_xl` also on `qd_d`.

Against the other references [Numerical].

- `hgb`: v3 is clearly better at all 6 cells, with errors 25–30% lower (`qd_c` 29–30%, `qd_d` 25–28%); mean $\log_2$ ratio −0.47.
- v1, the Part 8 portfolio: clearly better at all 6 cells, by 13.5–20.9% (`qd_c` 16.1%, 19.4%, 19.1% and `qd_d` 13.5%, 16.9%, 20.9% at $n = 1000$, 10000 and 60000); mean −0.28.
- v2: all 6 cells are ties. v3's error is lower at every cell, by 2.2–3.3% at $n = 1000$ and by 6.4–9.6% at $n \geq 10000$, but never by more than 10%. The mean $\log_2$ ratio is −0.10, and −0.125 at $n \geq 10000$.

The control and the context [Numerical].

- `mlp_pool` is `mlp_strong` given v2's pooled subset sums as extra inputs (seed 0, `qd_c` only). v3 is clearly better at both cells: 16.17% against 19.37% at $n = 10000$, and 12.76% against 14.48% at $n = 60000$ (16.5% and 11.9% lower). v2 ties it at both cells (17.79% and 13.63%, which are 8.2% and 5.9% lower), as it did on `qd_a` and `qd_b` in Part 9. This control matches v2's representation, not v3's: no MLP was given the bar sums (Section 5.3).
- `cnn`, a small convolutional network on the unpermuted pixels (Section 3.2), is context, not a reference. Its `qd_c` runs (seed 0) give 24.36% at $n = 1000$ and 14.84% at $n = 10000$, 8.4% and 8.2% below v3's errors (26.58% and 16.17%): two ties by the cell rule, but lower at both cells.

The best-learner rule [Numerical]. The pre-registered rule does not use cell verdicts (Section 3.4). It makes v3 the learner of this paper if two conditions hold:

- the mean $\log_2$ error ratio of v3 to v2 over the four held-out image cells with $n \geq 10000$ is below 0. It is -0.125: the cells give -0.138 and -0.095 on `qd_c` ($n = 10000$, 60000) and -0.122 and -0.146 on `qd_d`;
- the mean $\log_2 n^*$ ratio of v3 to v2 on `dlheld7` is at most $+0.1$. It is -0.004 (Section 4.1).

Both hold, so v3 replaced v2. The rule was applied by `study/part10/compare_part10.py` (output: `study/part10/RESULTS.md`) after every held-out cell of v3 and v2 was in, and the decision is recorded in `study/part10/PROTOCOL.md` (28 October, 04:25 UTC).

How firm the gain over v2 is. By the cell rule it is not established at any single cell. Its direction is consistent: in all 14 seed-paired fits (same training sample, same test set), v3's error is below v2's, by 0.47 to 1.86 points [Numerical]. This is a description made after the results were in, not a pre-registered test. The 14 fits share two test sets, so they are not independent.

5.2 The other image sets: development data for v3

Tables 20 and 21 in Appendix B.4 give v3 at the cells of its fixed development evaluation (one fit each, seed 0). These cells selected the image component (Appendix D.2), so they are in-sample for that choice; Appendix B.4 gives the detail. On `qd_a` and `qd_b`, held out in Part 9, v3 is clearly better than `mlp_strong`, `mlp_wide`, `hgb` and v1 at all 5 cells, than `mlp_xl` at `qd_a`, $n = 60000$ (10.82% against 12.46%), and than the control `mlp_pool` at the 3 cells where both ran; it ties v2 at all 5. The only learner better than v3 there is the `cnn` on unpermuted pixels: 12.00% against 14.22% at `qd_a`, $n = 10000$, an error 15.6% lower. On `qd_dev` v3 is clearly better than `mlp_strong`, `hgb` and v1 at both cells and ties v2. On Fashion-MNIST every cell is a tie with `mlp_strong`, v2, v1 and `hgb`; it is the set where the bars do not help.

5.3 Where the gain over v2 comes from

The same members, plus one [Numerical]. In each of the 14 held-out fits, v3's and v2's records give every shared member the same out-of-fold error, and `ggbt` the same number of boosting rounds. `cgbt` draws from its own random stream, so v3's other members are v2's fits. v3 and v2 therefore differ only by `cgbt` and by the stacking weights. The final refits also differ where a weight moves between zero and a positive value, because a member with weight 0 is not refitted. Table 10 gives the weights and the out-of-fold errors.

		v3: stacking weight					v2: stacking weight				out-of-fold error (%)			
set	n	cgbt	ggbt	gknn	knn	rest	ggbt	gknn	knn	rest	cgbt	ggbt	v3	v2
qd_c	1000	0.27	0.02	0.38	0.15	0.03	0.24	0.46	0.17	0.04	30.8	32.5	27.6	29.0
qd_c	10000	0.36	0.04	0.15	0.15	0.01	0.35	0.24	0.18	0.03	16.8	18.9	16.2	17.2
qd_c	60000	0.49	0.12	0.11	0.17	0.00	0.50	0.20	0.24	0.00	13.3	14.6	12.8	13.5
qd_d	1000	0.23	0.08	0.30	0.18	0.03	0.27	0.35	0.21	0.04	28.1	29.8	26.3	26.1
qd_d	10000	0.39	0.01	0.13	0.19	0.01	0.29	0.20	0.26	0.10	17.1	19.2	16.2	17.3
qd_d	60000	0.45	0.12	0.04	0.12	0.00	0.50	0.11	0.23	0.00	12.6	14.3	12.5	13.5

Table 10: What carries v3 and v2 on the held-out image sets [Numerical]. Left: stacking weight per member family (seed mean of 3 fits at $n \leq 10000$; one fit at $n = 60000$). `gknn`: nearest-neighbour votes on the pooled sums. `knn`: nearest-neighbour votes on the pixels. "rest": Part 8's `gbt`, `et` and `lin_C`; `lin_C` has weight 0 in every fit here. Right: out-of-fold error on the training rows of the two boosters and of the whole stack (5-fold at $n \leq 3000$; one holdout of 2000 rows at $n = 10000$ and 6000 rows at $n = 60000$). `ggbt`'s out-of-fold error is the same in v3 and v2 in every fit. The weights are not shares of the decision (Section 6.2.2). Source: `hyb_portfolio{3,2}__qd_{c,d}.info.json` in `study/standard_tasks/results/`.

- *`cgbt` is the better booster.* Its out-of-fold error is below `ggbt`'s in all 14 fits: by 0.2–4.0 points at $n = 1000$, and by 0.13–0.20 in $\log_2$ at $n \geq 10000$. It is the lowest of all members in 13 of the 14 fits. The exception is `qd_c`, $n = 1000$, seed 1, where nearest-neighbour votes on pooled sums are lower (30.1% against 31.8%).
- *It takes `ggbt`'s place in the stack.* At $n \geq 10000$, `cgbt` has the largest family weight in all 8 fits (0.32–0.49). `ggbt`'s weight falls from 0.25–0.50 in v2 to 0–0.12 in v3. At $n = 1000$, the nearest-neighbour votes on pooled sums keep the largest family weight in 5 of 6 fits, in v3 as in v2.
- *The stack improves.* v3's stacked out-of-fold error is below v2's in 12 of the 14 fits. At `qd_d`, $n = 1000$, it is equal in one fit and higher in one (27.2% against 26.3%).

The weights multiply log-probabilities of different calibration, and the convex stacker splits weight between two nearly collinear boosters, so "`ggbt` gets weight 0" describes one fit, not the learner (Section 6.2.2).

What `cgbt` changes: the bank, not the schedule [Numerical]. `cgbt` has `ggbt`'s tree settings and early-stopping rule. It differs from `ggbt` in three ways: its bank adds bar sums at two lengths, it drops the 300 prototype-pair projections, and its per-tree column fraction is 0.05 instead of 0.12 (Section 2.4). The builder's development runs separate these effects (one booster alone, $n = 5000$, validation rows, seed 0; `study/part10/p10_codes/DESIGN.md`, §4.2). On `qd_dev`:

- v2's bank gave 17.66% error, and dropping the pairs gave 17.20%;
- adding the bars at both lengths gave 15.92%, and the lower column fraction then gave 15.82%.

On `qd_a` the full change gave 17.68% → 16.78%. So on QuickDraw most of the gain is the bars. On Fashion-MNIST the bars did not help. In `cgbt`'s fitted trees the bars, two thirds of its columns, carry 0.72–0.80 of the split gain on QuickDraw and MNIST (development fits; Section 6.2.2). Split gain describes how the trees split, not what the predictions depend on.

Compared with Part 9 [Numerical]. In Part 9, v2 tied the CPU-matched `mlp_xl` at $n = 60000$ on `qd_a` and `qd_b`, and the control `mlp_pool` matched v2, so most of v2's large-n advantage over the MLPs was the pooled-sum representation, which an MLP uses as well as the trees do. The Part

		FLOPs / budget		CPU s per fit			MLP references: wall s per fit (FLOPs / budget)			
					of which					
set	n	v3	v2	v3	`cgbt`	v2	`mlp_strong`	`mlp_wide`	`mlp_xl`	`mlp_pool`
`qd_c`	1000	0.49–0.51	0.35	229–256	120–132	144–166	384 (2.0)	249 (5.0)	–	–
`qd_c`	10000	0.34–0.48	0.22–0.26	513–785	251–430	284–440	490 (2.0)	396 (5.0)	–	333 (5.9)
`qd_c`	60000	0.57	0.42	1361	668	1296	963 (2.0)	1168 (5.0)	2196 (9.2)	716 (5.9)
`qd_d`	1000	0.50–0.51	0.35	266–291	128–140	162–175	98 (2.0)	248 (5.0)	–	–
`qd_d`	10000	0.30–0.46	0.24–0.29	402–768	233–377	394–602	152 (2.0)	440 (5.0)	–	–
`qd_d`	60000	0.58	0.42	1464	760	1531	422 (2.0)	989 (5.0)	1335 (9.2)	–

Table 11: Compute per fit on the held-out image cells [Numerical]. Budget: three training runs of the baseline MLP (`common.budget_real`), 1.24, 2.42 and 6.21 $\times 10^{12}$ FLOPs at $n = 1000$, 10000 and 60000. FLOPs / budget: each learner's own count, everything in the fit included and prediction excluded. CPU s: process CPU seconds per fit, and the seconds v3 attributes to `cgbt`. A range is over the 3 seeds; $n = 60000$ has one fit. The MLP references record wall-clock seconds per fit on the shared 4-core machine, not process CPU time (the `cpu_seconds` field of their result files), with their FLOPs per fit in units of the budget in brackets. Sources: the result files and `.info.json` records in `study/standard_tasks/results/`.

10 held-out sets repeat this for v2: it ties `mlp_xl` on both sets (13.63% against 14.12%; 13.99% against 13.83%) and `mlp_pool` at both `qd_c` cells. v3 moves past the control but not past `mlp_xl` (Section 5.1).

Reading [Heuristic]. On images v3's gain over v2 is a better fixed representation for gradient-boosted trees: sums over short runs of pixels along lines recovered from pixel correlations, used at fixed positions. The gain is not evidence that discrete search beats gradient-based learning on images, for three reasons.

- *The control for v3's own representation was not run.* No MLP was given the bar sums as well as the pooled sums. Part 9's control matched v2 once it had v2's features, so part of v3's margin over the MLPs may again be available to a network given the same features.
- *Gradient-trained networks that learn local structure were not run.* For example, an MLP trained to learn sparse local connectivity [22] would be the information-matched comparison.
- *A network told the layout is better.* The small `cnn` (seed 0) has lower error at all three cells where both have run: clearly at `qd_a`, $n = 10000$, and within the tie band at both `qd_c` cells (Sections 5.1 and 5.2).

What v3 shows is narrower: a learner that never reads the pixel layout, and whose features are fixed subset sums chosen by rank statistics, reaches 12.6–12.8% error on fresh QuickDraw classes at $n = 60000$. That is 8.6–9.6% below a CPU-matched MLP ensemble, a tie by the cell rule, and clearly below an MLP given v2's representation (`qd_c`).

5.4 Compute

FLOPs [Numerical] (Table 11).

- *v3 and v2.* By its own count v3 uses 0.30–0.58 of the budget on the held-out cells, the most at $n = 60000$. v2 uses 0.22–0.42. `cgbt` itself is charged 0.14–0.22 of the budget; the net increase over v2 is 0.05–0.22, because `ggbt` is not refitted where its weight is 0.
- *Near the guard at $n = 3000$.* On the development sets, at $n = 3000$ (no held-out cell has this size), v3 used 0.85 of the budget on `qd_a` and on `qd_b`. That is close to the guard at 0.9,

which skips members whose projected cost would exceed it (Section 2.2). Part 9's compute audit found that the FLOP charge of Part 8's `gbt` member is too low by a factor of 1.14–1.46 at this size, and it projected v2's $n = 3000$ QuickDraw cells at about 0.72 (0.69–0.81) of the budget, against v2's own count of at most 0.62. v3 keeps that charge, and no compute audit was made of v3. Applying the audit's factor to `gbt` alone (0.28 of the budget by v3's count) puts these two cells at 0.88–0.98 [Heuristic]. So v3 is within the budget at $n = 3000$ by its own count, but not with a clear margin.

- *The MLP references.* Every MLP reference spends more FLOPs than the budget: `mlp_strong` 2.0 budgets per fit (six training runs), `mlp_wide` 5.0, `mlp_pool` 5.9 and `mlp_xl` 9.2. So by FLOPs v3 is cheaper than every MLP reference at every cell, by a factor of at least 3.4 (`mlp_strong` at `qd_d`, $n = 60000$).

CPU time [Numerical]. Equal FLOPs are far from equal CPU time.

- *v3 and v2.* A v3 fit takes 229–291 CPU s at $n = 1000$, 402–785 s at $n = 10000$, and 1361 and 1464 s at $n = 60000$. v2 takes 144–175, 284–602, and 1296 and 1531 s.
- *v3 against v2 at $n = 60000$.* v3 takes about as long as v2 (1.05× on `qd_c`, 0.96× on `qd_d`) although it fits a second booster, because it sets LightGBM's column-wise layout, which leaves the predictions unchanged and which Part 9 found but did not apply to v2. Like for like, the second booster about doubles the CPU time of a fit (Appendix B.5).
- *The MLP references.* At $n = 60000$ they take 963 and 422 s of wall time (`mlp_strong`), 1168 and 989 s (`mlp_wide`) and 2196 and 1335 s (`mlp_xl`); `mlp_pool` takes 716 s. So `mlp_xl` takes 0.9–1.6× v3's time (wall against CPU seconds), which is the sense in which it is CPU-matched. The other ensembles take 0.3–0.9 of v3's time. These are wall-clock seconds on a shared machine and vary with its load (Appendix B.5).

Prediction is not charged for any learner. A v2 query costs about 2.8×10^8 FLOPs at $n = 60000$, about 500× an MLP's, because the nearest-neighbour members scan the training set (Part 9). v3 adds `cgbt`'s bank projection, about 1.2×10^7 FLOPs per query, and up to about 4000 more tree traversals (`study/part10/p10_codes/DESIGN.md`, §6).

6 What it learns

This section describes what v3's fitted members contain on the toy targets (Section 6.1) and on images (Section 6.2). The analysis is descriptive: it says what structure the fits hold and when it appears, not why v3 does well, and nothing here is an ablation of the learner or a causal test. Every analysed fit is an instrumented seed-0 rerun of an evaluation fit, with the same training rows, column permutation and random streams. On the toy side every rerun reproduces the evaluation's seed-0 test error; on images, every rerun that has an evaluation cell reproduces it (Appendix D.4). Two independent verifiers per side checked the numbers and the readings, and their corrections are used throughout. Measured quantities are [Numerical]; descriptions of maps and explanations of mechanisms are [Heuristic].

Findings.

1. *DFG finds the exact signed blocks* on 49 of the 54 sum-structured targets (median $n_{\text{struct}} = 256$), and its mean output then follows $E[f \mid t]$ level by level (Section 6.1.1).

2. *Detection is cheap; discovery lags.* On the 25 pure symmetric targets the true block would pass DFG's test at $n = 128$ every time. The delay to discovery tracks the share of variance at Fourier degrees 1–2 (Spearman -0.86), and the 10% error point arrives with the block (Section 6.1.2). Since every seed of the block search starts from those statistics, this reads as the search, not the test, setting DFG's data cost [Heuristic].

3. *The residue classes mod 3 are never proposed.* They have $W_1 + W_2 \leq 0.0002$; the search's orders reach their block no better than chance and no piece of it carries signal, although at $n = 16384$ the true block would pass the test at 134–155 times the threshold (Section 6.1.3).

4. *Pooling sets and bars recover local pieces of the pixel grid.* On QuickDraw 98.3–99.9% of the other members of a 4-pixel set lie within one step, and bars are short runs along recovered lines; on Fashion-MNIST 7% of the 4-pixel sets include the mirror pixel (Section 6.2.1).

5. *The bar booster carries v3 on QuickDraw and MNIST.* `cgbt` has the lowest out-of-fold error of all 18 members in every analysed fit and takes `ggbt`'s stacking weight there; bars carry 0.77–0.80 of its split gain, 1.1–1.2 times their column share (Section 6.2.2).

6. *Much class evidence lies on blank pixels*: 0.56–0.69 of `cgbt`'s net TreeSHAP evidence, often where other classes have ink, though per pixel inked pixels carry more on QuickDraw and MNIST (Section 6.2.3).

Appendix C has the per-target detail, the models below the discovery size and on the other suites, and the caveats; Appendix B.2 says where v1.3's additions act in the fitted models.

6.1 Toy targets: the search, not the test, sets DFG's data cost

On toy inputs v3 is a stack of DFG v1.3 (a default variant and, at $n \leq 512$, a forced small-n variant) and the classical members (Section 2.2). The analysis looks at DFG's final model in each fit. It covers the 54 sum-structured targets of the six suites built around signed sums (`dlsymdev`, `dlfamdev`, `dlheld4`–`dlheld7`; the 14 non-symmetric controls are excluded) and, more briefly, the 70 targets of the other suites. The grid is $n = 16, 32, \ldots, 16384$, stopped after two consecutive sizes with test error ≤ 0.005.

Definitions.

- *Terms and shares.* The model's terms are its characters, sum blocks (Section 2.3), spectral features and the kernel background. A term's *share* is the standard deviation of its contribution to the model's output over the 4096 test inputs.

- *Exact structure.* The target's signed-symmetry classes are the sets of inputs it treats alike, up to sign. A class of size ≥ 3 is *matched exactly* when a sum block, a factor of a product block or the modulator χ_M of a modulated block has the same input set and the same signs up to a global flip. n_{struct} is the first grid size at which every class of size ≥ 3 is matched in the same model; we call it the *discovery size*. For a weighted target $u = s + a\tau$ the class is the unit-weight part s; the inner term is checked separately (Appendix C.1).

- *Detection.* n_{detect} is the first grid size at which DFG's own block test (3) would admit the *true* block (the largest class, with its signs) if it were proposed in round 0 on the seed-0 training rows.

- *Reach.* The fraction of the true class S among the first $|S|$ inputs of the better of the search's two eigenvector orders (Figure 5a).

	targets	exact structure	median n_{struct}	range of n_{struct}	detectable at $n = 128$	delay (doublings) < 0 / 0 / 1 / ≥ 2	never found
by mixture							
pure symmetric	25	22	256	64–8192	25	1 / 9 / 3 / 9	fd_10, f7_05, h6_00
with a parity	9	8	256	128–16384	9	0 / 3 / 3 / 2	h6_02
with a gate	5	5	256	128–1024	0	2 / 3 / 0 / 0	–
multiplexer	4	3	512	512–1024	0	0 / 2 / 0 / 1	f7_11
two sums	5	5	256	256–512	0	3 / 2 / 0 / 0	–
weighted sum	6	6	256	128–2048	5	0 / 1 / 3 / 2	–
by accept set (a two-block target counts once under each kind it has)							
threshold	11	11	256	128–2048	7	2 / 3 / 4 / 2	–
band	19	18	192	64–1024	10	4 / 11 / 1 / 2	f7_11
interval	10	9	256	128–1024	6	2 / 5 / 2 / 0	h6_02
arbitrary set	16	15	512	128–16384	13	0 / 3 / 2 / 10	f7_11
residue class	5	2	768	512–1024	3	0 / 1 / 0 / 1	fd_10, f7_05, h6_00
all	54	49	256	64–16384	39	6 / 20 / 9 / 14	fd_10, f7_05, f7_11, h6_00, h6_02

Table 12: Exact structure in v3's DFG model on the 54 sum-structured targets, seed 0 [Numerical]. "Exact structure": targets whose model matches every signed-symmetry class of size ≥ 3 at some analysed n; n_{struct} is the first such n. "Detectable at $n = 128$": the true block passes DFG's test (3) on the seed-0 training rows at $n = 128$. For the five weighted targets other than sumpar the tested block is the weighted one, with v1.3's larger threshold. For the gate and multiplexer mixtures the detection points are the verifier's, recomputed with the residual deflated by the modulator as DFG tests it; the analysis file has 4 more targets at $n = 128$. "Delay": $\log_2(n_{\text{struct}}/n_{\text{detect}})$ over the targets found. In the lower panel a target with two blocks counts once under each accept kind it has. Sources: `study/part10/interp/tables/toy_timing.csv` and `study/part10/interp/VERIFICATION.json`.

- *Low-degree share.* W_k is the target's Fourier weight at degree k, and $(W_1 + W_2)/(1 - W_0)$ is the share of its variance at degrees 1 and 2.

The exact-structure criterion checks the input set and the signs, not the learned function of the block or its fold. It still tracks the function closely: at n_{struct} the DFG member's test error is at most 0.03 for every target found. It is also strict: some models fit a target exactly with another representation, and those count as not found (below). n_{detect} uses the true block, an oracle: it measures what the test would accept, not what the search would propose. For mixtures it is a rougher proxy, because DFG tests a block in later rounds on a residual from which the gate or parity characters are already fitted out (Appendix C.2). Further caveats are in Appendix C.4.

6.1.1 Exact structure

Exact structure is recovered on 49 of 54 targets [Numerical]. Table 12 gives the counts.

- *When.* The median n_{struct} is 256 (range 64–16384). 15 targets have the exact structure by $n = 128$, 30 by 256 and 41 by 512.
- *By suite.* `dlsymdev` 8 of 8, `dlfamdev` 8 of 9, `dlheld7` 7 of 9, `dlheld6` 7 of 9, `dlheld5` 10 of 10, `dlheld4` 9 of 9.
- *Never found* (by the largest analysed n; f7_11's and h6_02's runs stopped at 4096 and 1024 because the portfolio's error was already ≤ 0.005): the three residue classes mod 3 of a 10–14-input count (fd_10, h6_00, f7_05), and two targets that DFG fits by another representation. On f7_11, a multiplexer between a 4-input and a 6-input block, each class is

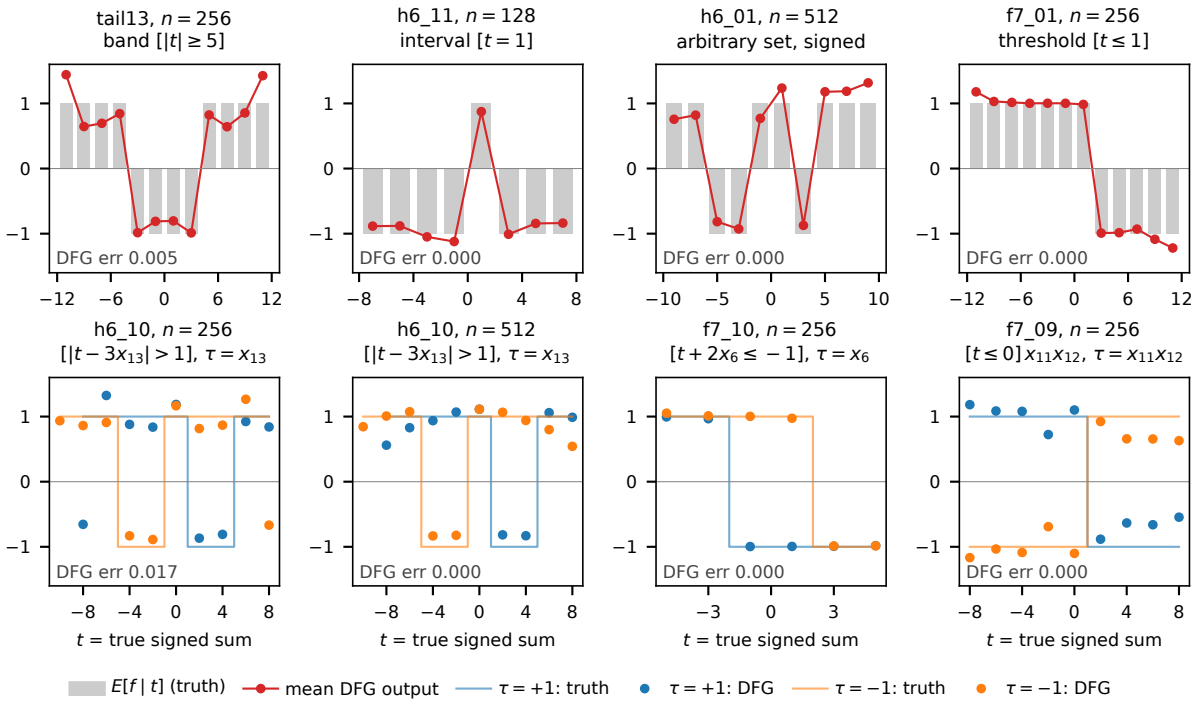


Figure 3: The learned function of the true block statistic t (seed 0) [Numerical]. Top: grey bars are $E[f \mid t]$ over the 4096 test inputs; red points are the mean output of the whole DFG model at each level of t. Bottom: the same split by the second term τ, from refits that reproduce the analysed cells. The plotted value is the conditional mean of the whole model, which includes spectral features, characters and stray product terms, not the block's own coefficients. Levels with fewer than 5 test points are not drawn; for h6_10 at $n = 256$ these include the cells $(t, x_{13}) = (-10, +1)$ and $(10, -1)$, where the model is also wrong. "DFG err": the DFG member's test error. Source: the structure runs and `toy_g` refits in `study/part10/interp/results/toy_final/`.

matched at some n (128 and 2048), but never both in one model; at $n = 4096$ the 4-input branch is carried by characters of degree 2–3. On h6_02 $= [1 \le s \le 5]\, x_{10}x_{11}$, the model is a modulated block plus characters at $n = 512$ (DFG error 0.002), and at $n = 1024$ the degree-11 character on $x_1, \ldots, x_{11}$ plus degree-3 characters.

- *Accept kinds.* Of the accept kinds learned on most of their targets, arbitrary sets need the most data (median 512; h5s7 8192, h5s1 16384). The only residue-class blocks that are learned are the 6-input multiplexer branches of f7_03 (mod 4, $n = 512$) and h6_08 (mod 3, $n = 1024$).

What the model looks like [Numerical]. Where the structure is found, the leading terms take a few recurring forms: one block on the class with its signs (folded for accept sets symmetric under $t \to -t$) for pure targets, degree-1 characters beside the block for thresholds, a modulated block for a parity times a symmetric function, a product block for two sums, and gates and selectors as characters and, in v3, as the inner literal of a weighted block. The five weighted targets other than sumpar are fitted to DFG error 0 at their largest analysed n, two of them through exact product forms rather than the weighted block. Appendix C.1 gives the examples and the inner terms.

The learned function of the block [Numerical] (Figure 3). Where the block is present, the model's mean output follows $E[f \mid t]$ level by level. This includes non-convex accept sets, such

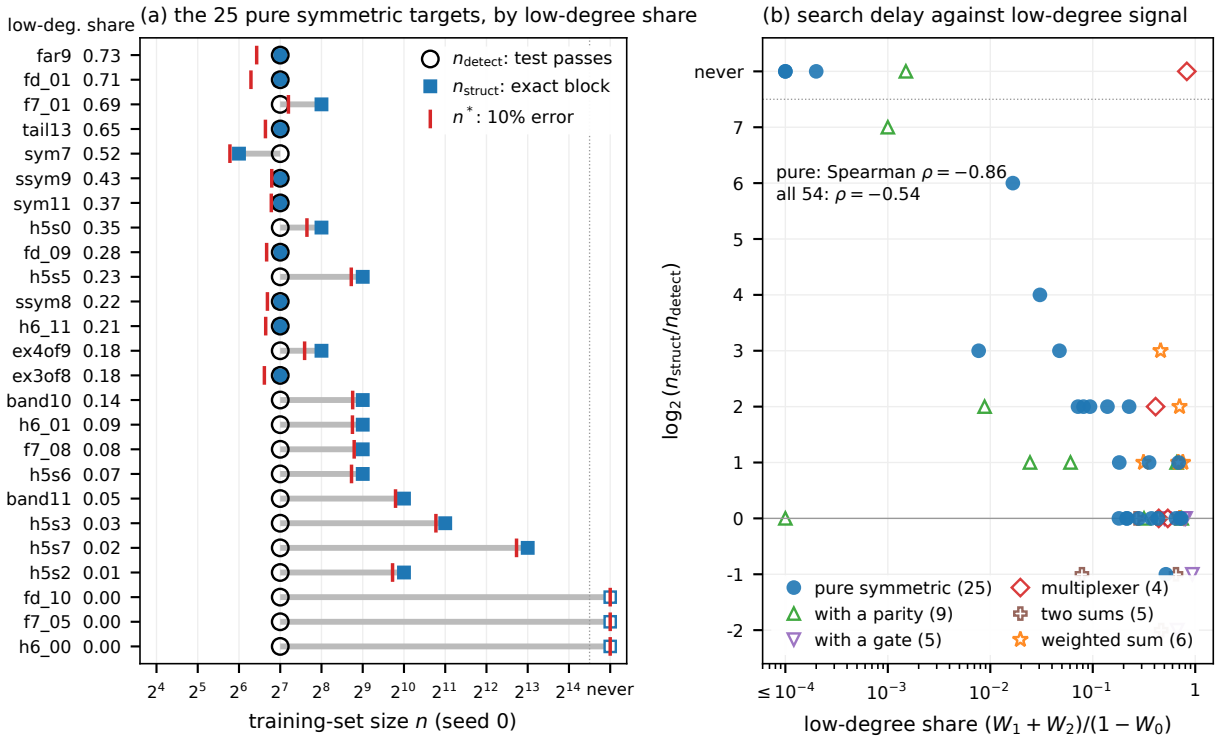


Figure 4: When the structure appears (seed 0) [Numerical]. (a) The 25 pure symmetric targets, sorted by their share of variance at Fourier degrees 1–2 (number after the name). Open circle: n_{detect}, the first grid size at which DFG's test would admit the true block. Square: n_{struct}, the first grid size at which the model holds it exactly (open at "never"); where $n_{\text{detect}} = n_{\text{struct}}$ the circle surrounds the square. Red tick: n^*, the 10% test-error point of the same seed-0 fits, interpolated in $\log n$. The grey bar spans detection to discovery. (b) The delay $\log_2(n_{\text{struct}}/n_{\text{detect}})$ against the low-degree share for all 54 sum-structured targets; filled: pure symmetric, open: mixtures. Targets never found are drawn in the top row. Shares below 10^{-4} are drawn at 10^{-4}, so h6_00 and f7_05 coincide at the top left. For the mixtures the detection points are the verifier's corrected ones (Table 12). Spearman ρ counts a target never found as $n_{\text{struct}} = 2^{15}$. No point is jittered. This figure replaces the analysis's `fig_toy_timing.pdf`, which used the uncorrected detection points. Sources: `study/part10/interp/tables/toy_timing.csv` and `study/part10/interp/VERIFICATION.json`.

as h6_01's $\{-9, -7, -5, -1, 1, 7, 9\}$, and both branches of a weighted or modulated target. The values are not ± 1: the logistic readout rescales them and overshoots at rare extreme levels (tail13 at $|t| = 11$). The panels add shape, not a new test. The share of test points at which the mean output has the sign of $E[f \mid t]$ is about one minus the DFG error by construction (0.983 for h6_10 at $n = 256$, 1.00 for the other seven panels). Below the discovery size the model has no block structure: for h6_10 at $n = 128$ the mean output is positive at every (t, x_{13}) level (0.01–0.90), so its sign agrees with $E[f \mid \cdot]$ only on the majority's 0.70 of the points, and the DFG error (0.39) is worse than a constant predictor's (0.30). The eight panels were chosen by hand; over all 53 exact cells of the 22 pure targets found, the agreement is at least 0.985 (`study/part10/interp/VERIFICATION.json`, toy reading, C8).

6.1.2 When the structure appears

Timing: the 25 pure symmetric targets [Numerical]. On these targets the analysis is cleanest: the target is a function of one signed sum, and the round-0 test on the true block is close to the test DFG applies. Figure 4a shows each target.

- *Detection is cheap.* The true block passes DFG's test at $n = 128$ for all 25 targets. The test statistic grows linearly in n: its ratio to the threshold is at most 0.89 at $n = 64$ and 1.07–1.78 at $n = 128$.
- *Discovery is later where the low-degree signal is weak.* 22 of the 25 are found. Against n_{detect}, n_{struct} is one doubling earlier for 1 target (sym7, held by the forced variant below the threshold), equal for 9, one doubling later for 3, and two or more later for 9 (band11, f7_08, h6_01, h5s2, h5s3, h5s5, h5s6, h5s7, band10).
- *The delay tracks the low-degree share* (Figure 4b). Spearman's ρ between the delay and $(W_1+W_2)/(1-W_0)$ is -0.86 over the 25 targets (a target never found counted as $n_{\text{struct}} = 2^{15}$), and -0.80 over the 22 found. The targets come from a few designed families, so a p-value that treats them as independent draws would overstate the evidence.
- *The readout is not the bottleneck.* For all 22 targets found, the seed-0 n^* falls in the doubling just below n_{struct}: the test error crosses 10% between the last grid size without the exact block and the first with it. Because n_{struct} is a grid point and n^* is interpolated, a finer statement cannot be made. In particular, the median offset of $\log_2(n^*/n_{\text{struct}})$ (-0.35 over all 49 targets found) reflects the grid, not the readout.

Reading [Heuristic]. For a function of one signed sum, DFG's data cost is set by its search, not by its test and not by its readout. Every seed of the block search reads the block's inputs off degree-1 and degree-2 statistics of the residual: prefixes of the top eigenvectors of the pair matrix, the degree-1 order and greedy growth from strong pairs (Section 2.3). The weaker those statistics, the more data the search needs before it proposes the block, although the test would accept the block much earlier. No intervention isolates the search (such as seeding it with the true block); acceptance in later rounds and the stacker's choice of variant could also contribute.

The 29 mixtures [Numerical]. For targets with a parity, gate, selector or second block, the round-0 test on the true block is a cruder proxy. With the verifier's corrected detection points (the modulated block tested on a residual already fitted on its modulator, as DFG tests it), 39 of the 54 targets are detectable at $n = 128$, the median delay over the 49 found is 0 doublings, and Spearman's ρ between delay and low-degree share over all 54 is -0.54 (Appendix C.2).

6.1.3 Where DFG fails: no low-degree foothold

The failures: residue classes mod 3 [Numerical]. Let $B = (s + k)/2$ count the k signed inputs of the block that equal $+1$. Then fd_10 $= [B \bmod 3 = 0]$ on $k = 10$ inputs, h6_00 $= [B \bmod 3 \neq 1]$ on 13 and f7_05 $= [B \bmod 3 \neq 0]$ on 14. Neither v3 nor v2 learns them (Section 4.3). Figure 5 shows why the search does not find them.

- *The model.* At every $n \geq 1024$ (fd_10 from 128, f7_05 from 256) the DFG model is the constant plus the kernel background. Its test error at $n = 16384$ (0.32–0.34) is the test set's minority rate. For fd_10 and h6_00 these errors come from out-of-fold DFG fits on 13107 rows, which the stacker gave no weight. At smaller n the forced variant adds 1–2 spurious

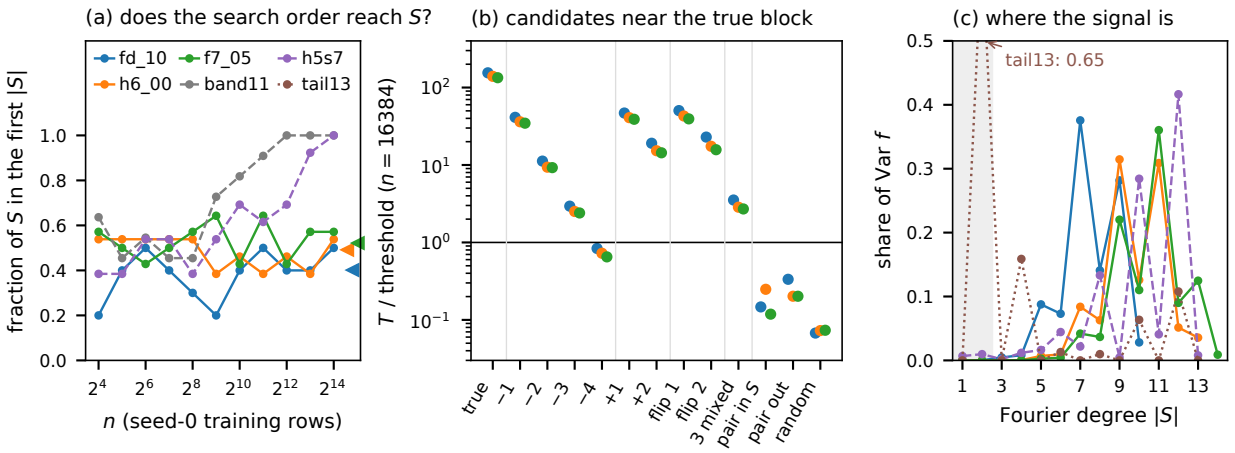


Figure 5: Why the residue classes mod 3 are never found (seed-0 training rows) [Numerical]. (a) The fraction of the true class S among the first $|S|$ inputs of the better of the search's two eigenvector orders. Triangles at the right edge: the value expected from the better of two random orders (0.40, 0.49, 0.52 for $|S| = 10, 13, 14$). band11 and h5s7 are dashed (h5s7 also in (c)); they are found at the first n at which all of S lies within the order's first $|S| + 5$ inputs (plotted fraction 0.82 and 0.92). The legend of (a) also serves (b) and (c). (b) DFG's test statistic over its threshold at $n = 16384$ for the true block and for candidates near it: medians over 200 random candidates with 1–4 inputs dropped, 1–2 added, 1–2 signs flipped, or 3 errors of mixed kinds; the best 2-input block inside S; the best pair among 12 inputs outside S; random blocks of the true size. Above 1 the candidate passes. (c) The share of each target's variance by Fourier degree; degrees 1–2 shaded. tail13, found at $n = 128$, has 0.65 at degree 2 (clipped). A compact version of the analysis's `fig_toy_failures.pdf`, with the chance level of panel (a) corrected for the better of two orders. Sources: `power/` and `toy_landscape/` in `study/part10/interp/results/toy_final/`.

characters or a spurious partial block (h6_00 at $n = 256$: $|-x_4 + x_{12} - x_{22} - x_{23}|$, 2 of its 4 inputs in S).

- *The test would accept the block.* The true block passes from $n = 128$ unfolded (512 folded). At $n = 16384$ it is 155, 139 and 134 times the threshold.
- *There is no low-degree signal.* For h6_00 and f7_05, $W_1 = W_2 = 0$ to five decimals; for fd_10, $W_1 = 4 \times 10^{-5}$ and $W_2 = 1.7 \times 10^{-4}$. The Fourier weight peaks at degrees 7, 9 and 11.
- *The search's orders do not reach the block* (Figure 5a). At $n = 16384$ the eigenvector orders put 0.50, 0.54 and 0.57 of S among their first $|S|$ inputs, against 0.40, 0.49 and 0.52 for the better of two random orders, and the fraction does not grow with n.
- *The basin is narrow* (Figure 5b). At $n = 16384$ the statistic falls about 4× per missing input (to 41, 36 and 35 times the threshold with one input dropped; 2.9, 2.5 and 2.4 with three; 0.84, 0.73 and 0.65 with four). The first wrong sign or extra input costs 3.1–3.4×, the second 2.2–2.7×. Every sampled candidate with three errors of mixed kinds still passes. But no piece of the block carries signal: the best pair inside S is at 0.15, 0.25 and 0.12 of the threshold, at the level of the best pair among 12 inputs outside S (0.33, 0.20, 0.20), and random blocks of the true size are at 0.07. The basin estimate rests on medians of sampled candidates; other 4-error mixes were not measured.
- *A needle.* Within three errors of the true block there are $\sum_{j \le 3} 2^j \binom{30}{j} = 34281$ candidates, against $\binom{30}{k} 2^{k-1} = 1.5 \times 10^{10}$, 4.9×10^{11} and 1.2×10^{12} blocks of the true size $k = 10, 13, 14$.

		pooling sets: members other than i				Isomap R^2		bars: exact lines	
set	n	$S_i(4)$ ≤ 1 step	$S_i(9)$ ≤ 1 step	$S_i(9)$ ≤ 2 steps	$S_i(9)$ > 2 steps	$k=4$ union	$k=9$ mutual	$L=2$	$L=4$
qd_dev	1000	0.983	0.823	0.985	79	0.84	0.87	0.33	0.16
qd_dev	3000	0.991	0.829	0.991	46	0.97	0.89	0.38	0.21
qd_dev	10000	0.994	0.834	0.994	31	0.97	0.92	0.42	0.24
qd_a	10000	0.999	0.817	0.989	59	0.99	0.83	0.40	0.25
MNIST	10000	0.976	0.787	0.966	186	0.92	0.97	0.20	0.06
Fashion	10000	0.881	0.633	0.840	1006	0.13	0.23	0.11	0.04

Table 13: The pooling sets and bars of v3's image members on the true pixel grid (seed 0) [Numerical]. Pooling sets: the fraction of the $k-1$ members of $S_i(k)$ other than i itself that lie within 1 or 2 Chebyshev steps of i, over all non-constant pixels i; "> 2 steps": the number of such members (of 5408 on QuickDraw at $k=9$). Isomap R^2: Procrustes R^2 between a 2-D Isomap embedding of the pooling-set graph (an edge from i to each member of $S_i(k)$; "mutual": only edges in both directions) and the true pixel coordinates. Bars: the fraction of bars of half-length L whose pixels lie on one horizontal, vertical or diagonal lattice line. MNIST's $k=9$ mutual graph covers 674 of its 682 non-constant pixels. Source: the analysis files in study/part10/interp/results/image_final/v3/.

Reading [Heuristic]. A residue class mod 3 of a 10–14-input count is detectable once proposed and invisible from every low-order foothold the search has. For h6_00 and f7_05 the single and pair statistics are zero in expectation, so more data cannot help the seeds. For fd_10 they are about 0.002, which the seeds could resolve only at n of order 10^6.

Late targets [Numerical]. The pure targets found late, band11, h5s2, h5s3 and h5s7 (found at 1024–8192, with $W_1 + W_2$ between 0.007 and 0.047), and h5s1, whose bottleneck is detecting its modulating parity, are consistent with the same mechanism in weaker form; h6_08 and sumpar are late for representational reasons. Each is one seed (Appendix C.3).

6.2 Images: what the boosters split on

The analysis covers v3 on qd_dev at $n = 1000$, 3000 and 10000, on qd_a, MNIST and Fashion-MNIST at $n = 10000$, and v2 on qd_dev at $n = 10000$. All four sets are development data for v3 (the held-out image sets are qd_c and qd_d), so the error comparisons below are in-sample; the held-out comparison is in Section 5.3. v2's analysis fit has test error 0.1467 against its multi-threaded evaluation cell's 0.1463 (Appendix D.3).

- *Which booster is analysed.* The final ggbt and cgbt of each fit. On qd_a and MNIST the stacker gave ggbt weight 0, so it was not refitted, and the booster analysed is its out-of-fold fit on 8000 (7998) rows.
- *Geometry.* It is measured on the true 28×28 grid, which the learner never sees; for QuickDraw the columns are mapped back through the stored permutation. Distances are Chebyshev steps, so the 8 neighbours of a pixel are 1 step away.

6.2.1 Pooling sets and bars on the true grid

Pooling sets are local neighbourhoods; on Fashion-MNIST some include the mirror pixel [Numerical]. Table 13 gives the geometry.

- *QuickDraw.* 98.3–99.9% of the other members of $S_i(4)$ lie within one step of i, and 98.5–99.4% of those of $S_i(9)$ within two. Only 8 pixels lie within one step, so for $k = 16$ at most $8/15 = 0.53$ of the other members can; 0.49 do. On `qd_dev` the members of $S_i(9)$ farther than two steps fall from 79 to 31 as n grows from 1000 to 10000. The sets overlap the k truly nearest pixels by 0.82, 0.88 and 0.78 for $k = 4$, 9 and 16 at $n = 10000$.
- *MNIST.* The 23 far members of $S_i(4)$ sit at 13 border pixels that are nonzero in at most 0.04% of the 10000 training images. Their correlations rest on a handful of images.
- *Fashion-MNIST.* At 57 of the 784 pixels (7%), $S_i(4)$ contains a far member, and all 65 far members are the centre's mirror pixel: the same row ± 1, with columns summing to 27 ± 2. At $k = 9$, 619 of the 1006 far members are mirror pixels.

Reading [Heuristic]. Garments are nearly left–right symmetric, so a pixel can correlate with its mirror image as strongly as with its neighbours. Sets at those pixels pool both sides of the garment.

The grid is recovered globally on QuickDraw and MNIST [Numerical]. The 2-D Isomap embedding [26] of the $k = 4$ pooling-set graph matches the true grid with Procrustes R^2 0.97–0.99 on QuickDraw at $n \geq 3000$ (0.84 at $n = 1000$), 0.92 on MNIST and 0.13 on Fashion-MNIST. Variants of the graph with a few long edges score much lower (Appendix C.5). The learner never builds a global embedding; it uses each set as a local sum. So R^2 summarizes how map-like the recovered neighbourhoods are, not a property the learner uses.

Bars are short, nearly straight runs along recovered lines [Numerical] (Table 13, Figure 6).

- *Shape on `qd_dev`.* At $n = 10000$, 42% of the 5-pixel bars ($L = 2$) are exact lattice lines, and 96% stay within one pixel of their principal axis. For the 9-pixel bars ($L = 4$) the figures are 24% and 63%, since each further step can turn.
- *Growth with n.* The exact-line fraction grows with n (0.33, 0.38, 0.42 for $L = 2$ at $n = 1000$, 3000, 10000; 0.16, 0.21, 0.24 for $L = 4$). The within-one-pixel fraction does not, and it is a weak criterion for 5 pixels.
- *Orientation.* On QuickDraw all four orientations occur about equally often; on Fashion-MNIST horizontal and vertical bars dominate (0.37 and 0.40 at $L = 2$).
- *Other sets.* On MNIST the bars are less straight (0.20 and 0.06 exact lines). On Fashion-MNIST they are rarely lines (0.11 and 0.04), and 27% ($L = 2$) and 30% ($L = 4$) contain a mirror pixel, against 2% and 9% on `qd_dev`. There a bar is often two short pieces on the two sides of the garment (Figure 6h).

Straightness is a property of the construction and the data, not of anything the booster learned.

6.2.2 Which members carry v3

`cgbt` is the member that carries v3 on QuickDraw and MNIST [Numerical] (Table 14).

- *The best member.* `cgbt` has the lowest out-of-fold error of all 18 members in every v3 fit. The margin over `ggbt` is within holdout noise on Fashion-MNIST and at `qd_dev`, $n = 1000$. It has the largest single-member weight everywhere except Fashion-MNIST, where Part 8's `gbt` has it.
- *Its weight.* Its share of the stacking weight grows with n on `qd_dev` (0.32, 0.40, 0.47) and is 0.56 on `qd_a`, 0.40 on MNIST and 0.10 on Fashion-MNIST.

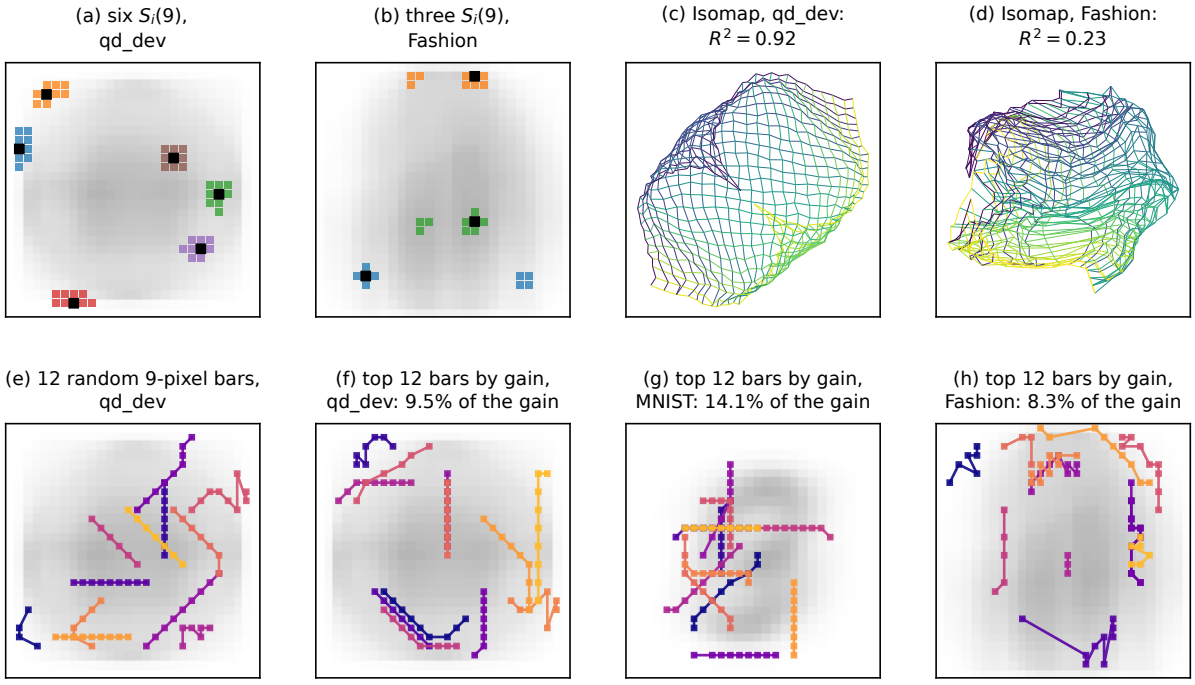


Figure 6: Pooling sets and bars of v3 on the true pixel grid, $n = 10000$, seed 0 [Numerical]. Background: the average of the ten class-mean training images. (a) $S_i(9)$ at six pixels of `qd_dev` (black: i). (b) $S_i(9)$ at three inked Fashion-MNIST pixels whose sets contain mirror pixels. (c, d) The 2-D Isomap embedding of the $k = 9$ mutual pooling-set graph, Procrustes-aligned to the grid and drawn as the true lattice: each pixel at its embedded position, joined to its horizontal and vertical grid neighbours (horizontal edges coloured by row, vertical edges by column). (e) Twelve randomly chosen 9-pixel bars ($L = 4$) of `qd_dev`. (f–h) The twelve 9-pixel bars with the most split gain in `cgbt`, with their combined share of `cgbt`'s total gain. In (e)–(h) colour only distinguishes bars (in (f)–(h) darker means more gain); a bar's pixels are joined in order along its principal axis, so a long segment joins two separate pieces of one bar. Simplified from `fig_img_topology.pdf` and `fig_img_bars.pdf` in study/part10/interp/figures/.

		share of the stacking weight						out-of-fold error		test error		
set	n	`cgbt`	`ggbt`	`gknn`	`knn`	`gbt`, `et`	`lin_C`	`cgbt`	`ggbt`	stack	`cgbt` alone	`ggbt` alone
`qd_dev`	1000	0.32	0.09	0.44	0.16	0.00	0.00	0.272	0.280	0.2118	0.2208	0.2478
`qd_dev`	3000	0.40	0.08	0.33	0.19	0.00	0.00	0.197	0.212	0.1753	0.1851	0.1968
`qd_dev`	10000	0.47	0.07	0.17	0.25	0.03	0.00	0.149	0.161	0.1330	0.1368	0.1578
`qd_a`	10000	0.56	0.00	0.15	0.24	0.06	0.00	0.161	0.172	0.1422	0.1442	0.1654[a]
MNIST	10000	0.40	0.00	0.23	0.24	0.13	0.00	0.031	0.043	0.0255	0.0278	0.0392[a]
Fashion	10000	0.10	0.16	0.00	0.04	0.50	0.20	0.126	0.127	0.1238	0.1216	0.1222
v2, `qd_dev`	10000	–	0.41	0.24	0.24	0.11	0.00	–	0.161	0.1467	–	–

Table 14: Which members carry v3 on images (seed 0, one fit each) [Numerical]. Left: each member family's share of the total stacking weight. `gknn`: k-NN votes on pooled sums; `knn`: k-NN votes on pixels; `gbt`, `et`, `lin_C`: Part 8's LightGBM, ExtraTrees and logistic regression on pixels. Middle: out-of-fold error of the two boosters (5-fold at $n \leq 3000$; one holdout of 2000 rows at $n = 10000$). Right: test error of the stacked learner (v3; v2 in the last row) and of each booster alone (its own argmax). [a] `ggbt` had stacking weight 0, so the booster shown is its out-of-fold fit on 8000 (`qd_a`) or 7998 (MNIST) rows; only the out-of-fold comparison is like for like there. Last row: v2's analysis fit. The weights are not shares of the decision (see the cautions in the text). Source: the analysis files in study/part10/interp/results/image_final/.

	cgbt				ggbt	
feature type	qd_dev	qd_a	MNIST	Fashion	qd_dev	Fashion
pixels	0.01 (0.1)	0.01 (0.1)	0.01 (0.1)	0.09 (1.0)	0.04 (0.1)	0.16 (0.7)
pooled sums, $k = 4$	0.02 (0.3)	0.03 (0.3)	0.05 (0.6)	0.10 (1.1)	0.10 (0.4)	0.15 (0.7)
pooled sums, $k = 9$	0.04 (0.5)	0.05 (0.5)	0.07 (0.9)	0.10 (1.2)	0.16 (0.7)	0.16 (0.7)
pooled sums, $k = 16$	0.15 (1.8)	0.12 (1.5)	0.07 (0.8)	0.12 (1.4)	0.32 (1.5)	0.15 (0.6)
bars, $L = 2$	0.24 (0.7)	0.25 (0.8)	0.30 (0.9)	0.25 (0.7)	–	–
bars, $L = 4$	0.53 (1.6)	0.54 (1.6)	0.51 (1.5)	0.35 (1.1)	–	–
prototype pairs	–	–	–	–	0.39 (4.1)	0.38 (4.4)
all bars	0.77 (1.2)	0.79 (1.2)	0.80 (1.2)	0.60 (0.9)	–	–
all pooled sums	0.22 (0.9)	0.20 (0.8)	0.19 (0.8)	0.31 (1.3)	0.57 (0.9)	0.46 (0.7)
columns	8220	8220	8286	9408	3112	3436

Table 15: Split gain by feature type in the final boosters, $n = 10000$, seed 0 [Numerical]. Each entry is the type's share of the booster's total split gain, with its ratio to the type's share of the columns in brackets (1.0: as much gain per column as the booster's average). Bars are two thirds of cgbt's columns; the 300 prototype pairs are about a tenth of ggbt's. LightGBM samples 5% (cgbt) or 12% (ggbt) of the columns per tree. ggbt on qd_a and MNIST, an out-of-fold fit, is not shown (pairs 0.30 and 0.41 of its gain). Split gain describes how the trees split, not what the predictions depend on. Source: the analysis files in study/part10/interp/results/image_final/v3/.

- *ggbt loses its weight.* At qd_dev, $n = 10000$, ggbt's share falls from 0.41 in v2's fit (0.37 in v2's evaluation cell) to 0.07 in v3. On qd_a and MNIST it is 0.
- *The rest of the weight.* Most of it goes to the k-NN members, on pooled sums and on pixels. On Fashion-MNIST half goes to Part 8's gbt and et and a fifth to logistic regression.
- *Alone.* On the test set cgbt alone beats ggbt alone on QuickDraw and MNIST (on qd_a and MNIST against ggbt's out-of-fold fit on fewer rows; the like-for-like out-of-fold errors, 0.161 against 0.172 and 0.031 against 0.043, point the same way): 13.68% against 15.78% at qd_dev, $n = 10000$. The two agree on 89–98% of test points. The stack is 0.2–1.0 points better than cgbt alone on QuickDraw and MNIST; the 0.2 points on qd_a are not resolved by the test set. On Fashion-MNIST the stack (12.38%) is not better than either booster alone (12.16%, 12.22%), a difference the test set does not resolve either.

Three cautions apply to the stacking weights, here and in Table 10.

- They act on log-probabilities of different calibration. Boosters stopped early on error give over-confident log-probabilities (clipped at 10^{-5}), while k-NN votes, ExtraTrees and logistic regression are smoother. Equal weights are not equal influence.
- The convex stacker splits weight between two nearly collinear boosters on one 2000-row holdout, so "ggbt gets weight 0" describes one fit. The 8 held-out fits at $n \geq 10000$ show the same shift (Section 5.3).
- The out-of-fold errors at $n = 10000$ rest on those 2000 rows. Unpaired, the standard error of an error near 0.15 is about 0.008; for two members on the same rows the paired one is smaller.

Split gain by feature type [Numerical] (Table 15).

- *The bars in cgbt.* The bars carry 0.77–0.80 of cgbt's split gain on QuickDraw and MNIST at $n = 10000$ (0.72 at qd_dev, $n = 1000$) and 0.60 on Fashion-MNIST. They are two thirds of its

columns, so this is 1.1–1.2 times their column share (0.9 on Fashion-MNIST): more than their share, not dominance. They also make 0.75 of the splits at `qd_dev`, $n = 10000$. Per column, the 9-pixel bars (1.5–1.6; 1.1 on Fashion-MNIST) and, on QuickDraw, the 16-pixel pooled sums (1.5–1.8) carry the most. The 5-pixel bars carry 0.7–0.9 of the average, and single pixels 0.1, except on Fashion-MNIST (1.0).

- *The pairs in `ggbt`.* The 300 prototype pairs carry about 0.4 of `ggbt`'s gain (0.30 in `qd_a`'s out-of-fold fit), about four times their column share. On `qd_dev` the 16-pixel pooled sums come next (1.5).
- *Use is wide, gain concentrated.* Almost every bar is split on at least once (at `qd_dev`, $n = 10000$: 2520 and 2596 of 2704 for $L = 2$ and 4). But the top 10% of bars carry 0.65–0.66 of the bar gain there (0.76 at $n = 1000$), 0.78–0.84 on Fashion-MNIST and 0.86 on MNIST. The twelve 9-pixel bars with the most gain carry 9.5%, 14.1% and 8.3% of `cgbt`'s total gain on `qd_dev`, MNIST and Fashion-MNIST (Figure 6f–h).

v3 against v2 at `qd_dev`, $n = 10000$ [Numerical]. The two fits have identical pooling sets and a bit-identical `ggbt`: the same gain vector, test predictions and maps, and 123 rounds in both. So, as far as checked, the fits differ by `cgbt`, the stacking weights, and which members are refitted. The errors are 0.1467 for v2 and 0.1330 for v3. 2.08% of the test points are right in v3 and wrong in v2, and 0.71% the reverse. Per-class detail is in Appendix C.7.

6.2.3 Class evidence maps

Class evidence maps: method and limits. For each class, 100 test points of that class are taken. LightGBM's TreeSHAP contributions [21] of each feature to that class's raw score are computed and spread over pixels. A pixel keeps its own contribution. A pooled sum or bar spreads its contribution equally over its pixels. A prototype pair $u = x_a - x_b$ spreads it over pixels j in proportion to $|u_j x_j|$. The maps are averaged over the 100 points. They can show where in the image the features that raise a class's score look. They cannot show more than that, for five reasons.

- *Own score only.* A map is the contribution to the class's own raw score in a 10-class softmax, not to its margin over the other classes.
- *No causal test.* Nothing was occluded or ablated, so no region is shown to be necessary.
- *The spreading rule.* Spreading a subset sum equally over its pixels books part of the evidence of a partly inked bar on blank pixels. On `qd_dev`, of the members of bars with at least one inked member, 43% ($L = 2$) and 52% ($L = 4$) are blank. So evidence on blank pixels marks where a feature looks, not necessarily what drove it.
- *Built-in agreement.* The pairs' maps follow the class's own ink by construction. `cgbt` and `ggbt` share the pixels, the pooled sums and the spreading rule, so part of their agreement is built in.
- *Post-hoc regions.* The regions named in the Reading and in Appendix C.6 were drawn after looking at the maps. Their shares describe the maps; they do not test them.

What the maps show [Numerical] for the shares (Figure 7).

- *Sign.* The mean maps are almost all positive: positive values make at least 0.96 of each map's absolute mass.

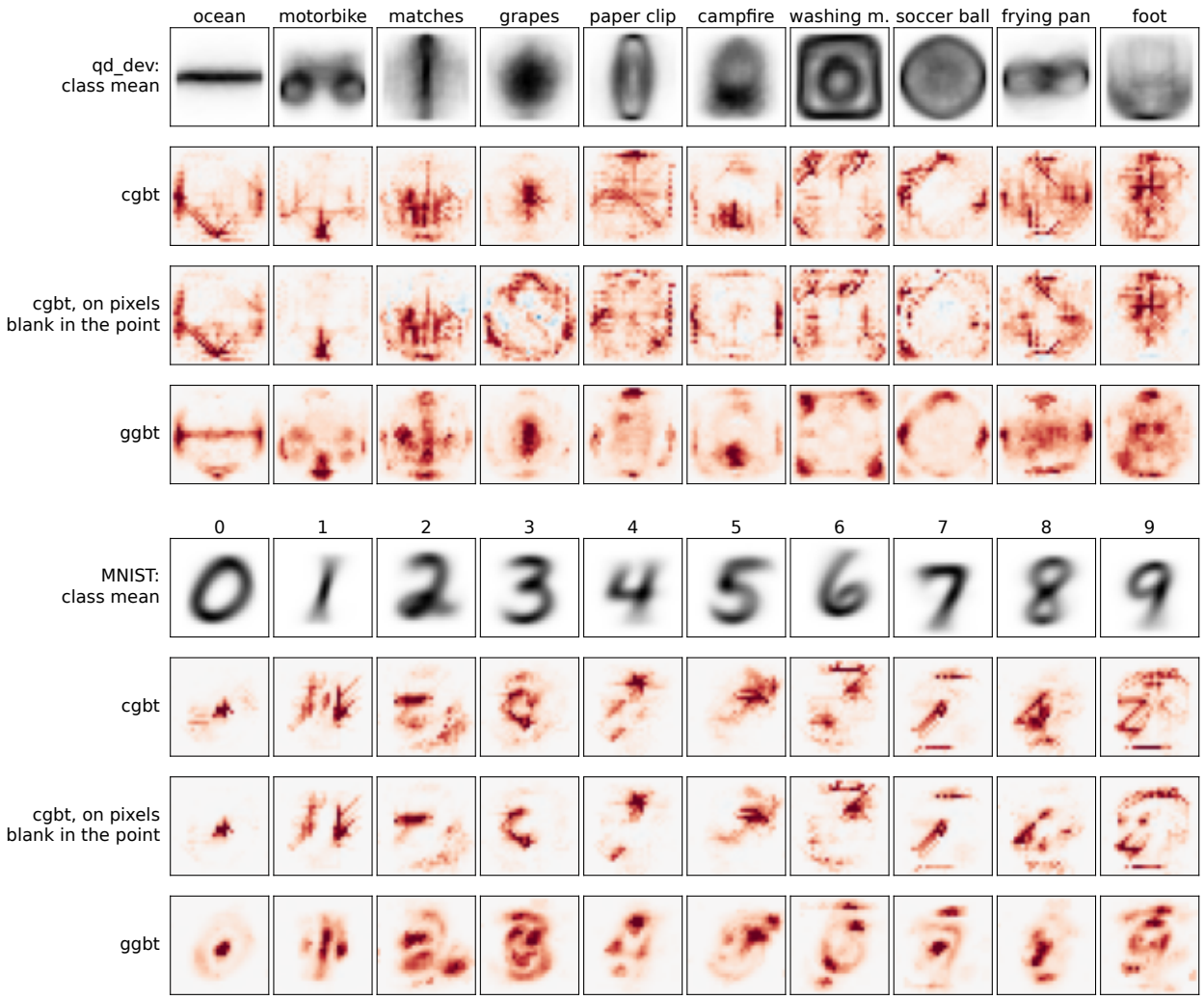


Figure 7: Class evidence maps of v3's two boosters on `qd_dev` and MNIST, $n = 10000$, seed 0 [Numerical]. Each map is the mean, over 100 test points of the class, of the TreeSHAP contributions to that class's own raw score, spread over pixels as described in the text; red is positive, and each map is scaled to its own 99th percentile of |value| (the top 1% saturate). Rows: the class's mean training image; `cgbt`; the part of `cgbt`'s map on pixels blank in the point ($x \leq 0.1$); `ggbt`, prototype pairs included. The maps are descriptive (see the text for what they cannot show). Simplified from `fig_img_contrib_qd_dev.pdf` and `fig_img_contrib_mnist.pdf` in `study/part10/interp/figures/`, which also show the pooled-sum and pair parts, and Fashion-MNIST and `qd_a`.

- *Blank pixels.* The share of the net evidence on pixels blank in the point, averaged over classes, is 0.61 for `cgbt` on `qd_dev`, 0.56 on Fashion-MNIST, 0.69 on MNIST and 0.62 on `qd_a`. For `ggbt` it is 0.39–0.44, and 0.53–0.68 once its pairs are removed. Blank pixels are 0.75, 0.73, 0.83 and 0.55 of these images (`qd_dev`, `qd_a`, MNIST, Fashion-MNIST), so per pixel, inked pixels carry more of the evidence on QuickDraw and MNIST, and Fashion-MNIST is at par. It is not so for every class: grapes, soccer ball and campfire (0.32–0.39 on `qd_dev`), pullover, shirt and coat (0.25–0.29), camouflage on `qd_a` (0.15) and the digit 8 (0.42) draw most of their evidence from ink.
- *Agreement.* The `cgbt` and `ggbt` maps of a class correlate at 0.63–0.94 on `qd_dev`, Fashion-MNIST and MNIST, and at 0.40–0.88 on `qd_a` (slightly less over non-constant pixels only: 0.60–0.94 and 0.33–0.87).
- *By feature type.* By absolute contribution, `cgbt`'s evidence on `qd_dev` is 0.78 bars, 0.20 pooled sums and 0.02 pixels. `ggbt`'s is 0.64 pooled sums, 0.29 pairs and 0.07 pixels.

Reading [Heuristic]. In these maps a sizable part of a class's evidence sits where the class has no ink and other classes do; for several digits the largest named region is an open one (0, 1, 3, 4, 7). Two examples (shares `cgbt` / `ggbt` of the class's net evidence): on MNIST the hole of the 0 carries 0.46 / 0.41 of the evidence on 3% of the image, 99% of it on blank pixels; on `qd_dev` the blank gap between a motorbike's wheels carries 0.20 / 0.16 (Appendix C.6 lists the others). `cgbt` does not look at different places from `ggbt`. It looks at the same places through thinner, oriented windows: lines and chains of short segments, where `ggbt`'s pooled sums give blobs. That difference in shape follows from the feature shapes and is not learned. What the fit chose is which segments carry weight.

7 Discussion

7.1 What kind of learner this turned out to be

v3 is not one mechanism but three learners under one convex stacker, and the input type decides which of them carries the fit (Section 2): enumerate-and-test on Boolean inputs, at about an Occam sample cost wherever its search can propose the right feature [Heuristic]; a fixed, data-derived hierarchy of subset sums read out by gradient-boosted trees on images, with nothing fitted inside the features; and the classical stack of Part 8 on other tabular data. That is the AutoML-style dispatch by input type of the Parts 6–9 paper (Section 2.1), and Part 10's two additions, a wider sum-block search for DFG and a second booster over a richer bank, sit inside the first two regimes. What the results add is where each regime's performance comes from. On Boolean inputs, the interpretability analysis places DFG's data cost in its search's low-degree seeds rather than in its test (Section 6.1; a reading of descriptive measurements, not an ablation). On images, the gain over v2 is a better fixed representation, which ties a CPU-matched MLP ensemble at $n = 60000$ (Section 5.3). On tabular data nothing has changed since Part 8 (Section 4.4).

7.2 Why it works where it works

Boolean targets: an Occam-like sample cost, when the search can propose [Heuristic]. DFG's test admits a feature once n exceeds a multiple of the log-size of its class over the squared residual correlation (Section 2.3). For a k-sparse character, $\ln N \approx k \ln d$; for a signed block on k of d inputs, $\ln N \approx \ln\binom{d}{k}2^{k-1}$. Either way the sample cost is logarithmic in the size of the dictionary: the Occam rate, paid for with FLOPs of enumeration and search that fit inside one network run at $d = 30$. Gradient-trained networks pay in samples instead. For parities the companion paper

measured rates close to the leap and correlational-SQ rate d^{k-1} [1–3, 6, 16]. For an even function of one signed sum, a network has to find the block's sign vector in the degree-2 statistic $\mathbb{E}[f\,xx^\top]$ against sampling noise. That is the information-exponent picture of single-index learning [4, 8]. Networks can trade data for width and compute along a frontier [9]; DFG sits at an extreme of that trade.

The interpretability analysis sharpens the picture in one respect (Section 6.1). On the pure symmetric targets the test is not the bottleneck, and the data cost is consistent with being set by when the search first *proposes* the block, which it seeds from degree-1 and degree-2 statistics of the residual (findings 2 and 3 of Section 6). So DFG relies on the same low-order statistics that a network's gradient sees. It needs far fewer samples because it rounds a coarse spectral estimate to an exact signed subset and then tests that subset with a union bound, where a network has to learn the direction to useful precision by noisy local steps [Heuristic].

Where neither has a foothold. The residue classes mod 3 of 10–14-input counts have almost no Fourier weight at degrees 1 and 2, and DFG's search never proposes their block, although the test would accept it long before $n = 16384$ (Section 6.1.3). The networks mostly fail too: none learns f7_05, and only the most regularized variant learns h6_00, at $n^* = 12304$ (fd_10, in `dlfamdev`, has no network runs). Their signal sits at high degree on an unknown subset of inputs, which learners that rely on low-order correlations should find hard [16]. Exhaustive search does not help within the budget: there are 1.5×10^{10} to 1.2×10^{12} blocks of the true size, each costing at least n operations to score, against 1.62×10^{11} FLOPs per toy fit. So this is a shared limit, not a loss specific to discrete search [Heuristic]. The only target on which a network is clearly better than v3 is of this kind.

Images: locality recovered from data, read out by trees [Heuristic]. On QuickDraw the most correlated pixels of a pixel are its neighbours, so the pooling sets are small blobs of the true grid and the bars short runs along it (Section 6.2.1). Recovering image topology from such correlations is an old idea [18], as is grouping inputs into receptive fields by correlation [7]. Boosted trees [15] over these fixed local sums then do what a first convolutional layer with a nonlinearity would do, but at fixed positions and without shared weights. The MLP references are denied this structure by design: they are invariant to the pixel permutation, and nothing in their training recovers it. So v3 beats MLP ensembles that spend 2.0 and 5.0 times its FLOP budget (`mlp_strong` at all six held-out cells, `mlp_wide` at five of six), and ties the 1024-wide ensemble, which spends 9.2 budgets and a comparable time per fit. The control and context runs bound how much of this is representation (Section 5.3): an MLP given v2's pooled sums ties v2 but not v3, no MLP was given the bars, and a CNN told the layout (seed 0) has lower error at all three cells where both ran, clearly at one. It does not show that discrete search beats gradient-based learning on images.

Tabular data [Heuristic]. Here the result is the known strength of tree ensembles and stacking on medium-sized tabular data against standard-recipe MLPs (Section 4.4); DFG runs on 2 of the 26 datasets and changes no cell there clearly.

7.3 Limits

- **Small held-out sets.** `dlheld7` has 12 targets from one family, and on 5 of its 10 wins default gradient boosting or logistic regression already beats the networks. `qd_c` and `qd_d` are two 10-class sets from one source. The 10 fresh tabular datasets of Part 8 give $p = 0.11$. A second family or source could change the picture.

- **The earlier suites are in-sample** (Section 3.4). Pooled numbers over all suites summarize the record; they are not held-out estimates. The margin against `nets_wd` is smallest on the two draws of the blind family, which is what selection on the other suites would produce, though the suites also differ in kind and the differences are within noise (Section 4.2).
- **One seed at $n \geq 30000$ on images.** The two cells that decide the comparison with `mlp_xl` rest on one fit each, and on the test set's binomial noise for their standard error. v3's gain over v2 is below the 10% threshold at every held-out cell; that v3 is lower in all 14 seed-paired fits is a description, not a pre-registered test (Section 5.1).
- **Targets without a low-degree foothold** are found late or never: the residue classes mod 3 never, h5s7 at $n = 8192$, and h5s1 in no learner's 3-seed mean (Sections 4.3 and 6.1.3). No cheap search for them is known (Section 7.4).
- **Locality is used in spirit.** The image members never read a pixel index, but the correlations give back neighbourhoods and lines, the information the permuted-pixel protocol was meant to withhold (Section 6.2.1). No MLP was given the bars, and a CNN told the layout has lower error wherever both ran (Section 5.3).
- **Equal FLOPs are not equal CPU time, and prediction is not charged.** v3 uses 0.30–0.58 of its FLOP budget on the held-out cells, but takes about as long per fit as `mlp_xl`, which it only ties. At $n = 3000$ it is close to the budget guard once Part 9's audited correction is applied, and no compute audit of v3 was made. A v2 query costs about 500× an MLP's, and v3 adds to it (Section 5.4).
- **Concrete references.** One toy network architecture at one width and one learning rate, and image MLPs without augmentation or architecture search. The hindsight minima favour the references, but stronger networks would plausibly narrow the margins.
- **No independent audit in Part 10.** Parts 8 and 9 had independent audits of leakage, compute, discreteness and reference strength [5]. Part 10 inherits their findings and fixes, but its own comparisons were not re-audited in that sense. The interpretability analyses were independently verified, and the paper's claims were checked against the result files and records by independent checkers (Appendix D).
- **The interpretability is descriptive**: seed 0, no ablation of the learner or causal test, mechanism labels for v1.3's changes read off fitted models, and stacking weights, split gain and contribution maps that describe the fits, not what their predictions depend on (Section 6).

7.4 What would test it further

1. *The information-matched image comparison.* Give an MLP v3's whole bank, pooled sums and bars, as `mlp_pool` did for v2's. Train a gradient-based network that learns sparse local connectivity [22], or a locally connected network on the recovered neighbourhoods. Either would say how much of v3's image margin is representation that a network can use as well.
2. *A second blind toy family*, written independently of the first, and the toy suites at larger d. At $d = 30$ the advantage on the blind family is a factor of about $2^{1.5} \approx 2.8$. For parities DFG's advantage grew with d in Part 6; whether it does for signed sums was not measured.
3. *Block search beyond degree 2.* A seed that reads higher-order statistics of the residual could reach the mod-q targets. The Part 10 builder found that the true block of fd_10 is detectable

only within about 3–4 errors of the truth, so that random restarts would need far more than 10^5 tries; a cheaper route is not known.

4. *Power and attribution.* More seeds at $n = 60000$; a pre-registered paired comparison with v2; image sets from another source; ablations of v1.3's three additions on held-out targets; and a compute audit of v3 that also charges prediction.

References

[1] Emmanuel Abbe, Enric Boix-Adserà, and Theodor Misiakiewicz. The merged-staircase property: a necessary and nearly sufficient condition for SGD learning of sparse functions on two-layer neural networks. In *Conference on Learning Theory (COLT)*, 2022. arXiv:2202.08658.

[2] Emmanuel Abbe, Enric Boix-Adserà, and Theodor Misiakiewicz. SGD learning on neural networks: leap complexity and saddle-to-saddle dynamics. In *Conference on Learning Theory (COLT)*, 2023. arXiv:2302.11055.

[3] Boaz Barak, Benjamin L. Edelman, Surbhi Goel, Sham Kakade, Eran Malach, and Cyril Zhang. Hidden progress in deep learning: SGD learns parities near the computational limit. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[4] Gérard Ben Arous, Reza Gheissari, and Aukosh Jagannath. Online stochastic gradient descent on non-convex losses from high-dimensional inference. *Journal of Machine Learning Research*, 22(106):1–51, 2021.

[5] Claude (Anthropic). Discrete learners against gradient-descent-trained networks: Boolean targets and standard benchmarks. Prompted by Kaarel Hänni. https://github.com/kaarelh/AI-works/tree/claude/nifty-mccarthy-j0lw2a/gd-circuit-growth/paper_discrete, 2026. Parts 6–9 of the study this paper continues.

[6] Claude (Anthropic). Gradient descent on Boolean targets as noise-limited greedy feature growth: experiments, discrete models, and generalization bounds. Prompted by Kaarel Hänni. https://github.com/kaarelh/AI-works/tree/claude/nifty-mccarthy-j0lw2a/gd-circuit-growth/paper, 2026. Parts 1–5 of the study this paper continues.

[7] Adam Coates and Andrew Y. Ng. Selecting receptive fields in deep networks. In *Advances in Neural Information Processing Systems 24*, 2011.

[8] Alex Damian, Loucas Pillaud-Vivien, Jason D. Lee, and Joan Bruna. Computational-statistical gaps in Gaussian single-index models. In *Proceedings of the Thirty-Seventh Conference on Learning Theory (COLT)*, 2024.

[9] Benjamin L. Edelman, Surbhi Goel, Sham Kakade, Eran Malach, and Cyril Zhang. Pareto frontiers in neural feature learning: data, compute, width, and luck. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2309.03800.

[10] Nick Erickson, Jonas Mueller, Alexander Shirkov, Hang Zhang, Pedro Larroy, Mu Li, and Alexander Smola. AutoGluon-Tabular: robust and accurate AutoML for structured data, 2020. arXiv:2003.06505.

[11] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[12] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.

[13] David Ha and Douglas Eck. A neural representation of sketch drawings, 2017. arXiv:1704.03477; ICLR 2018.

[14] Arlind Kadra, Marius Lindauer, Frank Hutter, and Josif Grabocka. Well-tuned simple nets excel on tabular datasets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[15] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: a highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[16] Michael Kearns. Efficient noise-tolerant learning from statistical queries. *Journal of the ACM*, 45(6):983–1006, 1998.

[17] Béatrice Laurent and Pascal Massart. Adaptive estimation of a quadratic functional by model selection. *Annals of Statistics*, 28(5):1302–1338, 2000.

[18] Nicolas Le Roux, Yoshua Bengio, Pascal Lamblin, Marc Joliveau, and Balázs Kégl. Learning the 2-D topology of images. In *Advances in Neural Information Processing Systems 20*, 2007.

[19] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[20] Nathan Linial, Yishay Mansour, and Noam Nisan. Constant depth circuits, Fourier transform, and learnability. *Journal of the ACM*, 40(3):607–620, 1993.

[21] Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.

[22] Behnam Neyshabur. Towards learning convolutions from scratch. In *Advances in Neural Information Processing Systems 33*, 2020.

[23] Randal S. Olson, William La Cava, Patryk Orzechowski, Ryan J. Urbanowicz, and Jason H. Moore. PMLB: a large benchmark suite for machine learning evaluation and comparison. *BioData Mining*, 10:36, 2017.

[24] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[25] Ravid Shwartz-Ziv and Amitai Armon. Tabular data: deep learning is not all you need. *Information Fusion*, 81:84–90, 2022.

[26] Joshua B. Tenenbaum, Vin de Silva, and John C. Langford. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000.

[27] David H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.

[28] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms, 2017. arXiv:1708.07747.

[29] Greg Yang and Edward J. Hu. Tensor programs IV: feature learning in infinite-width neural networks. In *International Conference on Machine Learning (ICML)*, 2021.

A Names used in this paper

Table 16 lists the names of learners, members, references, sets and terms that the paper uses, and Tables 17 and 18 define the targets of the sum-structured suites other than `dlheld7`, which Table 5 defines. Every target is defined in `study/boolfuncs.py`.

name	meaning
Learners	
v1, v2, v3	the study's portfolios of Part 8, 9 and 10: `hyb_portfolio`, `hyb_portfolio2`, `hyb_portfolio3` in the code (Table 2)
DFG	discrete feature growth, the Boolean member (Section 2.3); v1.1 in v1, v1.2 in v2, v1.3 in v3
`p10_sum`	the Part 10 toy builder's candidate (DFG v1.3); v3's code on toy inputs
`p10_codes`	the adopted Part 10 image candidate (adds `cgbt`); v3's code on images
`p10_hier`	the other Part 10 image candidate (a max-pooling level in `ggbt`'s bank); not adopted
`p9_sum`, `p9_gates`	v2's Part 9 components on toy inputs and on images (v2's code there)
Members of v3 (Table 2)	
`dfg`, `dfg_forced`	DFG and its small-n variant that forces the best candidate
`ggbt`	the blob booster: LightGBM on pixels, pooled sums and prototype pairs
`cgbt`	the bar booster: LightGBM on pixels, pooled sums and bar sums (new in v3)
`gknn_k`	nearest-neighbour votes on the 9-pixel pooled sums
`gbt`, `et`, `knn_<metric>_k`, `lin_C`	Part 8's LightGBM, ExtraTrees, nearest-neighbour votes and logistic regression on the raw columns
References (Section 3.2)	
`nets`, `nets_wd`, `nets_cewd`, `nets_cewd0.3`, `nets_cewd1.0`	the toy networks: square loss; weight decay 0.1; logistic loss with weight decay 0.1, 0.3, 1.0
`nets_best`	per target, the network variant with the smallest n^* among those run (hindsight)
`lin`, `hgb`	calibration references: cross-validated logistic regression on the inputs (a separate run, not the member `lin_C`); default histogram gradient boosting
`mlp`, `mlp_reg`	Part 7's baseline MLP (256-256) and its L2-regularized version
`mlp_strong`, `mlp_wide`, `mlp_xl`	ensembles of three MLPs at width 256, 512 and 1024; `mlp_strong1` and `mlp_wide1` are their single networks; `mlp_xl` is CPU-matched
`mlp_best`, `mlp_best+`	per-cell minimum over the Part 7–8 MLP variants (hindsight); `mlp_best+` adds the wider ones
`mlp_pool`	control: `mlp_strong` given v2's pooled sums
`cnn`	context: a small CNN on the unpermuted pixels
Sets (Section 3.1 and Table 4)	
`designed`, `random`, `dlheld`, `dlheld2`–`dlheld7`, `dlsymdev`, `dlfamdev`	the 11 toy suites; "the blind family" is the generator of `dlheld6`, `dlfamdev` and `dlheld7`
HELDOUT2; the Part 7 set	the 10 fresh tabular datasets of Part 8; the 11 held out in Part 7
`qd_dev`, `qd_a`–`qd_d`; Fashion	the five QuickDraw sets; Fashion-MNIST
target prefixes	f7_ `dlheld7`; h6_ `dlheld6`; h5 `dlheld5`; h4 (controls) `dlheld4`; fd_ `dlfamdev`; h2rf, h3rf, hrf random formulas of `dlheld2`, `dlheld3`, `dlheld`; rf `random`. `dlsymdev`, `dlheld4` and the earlier suites use descriptive names (sym7, band11, tail13, majsym, thr10, maj9, mux6, . . .)
Terms	
n^*	training-set size at which the seed-mean test error reaches 10% (Section 3.3)
cell; clearly better, tie	a (set, n) pair; the pre-registered verdicts (Section 3.3)
budget	one toy network run, or three runs of the baseline `mlp`, in FLOPs (Section 3.3)
character; block	$\chi_S(x) = \prod_{i \in S} x_i$; indicators of the groups of a signed sum $t = \sum_{i \in S} s_i x_i$, folded ($\lvert t \rvert$), modulated, product, split or weighted (Section 2.3)
sum-structured target	one of the 54 targets of `dlsymdev`, `dlheld4`–`dlheld7` and `dlfamdev` that is not a control
$S_i(k)$; bar	the k-pixel pooling set of pixel i; a sum along a recovered line, half-length L (Section 2.4)
n_{struct}, n_{detect}, share, reach	discovery size, detection size, a term's share and the search's reach (Section 6.1)
Part k	the k-th part of the study; Parts 6–9 are reported in [5], Parts 1–5 in [6]

Table 16: Names used in this paper.

suite	target	definition
`dlsymdev`	sym7	$\lvert x1 + \ldots + x7\rvert \le 1$
	ex3of8	exactly 3 of x1..x8 are +1
	band11	$\lvert x1 + \ldots + x11\rvert$ in {3, 5}
	far9	$\lvert x1 + \ldots + x9\rvert \ge 5$
	ssym9	$\lvert x1 - x2 + x3 - x4 + x5 + x6 - x7 + x8 + x9\rvert \le 1$
	symxpar	$[\lvert x1+\ldots+x6\rvert \le 2]$ xor (x7 x8)
	twosym	$[\lvert x1+\ldots+x5\rvert \le 1] == [\lvert x6+\ldots+x10\rvert \ge 3]$
	thrsum	sign(x1+...+x6 + 3 x7 x8 + 1/2)
`dlheld4`	sym11	$\lvert x1 + \ldots + x11\rvert \le 1$
	ex4of9	exactly 4 of x1..x9 are +1
	tail13	$\lvert x1 + \ldots + x13\rvert \ge 5$
	band10	$\lvert x1 + \ldots + x10\rvert = 2$
	ssym8	x1 + x2 - x3 + x4 - x5 - x6 + x7 - x8 = 0
	symxor	$[\lvert x1+\ldots+x7\rvert \le 1]$ xor x8
	symand	$[\lvert x1+\ldots+x6\rvert \le 2]$ & (x7 | x8)
	sumpar	sign(x1+...+x5 + 2 x6 x7 + 2 x8 x9 x10)
	majsym	MAJ($[\lvert x1+\ldots+x5\rvert \le 1]$, x6 x7, x8)
	dnf3x3	(x1 & ¬x2 & x3) | (x4 & x5 & ¬x6) | (¬x7 & x8 & x9) *(control)*
	h4rf0	random 9-gate formula on 7 inputs *(control)*
	h4rf1	random 9-gate formula on 7 inputs *(control)*
`dlheld5`	h5s0	$[\sum_{i\le 12} s_i x_i$ in [−10, −6, 0, 2, 4, 6, 8, 12]], s = [1, −1, 1, 1, −1, 1, −1, 1, 1, 1, 1, −1]
	h5s1	$[\sum_{i\le 9} s_i x_i$ in [−9, −5, −1, 3, 9]], s = [−1, 1, 1, 1, −1, 1, −1, −1, −1] xor chi(10, 11)
	h5s2	$[\sum_{i\le 7} s_i x_i$ in [−5, 1, 7]], s = [1, −1, −1, 1, 1, −1, −1]
	h5s3	$[\sum_{i\le 13} s_i x_i$ in [−7, −5, 1, 5, 9, 13]], s = [1, 1, 1, −1, 1, 1, 1, −1, 1, −1, 1, −1, 1]
	h5s4	$[\sum_{i\le 5} s_i x_i$ in [−5, −1, 5]], s = [−1, −1, −1, 1, 1] or ¬x6
	h5s5	$[\sum_{i\le 13} s_i x_i$ in [−11, −7, −5, −3, 1, 9, 11]], s = [1, 1, 1, 1, −1, −1, −1, −1, −1, 1, −1, −1, −1]
	h5s6	$[\sum_{i\le 9} s_i x_i$ in [−9, −5, −1, 1, 5]], s = [1, −1, 1, −1, −1, 1, 1, 1, 1]
	h5s7	$[\sum_{i\le 13} s_i x_i$ in [−11, −9, −5, −1, 1, 5, 7, 13]], s = [−1, −1, 1, −1, 1, 1, −1, 1, 1, −1, −1, 1, −1]
	h5t0	sign($\sum_{i\le 7} s_i x_i$ + 2 chi(8, 9) + 1/2), s = [−1, 1, 1, 1, 1, −1, 1]
	h5t1	sign($\sum_{i\le 5} s_i x_i$ + 2 chi(6, 7) + 1/2), s = [1, −1, 1, −1, 1]
	h5rf0	random 9-gate formula on 7 inputs *(control)*
	h5rf1	random 6-gate formula on 7 inputs *(control)*

Table 17: The targets of `dlsymdev`, `dlheld4` and `dlheld5`, as defined in `study/boolfuncs.py`; `dlheld6` and `dlfamdev` are in Table 18, `dlheld7` in Table 5. Conventions as there: $[\cdot]$ is ± 1, `*` multiplies ± 1 values, `c ? a : b` is a multiplexer, ¬ negates a literal, and in `dlheld5` s lists the signs s_i and $[t \in A]$ the accept set A. "(control)": one of the 14 non-symmetric controls; the other targets are the sum-structured ones, together with the 9 of `dlheld7`.

suite	target	definition
`dlheld6`	h6_00	[((s+13)/2) mod 3 != 1], s = x1+x2+x3+x4+x5+x6+x7+x8−x9−x10−x11−x12+x13
	h6_01	[s in {−9,−7,−5,−1,1,7,9}], s = −x1+x2−x3+x4+x5+x6−x7−x8−x9
	h6_02	[1 ≤ s ≤ 5] * x10 * x11, s = x1+...+x9
	h6_03	OR(AND(−x4,x5), AND(−x2,−x6), AND(x1,−x3,x4)) *(control)*
	h6_04	OR(−x1,AND(x2,AND(−x3,OR(x4,x5)))) *(control)*
	h6_05	if −x6 → +1; elif −x2 → −1; elif x8 → −1; elif −x4 → −1; elif −x3 → +1; elif x5 → −1; elif AND(−x7,−x1) → −1; else +1 *(control)*
	h6_06	[s in {−6,−2,0,6}] * x7 * x8, s = x1+...+x6
	h6_07	−x11 ? [s1 in {−2,4}] : [s2 ≤ 0], s1 = x1−x2−x3−x4−x5−x6, s2 = x7+x8+x9−x10
	h6_08	x11 ? [|s1| ≤ 2] : [((s2+6)/2) mod 3 == 0], s1 = x1+...+x4, s2 = −x5−x6+x7−x8+x9−x10
	h6_09	XOR([−4 ≤ s1 ≤ 0], [3 ≤ s2 ≤ 5]), s1 = x1+...+x6, s2 = −x7−x8+x9−x10−x11
	h6_10	[|u| > 1], u = s + 3*(−x13), s = x1+...+x12
	h6_11	[s == 1], s = x1+...+x7
`dlfamdev`	fd_00	if AND(−x2,x4) → −1; elif −x3 → +1; elif AND(−x6,−x1) → −1; elif AND(−x8,x7) → −1; elif −x5 → +1; else −1 *(control)*
	fd_01	[s ≥ 0], s = x1+...+x6
	fd_02	[|s| > 4] * x9 * x10, s = −x1+x2+x3+x4+x5+x6+x7−x8
	fd_03	OR(AND(−x1,−x2),AND(OR(x3,AND(−x4,−x5)),−x6)) *(control)*
	fd_04	[|s| ≤ 0] * x7 * x8 * x9, s = x1+...+x6
	fd_05	AND([s1 ≤ 0], [|s2| ≤ 3]), s1 = x1+...+x6, s2 = x7+...+x13
	fd_06	XOR([s1 ≥ 2], [−5 ≤ s2 ≤ 1]), s1 = −x1−x2+x3−x4, s2 = −x5+x6+x7+x8+x9−x10−x11
	fd_07	OR(AND(x1,−x6,x7,−x8), AND(x2,x3,−x5), AND(x1,−x4,−x8)) *(control)*
	fd_08	MAJ([−5 ≤ s ≤ 9], −x12, −x13), s = x1+x2−x3+x4+x5+x6−x7+x8+x9+x10−x11
	fd_09	[s in {2,6}], s = x1+...+x6
	fd_10	[((s+10)/2) mod 3 == 0], s = x1+x2+x3+x4−x5−x6+x7+x8+x9+x10
	fd_11	OR([|s| > 3], x10), s = x1+...+x9

Table 18: The targets of `dlheld6` and `dlfamdev`, as defined in `study/boolfuncs.py`. Conventions as in Table 17.

B Further results

This appendix holds the in-sample detail behind Sections 4 and 5: the toy suites target by target (Appendix B.1), what changed from v2 to v3 on toy inputs (Appendix B.2), the tabular datasets (Appendix B.3), the development image sets (Appendix B.4) and the compute of the image fits (Appendix B.5). None of it is held out for v3.

B.1 The toy suites in detail

Pooled over the held-out suites [Numerical]. Pooled over the seven held-out suites (`dlheld`, `dlheld2`, `dlheld3`–`dlheld7`; 77 targets), the mean $\log_2 n^*$ ratio against `nets_best` is -1.87 over the 74 targets both learn. v3 is clearly better on 72 targets and clearly worse on 1 (h6_00). This matches `study/part10/LEARNER_RESULTS.md`; the development check `random` is not included. The pooled number is a summary of the record, not a held-out estimate: 65 of the 77 targets were development data for v3, and only the 12 of `dlheld7` are held out. The held-out estimate is the `dlheld7` row alone.

The wins that need no feature search [Numerical]. Over all suites they are of three kinds.

- *Halfspaces.* Logistic regression alone is clearly better than `nets_best` on halfspace-like targets: dict, maj3, maj9, rand2 and thr10 on `designed`, maj15 and thr30 on `dlheld`, maj21 and thr20 on `dlheld2`, thrint on `dlheld3` and f7_10 on `dlheld7`. On these, v3 is within the tie band of `lin` (thr10 121 against 113, thr20 124 against 117, thr30 149 against 148, maj9 80 against 80, maj15 105 against 110, maj21 109 against 116, thrint 139 against 137, f7_10 82 against 87) or clearly below it (maj3 31 against 56, dict 16 against 27, rand2 16 against 50).
- *Formulas, decision lists and multiplexers.* `hgb` is clearly better than `nets_best` on six targets of `dlheld` (the random formulas hrf0, hrf1 and hrf2, the majority of majorities majmaj5, the multiplexer mux6 and the staircase stair6), on three of `dlheld2` (dnf4x2, h2rf0, h2rf1), on the five formula and multiplexer targets of `dlheld3` (cnf3, dlist8, h3rf0, h3rf1, mux11), on 8 of the 11 non-symmetric controls of `dlheld4`–`dlheld7`, on the multiplexer f7_11, and narrowly on h5s4, a signed-sum condition OR a literal (1465 against 1801).
- *Random formulas.* On the development check `random`, `hgb` is clearly better than `nets_best` on 9 of the 20 formulas and `lin` on 2 (rf06, rf10). v3 is clearly better than both on all 20 (Table 6).

Where the calibration references fail, they fail often: `hgb` never reaches 10% error on 5 of the 12 targets of `dlheld4` and on 6 of `dlheld5`, and `lin` on 11 of `dlheld4` and all 12 of `dlheld5`.

The next-smallest margins [Numerical]. After the five targets of Section 4.3, the smallest margin against `nets_best` is inter2 (`dlheld`, an AND of two halfspaces on 10 inputs: 346 against 454, -0.39). Then come the halfspace-like thrint (-0.46) and f7_10 (-0.53), where `lin` is already as good as v3, a random formula of `random`, rf00 (-0.62), and h5s2 (-0.68), an arbitrary accept set whose weight at degrees 1–2 is also small.

B.2 What changed from v2 to v3 on toy inputs

The change [Numerical]. The only difference between v2 and v3 on toy inputs is DFG: v1.3 replaces v1.2 (Section 2.3). v1.3 adds three things to the sum-block search, all of which act together:

suite	target	v2	v3	$\log_2$(v3/v2)	nets_best
designed	parmaj	142	110	−0.36	434
random	rf03	78	62	−0.32	276
dlheld	hrf0	46	62	+0.43	133
dlheld2	sym8	100	53	−0.92	377
dlsymdev	sym7	110	55	−1.00	329
	far9	87	71	−0.30	296
dlheld4	majsym	189	90	−1.06	595
	tail13	176	100	−0.81	420
	sym11	203	160	−0.34	374
dlheld5	h5rf0	268	203	−0.40	1547
dlheld6	h6_10	702	206	−1.77	451
	h6_07	678	447	−0.60	1812
dlfamdev	fd_09	171	101	−0.76	–
	fd_06	266	318	+0.26	–

Table 19: Every toy target on which v3's n^* differs clearly from v2's ($|\log_2 \text{ratio}| > 0.25$), on all suites with results for both [Numerical]. There are 14 such targets among the 138. None is on dlheld3 or on the held-out dlheld7. Source: study/part10/LEARNER_RESULTS.json.

- weighted blocks $u = t + a\tau$;
- forced blocks in the small-n variant at $n \geq 48$;
- a beam search at $n \leq 128$.

On the development suites the integration rule measured a mean $\log_2 n^*$ ratio of −0.066 against v2 over 40 targets, with the largest increase +0.26 on fd_06 (Appendix D.2). That is a small gain. Table 19 shows where it sits over all suites: 12 targets clearly better and 2 clearly worse, out of 138. Structure runs of both versions on the six sum-structured suites (68 targets with the controls) show where the changes act. Over the 64 of those targets with a finite 3-seed n^* for both, the mean $\log_2 n^*$ ratio of v3 to v2 is −0.116 (in-sample: every suite except dlheld7 is development data for v3, and the toy builder developed on dlheld6; on the held-out dlheld7 the mean is −0.004, Section 4.1).

Where n^* moved, and what the models show. The descriptions below come from the interpretability analysis (seed 0, Section 6.1) and report what the fitted DFG model contains [Numerical]. No ablation separates the three additions: the attribution of each change to weighted blocks or to the small-n search is read off what v3's model holds at its n_{struct}, so these labels are observations, not attributions [Heuristic].

- *Weighted blocks change what is learned* on four targets whose exact structure v2's models never match: h6_10 ($n_{\text{struct}} = 256$ in v3; n^* 702 → 206), h6_07 (512, with the selector as inner literal; 678 → 447), h6_08 (1024; 446 → 449) and sumpar (2048, with $(x_1 + \cdots + x_5 + 2x_6x_7)$; 164 in both).
- *h6_10* $= [|x_1 + \cdots + x_{12} - 3x_{13}| > 1]$ (dlheld6). v3's model holds the weighted block $|\pm(x_1 + \cdots + x_{12}) + 3x_{13}|$ from $n = 256$, with the right relative sign from $n = 512$. v2 does represent the target exactly from $n = 512$, by two folded 13-input blocks $|s + x_{13}|$ and $|s - x_{13}|$ that the matcher does not count, and which together determine the target (DFG error 0.000 at $n = 1024$). v3's model is nearly right one doubling earlier ($n = 256$: DFG error 0.017, with

the inner sign wrong, against v2's 0.33), and both are exact from 512. So the change is in how early the structure is found, not in what can be represented. This is the target the weighted blocks were designed for, on a suite the builder used, so it is not evidence for v3. It turns one of `dlheld6`'s two Part 9 losses to the networks (h6_10: v2 702 against `nets_best` 451) into a win (206); the other, h6_00, remains.

- *h6_07* (`dlheld6`), a multiplexer between conditions on two signed sums. The selector x_{11} enters the model as the heavy literal of weighted blocks: from $n = 512$ on the 4-input branch, $(x_7 + x_8 + x_9 - x_{10} - 3x_{11})$, and from $n = 1024$ on the 6-input branch, $(x_1 - x_2 - \cdots - x_6 + 3x_{11})$.
- *The beam and forced blocks move the first exact block one doubling earlier* on five pure targets (sym7 to 64; fd_09, tail13, sym11 and h6_11 to 128) and on the gate mixture majsym (to 128). On sym7 (`dlsymdev`, n^* 110 → 55) and majsym (`dlheld4`, 189 → 90) the block is in the forced variant, below the threshold (0.89 of it for sym7; 0.36–0.39 for majsym's plain block): forced, not admitted. On sym7 at $n = 64$ the stacker puts all its DFG weight on that forced member. For fd_09, tail13, sym11 and h6_11 the true block already passes the test at 128 (1.34–1.62 times the threshold; the unweighted test is the same in both versions), so the earlier find is a difference in the search, not attributable to the beam specifically.
- *Later or lost.* h5t1's n_{struct} moves from 256 to 512, with identical portfolio errors. f7_11 loses the joint structure: v2 holds both classes at 2048 and 4096, v3 never in one model (n^* 55 in both).
- *Not diagnosed:* parmaj, far9, h5rf0, sym8 and rf03. h5rf0 is a random-formula control (268 to 203), sym8 is on `dlheld2` (100 to 53) and rf03 is a random formula of `random` (78 to 62). On development data the toy builder traced parmaj to the beam at $n = 128$ and far9 to forced blocks at $n = 64$ (source: `study/part10/p10_sum/DESIGN.md`).
- *The two losses*, not diagnosed: hrf0 (`dlheld`, a random formula; 46 to 62, +0.43, the larger of the two), on a suite the toy builder did not use; and fd_06 (+0.26), an XOR of conditions on two signed sums.
- *The same n_{struct} is not the same fit.* 41 targets have the same n_{struct} in both versions. On them the 3-seed n^* still moves, from −0.30 (far9) to +0.26 (fd_06) in $\log_2$ (mean −0.005 over 40), and the portfolio's errors differ at some analysed n on 26 of the 41. They are identical at every analysed n on 20 of the 68 targets.

Where weighted blocks appear [Numerical]. The weighted class is used far more widely than for h6_10. At some n, 26 of the 68 targets of the six sum-structured suites have a weighted block with at least 20% of the largest term's share: 16 sum-structured targets and 10 of the 14 controls. Its inner term τ is one of five kinds.

- the true weighted term: h6_10, f7_10, h5t0, h5t1;
- a parity partner or one of its literals: symxpar $(+3x_7x_8)$, h6_06 (first $+2x_7$ at $n = 512$, then $\pm 3x_7x_8$), f7_07, symxor $(-2x_8)$;
- a multiplexer's selector: h6_07, h6_08, f7_03 $(\pm 3x_{11})$;
- the other inputs of a gate: h5s4 $(+3x_6)$, symand $(+3\,\text{AND}(-x_8, -x_7))$, majsym $(+2x_6x_7)$;
- an input or inputs of a true class: h6_09 $(+3\,\text{AND}(x_9, -x_{10}))$ and, at $n = 16384$ only, h5s7 $(+3x_{11}$, a literal of its own class).

The other 10 are controls (all but dnf3x3, f7_02, h4rf0 and h5rf1), mostly decision lists, DNFs and read-once formulas. There the inner term is an AND of two literals in 9 cases and a literal in 1, typically inside a weighted threshold, as in h6_04's $(x_1 + x_2 - 3\,\mathrm{AND}(x_4, -x_3))$. On 8 of the 26 targets (symxpar, f7_06, f7_10, h6_03, h5s7, h5t0, h5t1, majsym) the weighted block appears at a single analysed n. On most of the 26, n^* does not move clearly (Table 19).
Reading [Heuristic]. Any function of a block and one more ± 1 term that the block's partition cannot separate, such as a parity partner, a selector or a gate, can be written wholly or partly as a function of $s + a\tau$. The search finds these because τ is usually a strong single input or pair. The class was built for h6_10 and is used far more widely, mostly without a measurable change in n^*.

B.3 Tabular data in detail

kr_vs_kp [Numerical]. v3's own runs differ from v1's at 3 of 4 sizes, but no cell differs clearly (at $n = 1000$, 0.0075 against 0.0100; at the full 2397, 0.0029 against 0.0021).

Per dataset [Numerical]. The per-dataset picture is Part 8's (`study/hybrid/RESULTS.md`).

- *HELDOUT2.* The large margins against `mlp_best` are on ann_thyroid (-0.77), clean2 (-1.12, mostly from $n \leq 300$) and churn (-0.35). Three of the 8 datasets with lower error are within about ± 0.05. The losses are texture ($+0.61$; no cell resolves, since errors are below 1.6% from $n = 1000$) and krkopt ($+0.19$; clearly worse at $n \geq 10000$).
- *The Part 7 set.* The largest wins are dna (-0.77) and ring (-0.50). The losses are nursery ($+0.70$), pendigits ($+0.19$) and connect_4 ($+0.03$, worse at full n).

What Part 8 found [5].

- *DFG does not carry the tabular results.* DFG runs on none of the HELDOUT2 datasets. In Part 7, DFG's own dictionaries were about as good as a plain MLP on tabular data and worse than a regularized one.
- *No single member family matches the stack.* On Part 8's 48 ablation cells (HELDOUT2 and two image sets, $n \leq 3000$, seeds 0–1), LightGBM alone was $+0.08$ against the best MLP, where the full stack was -0.19. Different members carry different datasets: trees carry churn, ann_thyroid and clean2, k-NN carries banana, and logistic regression carries texture.
- *Default `hgb` alone does not beat the MLPs.* Against `mlp_best` it is $+0.08$ on HELDOUT2 and $+0.14$ on the Part 7 set (`study/hybrid/RESULTS.md`). The stack's mean $\log_2$ ratio against it is about -0.25 on both (Table 7).

B.4 The development image sets

Tables 20 and 21 give v3 at the cells of its fixed development evaluation. These sets are not held out for v3, and the cells are in-sample for a choice:

- *Tuning.* `p10_codes`'s builder tuned on validation rows of the training pools of `qd_dev`, `qd_a` and Fashion-MNIST (not `qd_b`; `study/part10/p10_codes/DESIGN.md`); `p10_hier`'s builder also used `qd_b`'s (`study/part10/p10_hier/DESIGN.md`). Neither tuned on a test set.
- *Selection.* After each freeze, the candidate's test error was computed once at 8 cells: `qd_dev` and Fashion-MNIST at $n \in \{1000, 10000\}$, and `qd_a` and `qd_b` at $n \in \{3000, 10000\}$. A pre-registered ninth cell, `qd_a` at $n = 60000$, checked error, budget and CPU time. The

set	n	v3	v2	v1	hgb	mlp_strong	mlp_wide	mlp_xl	mlp_best+	mlp_pool	cnn
(a) Part 9 held-out sets, development data for v3: v3 seed 0											
qd_a	3000	**17.48**	18.60°	21.28	24.31	23.02	22.67	–	22.67	–	–
qd_a	10000	**14.22**	15.00°	17.21	19.40	18.44	17.83	–	17.83	16.42	12.00*
qd_a	60000	**10.82**	11.97°	14.21	15.23	13.32	12.85	12.46	12.46	12.25	–
qd_b	3000	**21.76**	22.33°	25.13	27.96	26.41	26.13	–	26.13	–	–
qd_b	10000	**16.78**	17.95°	20.27	22.72	21.17	20.64	–	20.64	19.60	–
(b) development sets: v3 and v2 seed 0											
qd_dev	1000	**21.18**	22.22°	26.14	30.31	27.04	–	–	27.04	–	–
qd_dev	10000	**13.30**	14.63°	16.88	18.94	17.49	–	–	17.49	–	–
Fashion	1000	**17.97**	18.19°	18.29°	19.72°	18.60°	–	–	18.60°	–	–
Fashion	10000	**12.38**	12.59°	13.07°	12.94°	12.64°	–	–	12.64°	–	–

Table 20: Test error (%) at every development image cell with a v3 result [Numerical]. Conventions as in Table 8. Panel (a): v3 is seed 0 and the references are seed means. Panel (b): v3 and v2 are seed 0, and mlp_strong, hgb and v1 are seed means. "Fashion" is Fashion-MNIST. Source: study/part10/LEARNER_RESULTS.json.

	all n		$n \geq 10000$	
reference	cells (better/tie/worse)	mean $\log_2$ ratio	cells (better/tie/worse)	mean $\log_2$ ratio
(a) Part 9 held-out sets, development data for v3: qd_a, qd_b				
mlp_strong	5 (5/0/0)	−0.337	3 (3/0/0)	−0.337
mlp_wide	5 (5/0/0)	−0.302	3 (3/0/0)	−0.291
mlp_xl	1 (1/0/0)	−0.204	1 (1/0/0)	−0.204
mlp_best	5 (5/0/0)	−0.337	3 (3/0/0)	−0.337
mlp_best+	5 (5/0/0)	−0.294	3 (3/0/0)	−0.276
hgb	5 (5/0/0)	−0.443	3 (3/0/0)	−0.460
v2	5 (0/5/0)	−0.089	3 (0/3/0)	−0.107
v1	5 (5/0/0)	−0.286	3 (3/0/0)	−0.314
mlp_pool (control)	3 (3/0/0)	−0.204	3 (3/0/0)	−0.204
cnn (context)	1 (0/0/1)	+0.245	1 (0/0/1)	+0.245
(b) development sets: qd_dev, Fashion				
mlp_strong	4 (2/2/0)	−0.207	2 (1/1/0)	−0.212
mlp_best	4 (2/2/0)	−0.207	2 (1/1/0)	−0.212
mlp_best+	4 (2/2/0)	−0.207	2 (1/1/0)	−0.212
hgb	4 (2/2/0)	−0.306	2 (1/1/0)	−0.287
v2	4 (0/4/0)	−0.062	2 (0/2/0)	−0.081
v1	4 (2/2/0)	−0.188	2 (1/1/0)	−0.211

Table 21: v3 against each image reference on the development sets [Numerical]. Conventions as in Table 9. Both panels compare single v3 fits (seed 0) with the references' seed means where 3 seeds exist (v2 in panel (b), mlp_xl, mlp_pool, cnn and all $n = 60000$ cells are seed 0). Source: study/part10/LEARNER_RESULTS.json.

integration rule then used these test errors to qualify the two candidates and choose between them; it took the one with the lower mean $\log_2$ ratio against v2 (Appendix D.2).

The v3 numbers here are therefore the errors of the candidate that these very numbers selected. With two candidates this close the selection effect is small, but these cells are not a test of v3. Each v3 entry is one fit (seed 0).

`qd_a` and `qd_b` (held out in Part 9) [Numerical].

- *References.* v3 is clearly better than `mlp_strong`, `mlp_wide`, `hgb` and v1 at all 5 cells. It is also clearly better than `mlp_xl` at `qd_a`, $n = 60000$ (10.82% against 12.46%, 13.2% lower), and than `mlp_pool` at the 3 cells where both were run (at `qd_a`, $n = 60000$: 10.82% against 12.25%).
- *v2.* All 5 cells are ties, with v3 2.6–9.6% lower (mean $\log_2$ ratio -0.089). Here v2's entries are seed means. Against v2's seed-0 fits, as the integration rule compared them, the margins are similar: for example 17.48% against 18.75% at `qd_a`, $n = 3000$.
- *The context.* The `cnn` on unpermuted pixels is the only learner better than v3: 12.00% against 14.22% at `qd_a`, $n = 10000$, an error 15.6% lower. Among v3's cells, only the one at $n = 60000$ (10.82%) is below it.
- *Comparison with Part 9.* At $n = 60000$, v2 tied `mlp_xl` on both sets (11.97% against 12.46% on `qd_a`; 14.88% against 14.79% on `qd_b`). The control `mlp_pool` matched v2 (12.25% and 14.74%). v3's `qd_a` cell (10.82%) is clearly better than both. v3 was not run at `qd_b`, $n = 60000$.

`qd_dev` and Fashion-MNIST (development throughout) [Numerical].

- *`qd_dev`.* v3 is clearly better than `mlp_strong`, `hgb` and v1 at both cells, for example 13.30% against `mlp_strong`'s 17.49% at $n = 10000$. It ties v2 at both cells (21.18% against 22.22%, 13.30% against 14.63%). v2's cells here were run with multi-threaded BLAS, unlike every v3 and held-out run (Appendix D.3).
- *Fashion-MNIST.* Against `mlp_strong` (= `mlp_best`), v2, v1 and `hgb` every cell is a tie. v3 has 17.97% and 12.38% against `mlp_strong`'s 18.60% and 12.64% (3.4% and 2.0% lower). Only the weaker baseline `mlp` (at $n = 10000$, 14.45%) and `mlp_reg` (20.32% and 14.17%) are clearly worse than v3. This is the set where the bars do not help. On the builder's validation rows, a booster on `cgbt`'s bank was worse there than one on v2's bank (18.30% against 17.74% at $n = 1000$; `study/part10/p10_codes/DESIGN.md`, §4.2). `cgbt`'s raw stacking weight is 0.05 and 0.07 there (shares of the total weight 0.07 and 0.10), against 0.23–0.50 (shares 0.28–0.68) on the QuickDraw development cells.
- *Larger n.* v3 was not run on Fashion-MNIST beyond $n = 10000$. v2's image path (`p9_gates`, Part 9, seed 0) reached 9.64% at $n = 60000$, against `mlp_strong`'s 9.66%.

B.5 Compute of the image fits in detail

CPU time [Numerical]. Three details of Section 5.4:

- *Why v3 is not slower at $n = 60000$.* There v3 takes about as long as v2 (1.05× on `qd_c`, 0.96× on `qd_d`), although it fits a second booster. `cgbt` takes 668 and 760 s. But v3's `ggbt` takes 482 and 474 s against v2's 1091 and 1305 s. The reason is that v3 sets LightGBM's column-wise

layout, which leaves the predictions unchanged (`study/part10/p10_codes/DESIGN.md`, §1). Part 9 found this fix but did not apply it to v2.

- *A like-for-like estimate.* The comparison with v2 is not like for like. v3's time minus `cgbt`'s, 693 and 704 s, estimates v2 with the same layout. By that estimate the second booster about doubles the CPU time of a fit at $n = 60000$.
- *How rough the MLP times are.* They are wall-clock seconds on the shared machine, and they vary with the load from concurrent jobs. The same `mlp_strong` recipe took 384 s per fit at `qd_c`, $n = 1000$, and 98 s at `qd_d`, $n = 1000$. In Part 9 it took 28 s at `qd_a`.

C Interpretability: further detail

This appendix holds the detail behind Section 6: the forms of DFG's models (Appendix C.1), the timing of the mixtures (Appendix C.2), the late targets (Appendix C.3), the models below the discovery size and on the other suites, with the caveats (Appendix C.4), and on images the long edges, the bars used, the prototype pairs (Appendix C.5), the map examples (Appendix C.6) and the per-class comparison with v2 (Appendix C.7). Where v1.3's additions act is in Appendix B.2.

C.1 What DFG's model looks like when it has the structure

The forms [Numerical]. The leading terms at each target's largest analysed n take a few recurring forms.

- *Pure symmetric targets.* One block on the class with its signs, folded ($|t|$) for an accept set symmetric under $t \to -t$ and unfolded otherwise. Examples: band11, $|x_1 + \cdots + x_{11}|$ with share 0.97; h6_01, $(-x_1 + x_2 - x_3 + x_4 + x_5 + x_6 - x_7 - x_8 - x_9)$ with share 0.91. One exception: h5s6, with the asymmetric set $\{-9, -5, -1, 1, 5\}$, gets a folded block. In 11 of the 22 pure targets found, a spectral feature whose support contains the block's inputs (and 9–17 others) sits next to the block.
- *Thresholds.* Degree-1 characters carry as much as the block. On f7_01 the 13-input block has share 0.26, next to $\chi_{\{4\}}$ and $\chi_{\{1\}}$ at 0.19 each.
- *A parity times a symmetric function.* The parity enters first as a character and then modulates a block: fd_04 is $\chi_{\{7,8,9\}} \cdot |x_1 + \cdots + x_6|$, share 1.06.
- *Gates and multiplexer selectors.* They enter as characters and, in v3, as the heavy inner literal of a weighted block: $(x_1 - x_2 - \cdots - x_6 + 3x_{11})$ on h6_07 (share 0.40, from $n = 1024$), $(x_1 + \cdots + x_6 + 3x_{11})$ on f7_03 (0.24).
- *Two sums.* A product block over the joint cells, as on twosym ($|x_1 + \cdots + x_5| \times |x_6 + \cdots + x_{10}|$, share 0.72), or a block over the union with one side's signs flipped.
- *A quirk.* In 15 cells (13 targets) the true block appears inside a product with a factor on two irrelevant inputs (shares 0.13–0.40), for example $|x_1 + \cdots + x_7| \times |x_{15} - x_{16}|$ on sym7 at $n = 128$ (share 0.40). The test error there is at or near 0. The product test scores the whole joint partition, not its increment over the stronger factor, so a strong block carries any second factor through the test (confirmed in the code).

Weighted targets: is the inner term right? [Numerical]. Six targets are weighted sums $u = s + a\tau$.

- *h6_10* $= [|s-3x_{13}| > 1]$, $s = x_1+\cdots+x_{12}$. At $n = 256$ the weight is right and the relative sign wrong (Appendix B.2); the DFG error, 0.017, falls entirely in the four cells $(t, x_{13}) = (\pm 8, \mp 1)$ and $(\pm 10, \mp 1)$, 68 of 4096 test points.
- *f7_10* $= [s + 2x_6 \leq -1]$: a block modulated by $\chi_{\{6\}}$ at $n = 128$, the exact $(x_1 + \cdots + x_5 + 2x_6)$ at 256.
- *h5t0*: a product with $|x_8 + x_9|$ at 256, the exact weighted block at 512.
- *thrsum and h5t1* are fitted through a product with a factor on the parity's two inputs. In h5t1 the weight is 3 where the truth has 2.
- *sumpar* has a degree-3 inner term, outside v1.3's class. Characters carry it: DFG error 0 at $n = 256$, 0.013 at 512. Both classes are matched in one model only at 2048.

A product with $|x_j \pm x_l|$ represents every function of $(s, x_j x_l)$, so the product forms are exact representations, though not the weighted ones. The five targets other than sumpar are fitted to DFG error 0 at their largest analysed n.

C.2 The 29 mixtures

Timing [Numerical]. For targets with a parity, gate, selector or second block, the round-0 test on the true block is a cruder proxy for what DFG does.

- *A corrected detection test.* The analysis script tested a modulated block on a residual that was neither centred nor fitted on the modulator, and so it counted the modulator character's own signal as block signal. DFG tests modulated blocks on a residual already fitted on χ_M. A verifier recomputed the statistic that way. For pure parity mixtures nothing changes. For seven gate and multiplexer targets the detection point moves later: fd_11 and h5s4 from 128 to 256, f7_03 from 128 to 512, f7_11 from 128 to 1024, h6_07 from 256 to 512, and majsym and fd_08, whose modulated statistic never passes, to the points where their plain block passes (512 and 2048).
- *With the corrected points*, 39 of the 54 targets are detectable at $n = 128$ (43 in the analysis file), and the median delay over the 49 found is 0 doublings (1 in the analysis file). Spearman's ρ between delay and low-degree share is -0.54 over all 54 (-0.44 with the uncorrected points).
- *Where the proxy fails.* fd_08 is found at $n = 1024$, before its plain block passes the round-0 test at 2048: in DFG's own run the gate's characters are fitted first. For the weighted targets detection uses the weighted block but n_{struct} the unit-weight class, so h5t1 counts as two doublings late only because of the definitions.
- *Search orders.* For 8 of the 49 targets found, the search's spectral and degree-1 orders never reach the block on the plain residual: the 7 parity mixtures fd_02, fd_04, symxor, h5s1, h6_06, f7_09, symxpar, and the multiplexer h6_07. On 22 targets the block is found before those orders list its inputs, presumably through greedy growth, local refinement or, at $n \leq 128$, the beam (not measured). The reach measure covers only the eigenvector and degree-1 orders.

C.3 Late targets

The targets [Numerical]. The pure targets found late ($n_{\text{struct}} \geq 1024$, two or more doublings after detection), and h5s1, are consistent with the same mechanism in weaker form. (h6_08 and sumpar are also late, for representational reasons; sumpar's inner term is outside v1.3's class.) Each is one seed, so the coincidences below are suggestive, not established.

- *band11* $= [|x_1 + \cdots + x_{11}| \in \{3, 5\}]$ ($W_1 = 0$, $W_2 = 0.047$): detectable at 128, found at 1024, the first n at which the top eigenvector order lists all 11 inputs among its first 16. At $n = 512$ one sign flip drops the statistic from 5.4 to 0.61 of the threshold, and pairs inside S are at noise level (0.27).
- *h5s7*, an arbitrary set of a 13-input signed count ($W_1 + W_2 = 0.015$): detectable at 128, found at 8192, when the eigenvector order covers 0.92 of S. At $n = 2048$ the true block is 18.6 times the threshold, but only 12% of candidates with three mixed errors pass.
- *h5s1*, a signed 9-input count times the parity $x_{10}x_{11}$: 92% of its variance is on the single degree-11 character $x_1 \cdots x_{11}$. The plain block carries no signal. The modulated block passes from $n = 128$, but DFG takes modulators only from characters already in its model, and $\chi_{\{10,11\}}$ has coefficient -0.031. So the bottleneck is detecting the modulator. It is found at $n = 16384$ on seed 0 (whose own 10% error point is at $n \approx 14000$), and the 3-seed n^* is "never".
- *h5s2, h5s3* (arbitrary sets; $W_1 + W_2 = 0.007$ and 0.03): found at 1024 and 2048.

C.4 Below the discovery size, the other suites, and caveats

Below the discovery size [Numerical]. 216 analysed cells of the 49 found targets lie below n_{struct}.

- *Contents.* The summarised model is the forced variant in 177 cells and the default in 39. The default variant is empty (constant and kernel background) in 26 of its 39 cells; the others hold characters of degree 1–5, true or spurious. The forced variant holds 1–2 forced characters of degree 1–4 (up to 7 from the subsampled high-degree search). They are spurious, outside the relevant inputs, in 152 of the 177 cells, and 87 cells also hold a true character such as a parity partner, a gate literal or a threshold's input. Partial blocks appear in 21 cells, 14 of them one doubling before n_{struct}. In 83 of the 216 cells the summary is an out-of-fold fit, because the stacker gave DFG no weight.
- *Error.* The DFG member's median test error is 0.41 below n_{struct} (the portfolio's 0.37) and 0 from n_{struct} on.
- *Stacking weight.* The two DFG members' median share of the stacking weight is 0.21 below n_{struct} (mean 0.33) and 0.89 from n_{struct} on (mean 0.87). DFG has nonzero weight in 62% of the cells below. By $\log_2(n/n_{\text{struct}})$ the median share is 0.03 at -3, 0.21 at -2, 0.44 at -1, 0.87 at 0 and 0.92 at $+1$. Below n_{struct} the total weight is small (median 0.35, against 1.52 from n_{struct} on; all weights are 0 in 17 cells).

Weight shares multiply log-probabilities and are not contributions to the prediction. What they show is that the stacker's trust in DFG changes abruptly, within one or two doublings of n_{struct}, and that below that point the DFG members get a modest share (median 0.21) of a total weight that is itself small (median 0.35, against 1.52), so their spurious characters have little absolute weight [Heuristic].

The other suites [Numerical]. On the 70 targets of `designed`, `random` and `dlheld`–`dlheld3`, every model reaches test error ≤ 0.005 at its largest analysed n. These runs, too, reproduce the evaluation's seed-0 errors at every cell (Appendix D.4), and their n_{struct} is seed 0. Parities are their own characters, and small juntas their full Fourier expansions. Majorities and symmetric targets get one block (maj9, maj15, maj21 and sym9 at $n_{\text{struct}} = 128$, sym8 at 64). Halfspaces with unequal weights are carried by degree-1 characters (thr10: 209 of them at $n = 4096$). Formulas are carried

mostly by characters of degree 1–4: the characters present in the model cover a per-suite median of 0.77–0.88 of the target's non-constant Fourier weight, and at least 90% on 23 of the 67 targets with an exact Fourier table (presence only; coefficients not compared). 25 of the 70 models have a sum block among their three leading terms, 9 of them weighted. Outside the sum-structured suites the model is thus what Part 6's DFG learned, a sparse set of low-degree characters plus the linear block for halfspaces [5], with sum blocks where a target has a symmetric class of five or more inputs [Heuristic].

Caveats.

- The summarised model depends on the stacker. When both DFG variants are kept, only the forced one is seen; when neither is kept, the summary is the last out-of-fold DFG fit, on fewer rows.
- Every structure measurement in Section 6.1 and here is seed 0; n^* values marked 3-seed are means over seeds 0–2. n_{struct}, n_{detect} and the model contents can differ on other seeds.
- The symmetry classes are found from 4096 random inputs. Several controls have small classes, which is why controls are excluded from every summary of sum-structured targets.
- The structure runs store only the 10 leading terms per cell, while n_{struct} uses all terms; on 8 v3 cells the matching term is beyond the stored ten.
- The exact-structure match accepts a modulator χ_M, which carries no signs, for a signed class. It changes no n_{struct} reported here.

C.5 Images: long edges, the bars used, and the prototype pairs

Long edges in the pooling-set graphs [Numerical].

- *Variants collapse when a few edges are long.* `qd_a`'s $k = 9$ union graph is about as local as `qd_dev`'s (0.989 and 0.994 of the other members within two steps), but it has 10 edges longer than 5 steps, and its R^2 is 0.24 against 0.86. A verifier removed the edges longer than 5 steps (an oracle ablation that uses the true grid). R^2 then rose to 0.96–0.99 on every QuickDraw and MNIST graph, and to 0.95 on Fashion-MNIST's $k = 9$ union graph.
- *Not always.* MNIST's $k = 4$ graph has 23 edges longer than 5 steps and still scores 0.92.

Which bars are used [Numerical].

- *Straightness.* On QuickDraw the bars weighted by their gain are straighter than the average bar at $L = 4$ (0.32 exact lines against 0.24 at `qd_dev`, $n = 10000$), but not at $L = 2$ (0.43 against 0.42). Straightness correlates with how often the bar's centre is inked (+0.48 for $L = 4$), and the ink-frequency mix of the used bars alone predicts 0.30 of the 0.32. On MNIST a preference remains after that correction ($L = 2$: 0.36 weighted, 0.20 for all bars, 0.27 predicted from ink frequency).
- *Orientation.* On QuickDraw the gain splits over the four orientations as the bars do. On MNIST horizontal bars carry 0.40 of the bar gain at both lengths, against 0.27 and 0.26 of the bars. But several of MNIST's top bars lie in margins that are rarely inked; the top $L = 2$ bar (4 pixels), centred at row 1, column 18, has pixels inked in 0–7.5% of training images. So they look more like detectors of a digit's extent than of strokes.

These gain-weighted statistics rest on a few bars of one fit.

ggbt's prototype pairs [Numerical]. The 300 pairs carry 0.38–0.41 of ggbt's gain (0.30 on qd_a), and almost all are split on, but the top 10 carry 0.29 (qd_dev), 0.50 (Fashion-MNIST) and 0.36 (MNIST) of the pair gain. In several of the class pairs with the most gain one drawn pair carries most of it (grapes–soccer ball 0.85, motorbike–frying pan 0.85, shirt–sneaker 0.89, MNIST 0–1 0.90), but not in all (trouser–boot 0.59, MNIST 1–2 0.69). Per drawn pair, relative to the average pair, on Fashion-MNIST the most gain goes to contrasts of very different silhouettes (trouser–boot 5.8×, shirt–sneaker 4.9×), and similar classes get almost none (T-shirt–dress 0.01, sandal–boot 0.02, T-shirt–shirt 0.04). Elsewhere there is no such pattern: qd_dev's top class pair is grapes–soccer ball (5.3×), two round shapes and its third most confused pair. MNIST's pairs (1–2 4.0×, 0–1 3.5×) are from ggbt's out-of-fold fit, which has stacker weight 0. Over class pairs, gain per pair correlates with ggbt's own confusion between the two classes at −0.59 on Fashion-MNIST and at −0.34 to +0.07 elsewhere. Which class pairs win depends on which drawings happened to be drawn: on qd_dev the top one changes between $n = 3000$ and 10000. *Reading* [Heuristic]: a projection on $x_{\text{trouser}} - x_{\text{boot}}$ is a coarse template that separates two groups of silhouettes, which the trees refine with local features.

C.6 Images: examples from the evidence maps

Examples (Figure 7). The shares are cgbt / ggbt, as parts of the class's net evidence; the descriptions are [Heuristic], and the regions were named after looking at the maps.

- *MNIST.* 0: the hole carries 0.46 / 0.41 of the evidence, less than half, on 3% of the image and 99% on blank pixels. 1: strips beside the stroke (0.46 / 0.32), which cgbt draws as bars parallel to it. 7: a '/' band of blank pixels left of the stem (0.30 / 0.23). 3: the open left side (0.32 / 0.23). 4: the open top (0.28 / 0.25). 8: the central crossing, on ink (0.26 / 0.30).
- *qd_dev.* Motorbike: the blank gap between the wheels at the bottom centre (0.20 / 0.16 on 2.6% of the image, where the class's mean ink is 0.04 and the other classes' 0.28). Ocean: for ggbt the horizon line (0.40 on 11% of the image). For cgbt, the horizon's ends (0.17) and a diagonal band of mostly blank pixels from the left edge to the bottom centre (0.26 on 8% of the image, 84% on blank pixels), where ocean has little ink (0.13, against 0.26 for the other nine classes) and soccer ball, campfire, grapes and washing machine have the most (0.37, 0.33, 0.30, 0.29). Campfire: the base of the fire, on ink (0.36 / 0.32). Soccer ball: cgbt follows the upper-left arc as a chain of diagonal bars (0.18 against 0.12). Foot: the blank middle (0.22 / 0.17).
- *Fashion-MNIST* (not in the figure). Trouser: the gap between the legs (0.26 / 0.31). Boot: a blank strip left of the shaft (0.41 / 0.25; the class's mean ink there is 0.001, the others' 0.34). Sandal and sneaker: a blank upper-left region (0.39 / 0.30 and 0.36 / 0.26).

C.7 Images: v3 against v2 by class

v3 against v2 at qd_dev, $n = 10000$: per class [Numerical]. The two fits share a bit-identical ggbt (Section 6.2.2).

- *Per class, v2 → v3.* Campfire 0.200 → 0.158, grapes 0.280 → 0.243, matches 0.269 → 0.247, motorbike 0.110 → 0.095, soccer ball 0.082 → 0.075, washing machine 0.060 → 0.055, paper clip 0.138 → 0.134, ocean 0.087 → 0.084, frying pan 0.121 → 0.119, foot 0.120 → 0.120.
- *Where cgbt beats ggbt.* The largest gains are mostly on the classes where cgbt alone beats ggbt alone most: campfire (0.155 against 0.201), matches (0.254 against 0.288), grapes (0.229 against 0.262). But the match is not one for one. Frying pan has the fourth-largest cgbt

advantage (0.026) and almost no gain (0.002), and the rank correlation over the ten classes is about 0.79. Foot is the only class where `cgbt` alone is worse (0.123 against 0.121). No measurement ties the per-class gains to particular bars.

- *Other sets.* On Fashion-MNIST, v2's cell has 0.1259 and v3 0.1238, but the two runs' `ggbt` differ (162 against 242 rounds; out-of-fold 0.1321 against 0.1271), and `cgbt`'s weight share there is 0.10. So the difference cannot be attributed to `cgbt`. On `qd_a` v2's and v3's cells give 0.1509 and 0.1422, and `ggbt` stops at 206 rounds with the same out-of-fold error in both. On MNIST v2's image path gives 0.0299 (Part 9's evaluation) against v3's 0.0255 here.
- *Cost.* `cgbt` takes 302 of the 530 CPU seconds of v3's evaluation fit at `qd_dev`, $n = 10000$.

On Fashion-MNIST, where the bars are rarely lines and the evidence is mostly the silhouette's outline, the two boosters are about equal, and the stack gains nothing measurable from `cgbt` [Heuristic].

D How Part 10 was run

Part 10 and this paper were produced by an AI system, Claude (Anthropic), on 27–28 October 2026. Kaarel Hänni posed the request: "a couple more hours" on a better discrete learner, then interpretability of the best learner so far, and then a paper about it that explains what kind of learner it is, how well it does against the baselines, and what the interpretability shows. No human has reviewed the code, the results or this text. Times are UTC. The records are:

- the protocol `study/part10/PROTOCOL.md`, written before any design work and amended only by dated decision entries and notes;
- the builders' notes `study/part10/p10_sum/DESIGN.md`, `study/part10/p10_hier/DESIGN.md` and `study/part10/p10_codes/DESIGN.md`;
- the assembly record `study/part10/hyb_portfolio3/ASSEMBLY.md`;
- the verification record of the interpretability analyses, `study/part10/interp/VERIFICATION.json`.

Organisation. One orchestrating session wrote the protocol, fixed the held-out sets, ran every held-out evaluation and reference run, applied the rules and assembled v3. It delegated five kinds of work to separate agents with their own instructions: the three builders; two interpretability analysts; four verifiers of the analyses; the drafting of sections of this paper; and three checkers of its claims (Appendix D.5). The builders worked on development data only. The orchestrator read no held-out result of a Part 10 learner before that learner was frozen (Appendix D.3 lists what it did read).

D.1 Protocol and builders

The protocol was committed at 18:50 on 27 October (`dc70266c`), before any design work. It fixed the held-out sets `dlheld7`, `qd_c` and `qd_d` and the development draw `dlfamdev`; the budgets; the three builders' tasks and their fixed development evaluations; the integration rule; the held-out evaluation with its references; the best-learner rule; and the plan for the interpretability analysis and this paper (Section 3.4).

builder	component	development	frozen (md5)	fixed development evaluation
T, `p10_sum`	DFG v1.3: weighted sum blocks, forced blocks at $n \geq 48$, beam at $n \leq 128$	18:52–19:44 (52 min)	19:44 (`5ee995ad`; `dfg13.py 69cabe5c`)	20:00–21:35 on `designed`, `dlsymdev`, `dlfamdev`, 3 seeds; kr_vs_kp at $n = 1000$
I1, `p10_hier`	a max-pooling second level in `ggbt`'s bank; pairs dropped	18:52–19:45 (53 min)	19:45 (`1da21bf7`)	19:45–21:35 on the 8 image cells and `qd_a` at $n = 60000$, seed 0
I2, `p10_codes`	a second booster `cgbt` on pooled sums and bars	21:37–22:10 (33 min)	22:10 (`fd688acd`)	22:10–23:43, the same 9 cells; interrupted at 22:40 and resumed (below)

Table 22: The three Part 10 builders, 27 October. Times from each builder's `DESIGN.md`; the evaluation's end for `p10_codes` is the commit that recorded it (`16ef5fde`). Every evaluation cell records the md5 of the file it ran.

The builders are summarized in Table 22. Each changed one component of v2, delivered a full learner file, a `DESIGN.md` and an md5 at its freeze, and then ran the fixed development evaluation once. The image builders used validation rows of the development sets only. The toy builder scored its DFG-alone checks on fresh validation draws, and its portfolio checks on the toy harness's test points of the development suites (`designed`, `dlsymdev`, `dlfamdev`, `dlheld6`). No builder used a held-out set. Three events during the evaluations are recorded in the notes:

- *`p10_sum`.* Its evaluation was started at 19:44 as 8 shards in one shell session. The machine's autogroup scheduler gave the shards together about 0.55 cores, so at 20:00 they were stopped, before any result was written, and restarted unchanged in four sessions. Only the restarted run produced results.
- *`p10_hier`.* Its `qd_dev` process was killed by the shared memory limit at about 20:29 and restarted; the finished $n = 1000$ cell was kept and $n = 10000$ refitted with the same random streams. Its $n = 60000$ cell waited 1 h 28 min for the shared large-fit lock and ran 21:13–21:35.
- *`p10_codes`.* It started after the other two builders had finished. The container restarted at 22:40, with 2 of its 9 cells done, and the workflow that ran the builders was lost. The orchestrator resumed the evaluation with the builder's own script. The cells are resumable and record the learner's md5, which was unchanged. The orchestrator took the builder's summary from its `DESIGN.md`, whose evaluation and cost placeholders (§3, §5) the builder never filled.

D.2 The integration and best-learner decisions

The rules (Section 3.4) were applied by `study/part10/compare_part10.py` from the result files, and each decision was entered in the protocol when it was made. The numbers are in `study/part10/RESULTS.md`.

- *Toy component: adopted* (21:35; commit `33df158e`, 21:50). Against v2 on `designed`, `dlsymdev` and `dlfamdev` (41 targets, inputs permuted, 3 seeds), `p10_sum`'s mean $\log_2 n^*$ ratio is -0.066 over the 40 targets both learn: -0.028 on `designed`, -0.178 on `dlsymdev` and -0.055 on `dlfamdev`. The largest increase is $+0.26$ (fd_06). The rule asked for a mean below -0.05 and no increase above 0.5. v2's development fits are `p9_sum`'s on `designed` and `dlsymdev`, the same code on toy inputs (110 of 110 seed-0 cells on `designed` checked equal), and v2's own on `dlfamdev`.

- *Image component: `p10_codes` adopted* (23:40). Both candidates qualified, and the rule takes the one with the lower mean $\log_2$ error ratio against v2 over the 8 development cells. `p10_codes` had -0.078; at `qd_a`, $n = 60000$, 10.82% error (limit 12.57%), 1453 CPU s (limit 2636 s) and 0.58 of the FLOP budget. `p10_hier` had -0.071, 11.08%, 924 s and 0.44. The builders' own numbers used Part 9's `p9_gates` development errors for v2 on `qd_dev` and Fashion-MNIST (so `p10_hier`'s note reports -0.068); the rule used v2's own runs on those cells, made for Part 10.
- *Assembly and freeze* (23:45; commit `20003f85`, 23:50). The script `study/part10/assemble_v3.py` built v3 from the learner file of `p10_codes`, with DFG v1.3. `learner.py` has md5 `b045fafe1e24418620bf8c503aa601c6` and `dfg13.py` `69cabe5c50345cb24079198ed52b562f`. All 15 identity checks on development data gave identical predictions: v3 against `p10_sum` on 12 toy fits and on kr_vs_kp, against `p10_codes` on a `qd_dev` fit, and against v2 on optdigits. The files were made read-only.
- *Scheduling.* Neither image candidate changes any non-image code path, so v3's behaviour on toy inputs was fixed by the toy decision alone. v3's held-out toy evaluation therefore started after the toy decision, as `p10_sum` on `dlheld7`. Its results were not read before v3 was frozen, and v3 $\equiv$ `p10_sum` on toy inputs was checked at assembly.
- *Best-learner rule: v3* (commit `adcbfd0c`, 04:20 on 28 October; the protocol entry says 04:25). It was applied after every held-out cell of v3 and v2 was in. The mean $\log_2$ error ratio of v3 to v2 over the four held-out image cells with $n \geq 10000$ is -0.125 (rule: below 0), and the mean $\log_2 n^*$ ratio on `dlheld7` is -0.004 (rule: at most $+0.1$).

D.3 Disclosures and notes

What was seen of held-out data. The protocol and the builders' notes record three items.

- *The `p10_hier` builder*, about 15 minutes after its freeze, printed the last 5 lines of the orchestrator's log of the baseline `mlp` on `qd_c` while finding out why its $n = 60000$ cell was waiting for the shared lock. Those lines are the baseline MLP's `qd_c` test errors at $n = 100$–3000. Nothing about a discrete learner on `qd_c` was shown, and the learner file was already frozen and read-only. `p10_hier` was not adopted.
- *The `p10_sum` builder* recorded incidental exposures: a process listing that showed the command lines of reference jobs on `qd_c`, and a one-line function signature `make_dl_heldout7_suite` printed at the end of a source range, with no body. Nothing about `dlheld7`'s targets was seen.
- *The orchestrator*, before v3's freeze, read reference results on held-out data: the `nets_wd` training logs of f7_00–f7_04 on `dlheld7` and the MLP references' `qd_c` progress lines. It read no result of v2 or of a Part 10 learner on `dlheld7`, `qd_c` or `qd_d`. v2's `dlheld7` results and v3's toy results there were committed unread.

BLAS threading. The image analyst found at about 02:00 on 28 October that v2's own development cells on `qd_dev` and Fashion-MNIST had been run without forcing single-threaded BLAS, while the candidates' cells and every reference lane force it. Thread count changes summation order, which can move nearest-neighbour ties and booster early stopping: v2 at `qd_dev`, $n = 10000$, seed 0 gives 0.1463 multi-threaded and 0.1467 in a single-threaded analysis refit (`ggbt` stopped at 166 and 123 rounds). An earlier analysis refit, with unrecorded thread settings, reproduces the cell's 0.1463 exactly, so the BLAS explanation fits the evidence but was not verified. The integration rule

compared both image candidates with the same v2 numbers, so the choice between them (-0.078 against -0.071) is unaffected, and both are far from the -0.05 threshold. Every Part 10 held-out run, of v3, v2 and the references, is single-threaded. The note was added to the protocol at 02:03 (`a2b8470e`).

Memory and job control. Up to about ten jobs shared 4 cores and 16 GB. The builders ran at nice 0. Every reference and analysis process ran at nice ≥ 5, was resumable (per cell for the image evaluations, per target for the networks), and had a raised `oom_score_adj`, so that under memory pressure the kernel would kill it before a builder's process or v3's evaluation. Large fits ($n \geq 30000$) took a shared lock. The out-of-memory kills were:

- the baseline `mlp` on `qd_c`, at about 20:00–20:15 during the builders' evaluations, after its $n \leq 10000$ cells had finished;
- `p10_hier`'s `qd_dev` evaluation process (about 20:29, above);
- v2's held-out image evaluation, more than once between about 22:00 and 00:30. One kill, at about 00:24, happened while two image analysis fits were running; the analyst then held its own processes at the highest kill priority, and v2's held-out cells were moved to nice 0 and out of the killer's preference, since they feed the best-learner rule;
- reference processes while v3's $n = 60000$ cells ran: v1 on `qd_c`, `mlp_xl` on `qd_d`, `mlp_pool` and `cnn` on `qd_c`;
- `nets_cewd0.3` and `nets_cewd1.0` on `dlheld7` at about 02:40, with 6 of 12 targets saved each (both have since completed all 12).

Every killed job was rerun with the same code and seeds, keeping its finished cells or targets. The container restart at 22:40 also stopped every job lane; the lanes were resumed the same way.

D.4 The interpretability workflow

Analysts. Two analysts, one for the toy side and one for images, ran instrumented reruns of v3's (and, for comparison, v2's) seed-0 evaluation fits, with the learner's internals recorded (Section 6). The protocol fixed the analysis as analysis of the chosen learner, to be run after the held-out results were in; in practice it ran in parallel with the held-out evaluation, on v3 before the best-learner rule had chosen it. The image analysis used `qd_dev`, `qd_a`, MNIST and Fashion-MNIST and opened no result on `qd_c` or `qd_d`. The toy analysis covers every toy suite, and the six sum-structured ones in detail, `dlheld7` included; its `dlheld7` runs were made from 00:16 on 28 October, after v3's freeze. The work ran in parallel with the held-out evaluation and was committed at 03:54 on 28 October (`9214cfde`), before the best-learner rule was applied at 04:20. The rule is mechanical, and the analysis did not touch its inputs. The protocol fixed the analysis as descriptive, not as evidence that the learner is better than anything.

Identity of the analysed fits. Every analysed fit is seed 0, and each (target, n) or (set, n) pair is one fit. The fits are instrumented reruns of the evaluation's own fits: the same training rows, column permutation and random streams, with the learner's internals recorded.

- *Toy.* At all 875 cells of v3 (every toy suite) and all 477 cells of v2 (the six sum-structured suites) that have an evaluation result, the reruns' test errors equal the evaluation's seed-0 errors. Equal errors at every cell are strong evidence that the fits are the same, but not a proof. The analysis's own check (`check_identity` in `study/part10/interp/toy_interp.py`) covered

587 of the v3 cells, because v3's result files on `random` and `dlheld`–`dlheld3` were incomplete when it ran. `paper_learner/check_toy_identity.py` repeats the same comparison on the final result files (output: `check_toy_identity.json`); it adds the 288 cells of those four suites, all equal.

- *Images.* The four v3 fits that have an evaluation cell (`qd_dev` at $n = 1000$ and 10000, `qd_a` and Fashion-MNIST at $n = 10000$) reproduce its test error, stacking weights, `ggbt`'s number of rounds and every member's out-of-fold error. The fits at `qd_dev`, $n = 3000$, and MNIST, $n = 10000$, have no cell and are unchecked. No result on `qd_c` or `qd_d` was opened. v2's analysis fit at `qd_dev`, $n = 10000$, differs from its evaluation cell by BLAS threading (Appendix D.3).

Verifications. The analyses are `study/part10/interp/TOY_INTERP.md` and `study/part10/interp/IMAGE_INTERP.md`, with their scripts, tables and result files in the same folder; the verifications are in `study/part10/interp/VERIFICATION.json`. Each side had two independent verifiers. One recomputed every number from the files with its own scripts; the other reviewed the readings against the files and the code. Where a verifier corrected a number or a reading, Section 6 uses the corrected one. Table 23 counts the verdicts. "Corrected" means that the verifier changed a number, narrowed a statement or rejected a reading. Each verifier also listed further problems that were not tied to one claim.

side	verifier	claims	confirmed	corrected	further problems listed
toy	numbers	13	8	5	12
toy	reading	13	2	11	13
images	numbers	11	10	1	12
images	reading	11	2	9	11

Table 23: The verifications of the interpretability analyses. Source: `study/part10/interp/VERIFICATION.json`.

The corrections that change numbers in Section 6 are these:

- *The detection test for modulated blocks*, recomputed as DFG tests it (Appendix C.2): seven gate and multiplexer targets became detectable later, so the targets detectable at $n = 128$ went from 43 to 39, the median delay over the targets found from 1 doubling to 0, and Spearman's ρ over all 54 from -0.44 to -0.54.
- *Counts and scopes.* The beam and forced blocks move 5 pure symmetric targets plus one gate mixture, not 6 pure targets. The pooling-set locality fractions exclude the centre pixel. The share of evidence on blank pixels is not above one half for every class; several classes are listed as exceptions.

The corrections of readings are these:

- the "learned function of the block" is the whole DFG model's conditional mean, not the block's own coefficients;
- for fd_10 the pair statistics are small (about 0.002) but not zero;
- stacking-weight shares are not contributions to the prediction;
- split-gain shares must be read against column shares;
- the bars' straightness is a property of the construction and the data, not of the booster;

- the regions named in the contribution maps were chosen after looking at them.

Section 6 uses the corrected numbers and wording; where the claims check below found a correction that had not been carried over, it was applied. `make_interp.py` hard-codes the verifier's corrected detection points and regenerates every interpretability figure and table from the analysis files.

D.5 This paper and reproducibility

This paper. Its sections were drafted by the orchestrating session and by separate writing agents, each from the records it names, and then compiled together. Every table and figure is generated from the result files: `make_tables.py` and `make_figures.py` from `study/part10/LEARNER_RESULTS.json` and the result files it reads, and `make_interp.py` from the analysis files, applying the verifiers' corrections. The plotted values are in `figures/*.json` (`fig_toy_nstar.json`, `fig_images.json`, `fig_interp.json`). `study/part10/LEARNER_RESULTS.json` is written by `study/part10/compare_learner.py`, whose docstring says which result file stands for v3 on each input type.

Which result files stand for v3.

- *Toy.* On toy inputs v3 runs exactly the code of the Part 10 toy candidate `p10_sum`; its predictions were checked identical at assembly (`study/part10/hyb_portfolio3/ASSEMBLY.md`). v3's toy results are therefore the `p10_sum-perm` result files in `study/hybrid/results/`. v2's are the `hyb_portfolio2-perm` files, and on `designed` and `dlsymdev` the `p9_sum-perm` files (the same code on toy inputs).
- *Images.* On `qd_c` and `qd_d` they are v3's own runs (`hyb_portfolio3`). On the development sets it is the frozen image candidate `p10_codes`, whose predictions on images are identical to v3's (checked identical on one `qd_dev` fit at assembly, $n = 300$; `study/part10/hyb_portfolio3/ASSEMBLY.md`).

Claims check. After drafting, three independent AI checkers, each with its own instructions, checked the paper's claims on 28 October. One covered the two results sections, one the interpretability section, and one the abstract, the introduction, Sections 2, 3 and 7 and this appendix. Each compared the text with the result files, the records named in it and the learner code, recomputed what it could, and listed each problem with a severity (error, misleading, minor) and a proposed correction. They found stale counts from runs that had landed after drafting, a few arithmetic and transcription errors (for example the `cnn`'s relative error on `qd_a`, and the order of HELDOUT2 and v1's freeze), several verifier corrections of the interpretability analyses that had not been carried over, statements broader than the evidence (for example "fail because" for a descriptive seed-0 analysis), and details of the learner description that did not match the code. Their corrections were applied before the final regeneration of every table and figure, with counts updated for the runs that landed in the meantime (`random` for v3, the remaining `dlheld7` network targets, and the image references on `qd_c` and `qd_d`). The check is not an audit of leakage, compute or reference strength of the kind made in Parts 8 and 9.

Editorial revision. After the claims check, three reviewing agents (on readability, structure and layout) commented on the presentation, and an editing agent reorganised the paper: it added the comparison with familiar learners and the hypothesis class of Section 2, Table 1, the glossary and target definitions of Appendix A, and moved in-sample and per-target detail to Appendices B and C. A further agent compared every number of the revised text with the checked text, and the few new

ones (the spread of the `dlheld7` ratios, the split of the 30 calibration wins, the `cnn`'s recipe) with the result files and code.

Regenerating the numbers. From `gd-circuit-growth/`:

- `cd study/part10 && python3 compare_part10.py` (the rules; writes `RESULTS.md`);
- `python3 compare_learner.py` (every comparison; writes `LEARNER_RESULTS.{md,json}`);
- `cd ../../paper_learner`, then `python3 make_tables.py`, `python3 make_figures.py`, `python3 make_interp.py` and `python3 check_toy_identity.py`;
- `latexmk -pdf main.tex`.

The result files are:

- `<learner>__<dataset>.json`, with `.info.json` records, in study/standard_tasks/results/;
- `toy__<learner>__<suite>.json` in study/hybrid/results/.

What is not in the repository. The job drivers and queue scripts of Part 10, the builders' development scripts and logs, and the analyses' raw outputs stayed in the session scratch directory. The drivers only wrap the committed scripts, the builders' notes give the commands of their evaluations, and copies of the analysis outputs that the paper uses are in study/part10/interp/results/.